

Introduction to DBMS and ER Data Model

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Database Management System

Database: A collection of related data. It should support

- Definition
- Construction
- Manipulation

Database Management System: A collection of programs that enable the users to create and maintain a database.

- Objective of DBMS is to provide a convenient and effective method of defining, storing and retrieving the information which are stored in a database.

Difference between Data and Information

Data	**Information**
Data is unorganised and unrefined facts	Information comprises of processed, organised data presented in a meaningful context
Data does not depend on information	Information depends on data
Example: product name, name of student	Example: report card of a student
Data is an individual unit that contains raw materials which do not carry any specific meaning	Information is a group of data that collectively carries a logical meaning
Understanding is difficult	Understanding is easy

Features of Database

1. Data storage, retrieval, and update:The ability to store, retrieve, and update the data that are in the database.

2. User-accessible catalog: where descriptions of database components are stored and are accessible to the users

3. Shared update support: A mechanism to ensure accuracy when several users are updating the database at the same time

4. Backup and Recovery Services: Mechanisms for recovering the database in the event that a database is damaged somehow.

5. Security Services: Mechanisms to ensure that certain rules are followed with regard to data in the database and any changes that are made in the data. (collective measures used to protect and secure a database or Database management software from illegitimate use and malicious cyber threats and attacks)

Features of Database

5. Integrity services: Mechanisms to ensure that certain rules are followed with regard to data in the database and any changes that are made in the data.

6. Data Independence: Facilities to support the independence of programs from the structure of the database.

7. Replication support: A facility to manage copies of the same data at multiple locations.(it is the frequent electronic copying of data from a database in one computer or server to a database in another)

5. Utility Services: DBMS provided services that assist in the general maintenance of the database. (it also provides a set of utilities for managing and controlling database activities such as reorganization, back up and copy, recover, integrity check, load data, unload data and repair database)

- Multiple users are making updates to the database at the same time.

Problem:

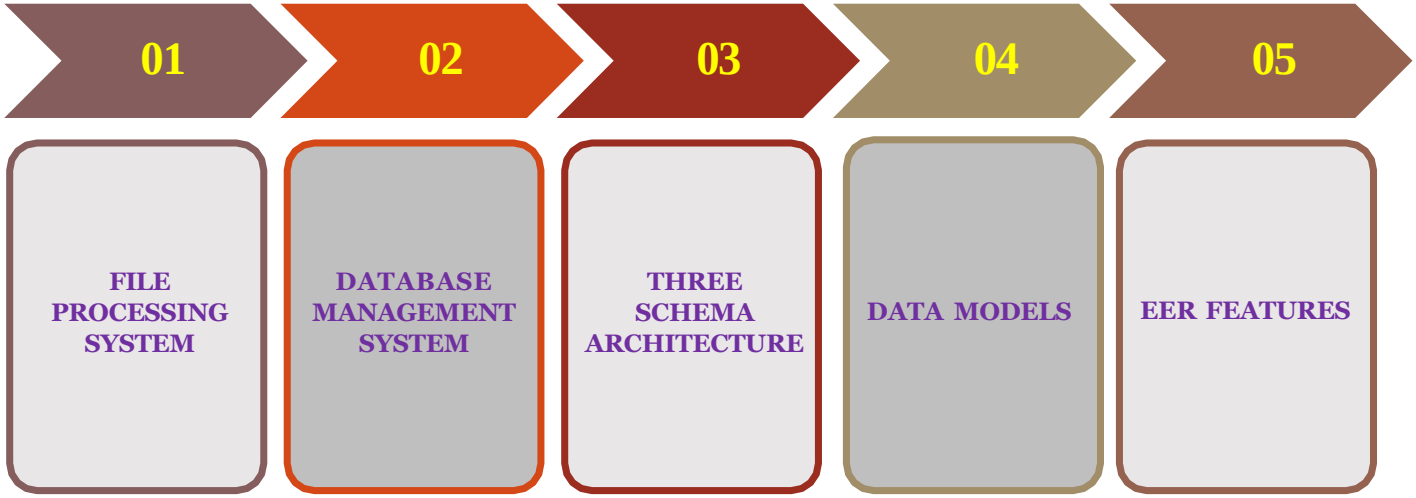
- Multiple people updating the database simultaneously can override each other

Example:

- Agents T1 &T2 simultaneously read the seats reserved on Flight 890 i.e. 80
- T1 cancels 5 seats updating the seats reserved on Flight 890
- T2 reserves 4 additional seats on the flight and updates the seats reserved on Flight 890
- If T1 updates the database before T2. T2 will override T1's change
- Similarly if T2 updates before T1 then also problem occurs.

- Batch Processing
  - Allow multiple users to retrieve data simultaneously
  - Updates are added to a batch file which does the appropriate processing
  - Does not work for real time situations
- Locking
  - Restrict access to the record being updated by a user till the transaction is complete.

Today's Discussion



Difference Between File System Approach and DBMS Approach

Basis	DBMS Approach	File System Approach
Meaning	DBMS is a collection of data. In DBMS, the user is not required to write the procedures.	The file system is a collection of data. In this system, the user has to write the procedures for managing the database.
Sharing of data	Due to the centralized approach, data sharing is easy.	Data is distributed in many files, and it may be of different formats, so it isn't easy to share data.
Security and Protection	DBMS provides a good protection mechanism.	It isn't easy to protect a file under the file system.
Redundancy and Inconsistency	Due to the centralization of the database, the problems of data redundancy and inconsistency are controlled.	In this, the files and application programs are created by different programmers so that there exists a lot of duplication of data which may lead to inconsistency.
Examples	Oracle, SQL Server, Sybase etc.	Cobol, C++ etc.
Integrity Constraints	Integrity Constraints are easy to apply.	Integrity Constraints are difficult to implement in file system.

File System

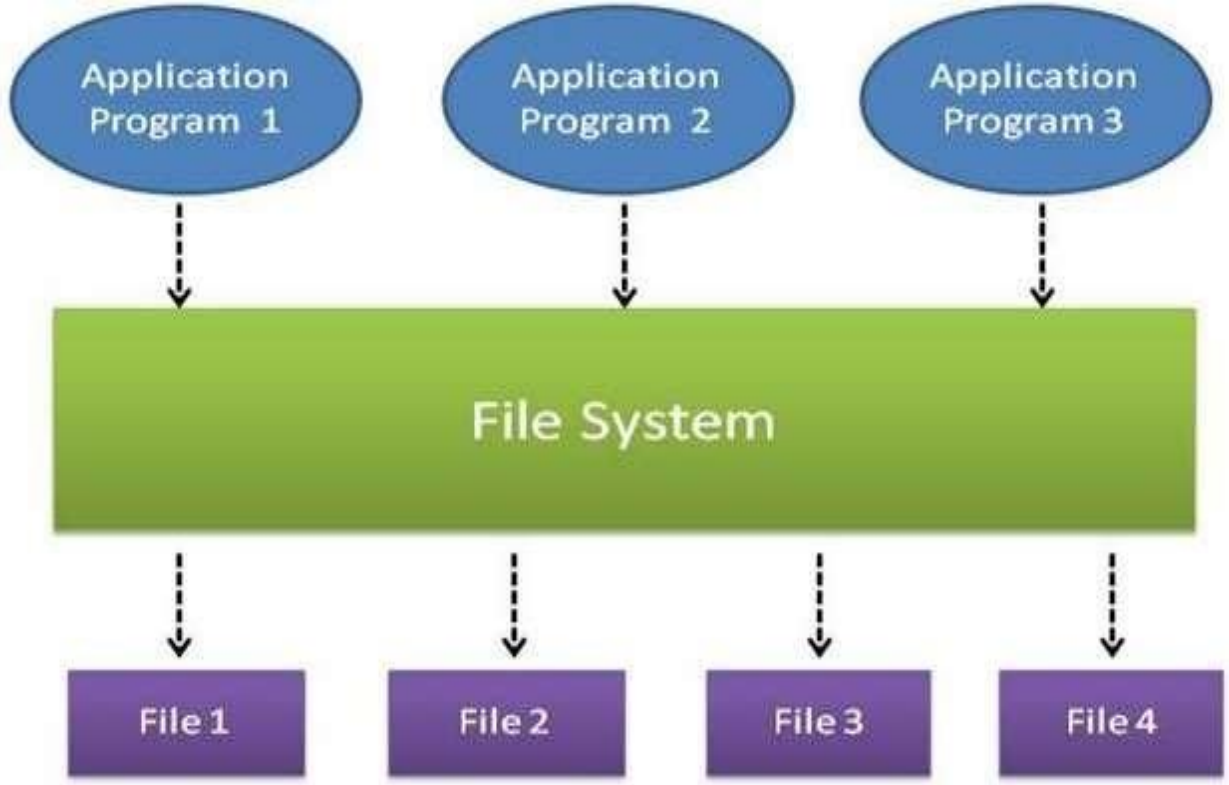
- An early attempt to computerize the manual filing system.

- Method of storing and organizing computer files and the data to make it easy to find and access.

- Use storage devices like:
- **Primary Storage**
- **Secondary Storage**
- **Tertiary Storage**



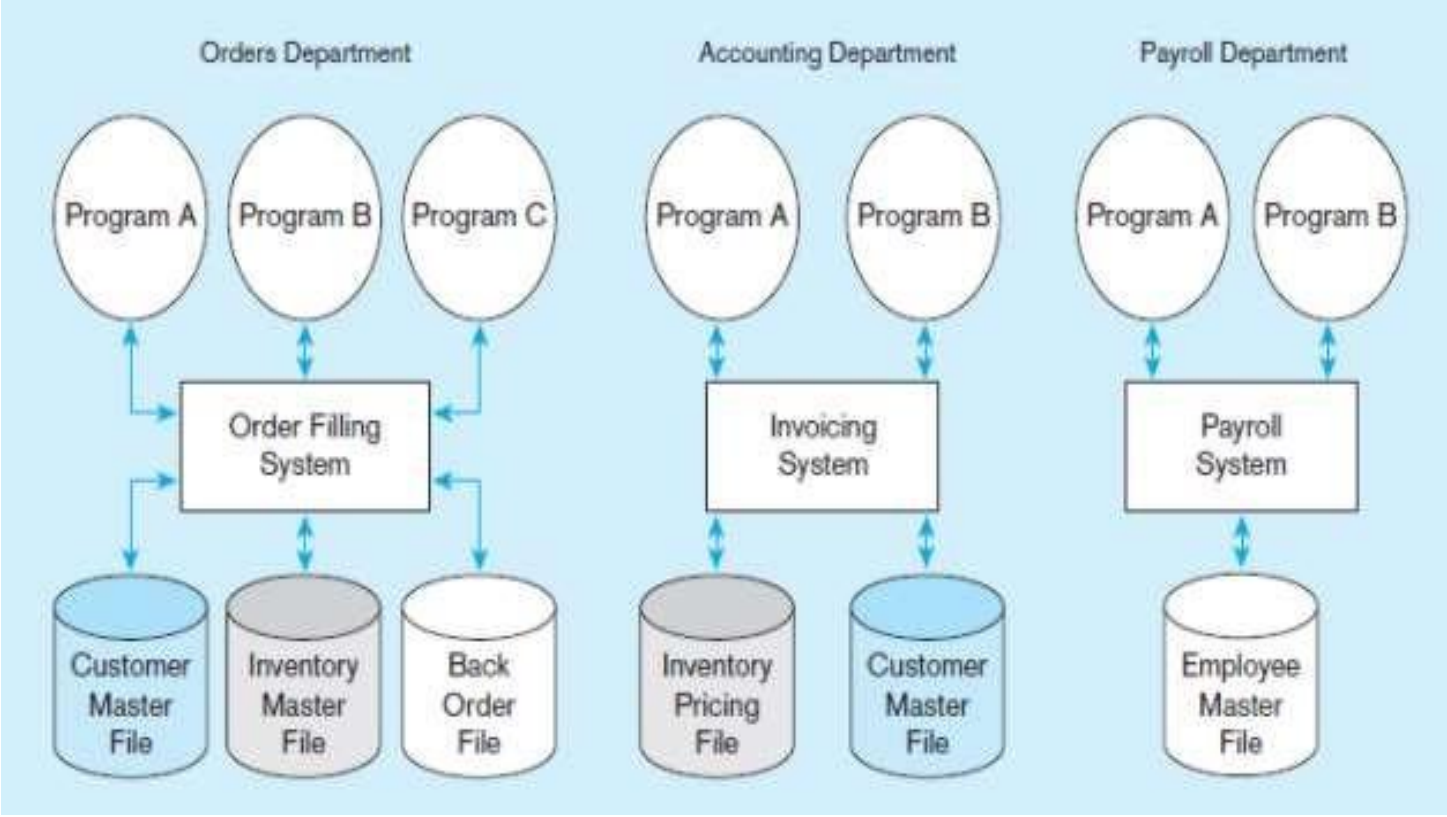
File processing system



File processing system

- In this traditional file system, each file is independent of other file and data in the different file can be integrated only by writing an individual program for each application.
- The data and application program that uses the data are arranged that any change to data requires modification of all the programs that use the data. Sometimes, it is not possible to identify all the programs using data and identified on trial and error basis.
- All functional areas in the organization create, processes its own files.
- The files such as inventory and payroll generate separate files and do not communicate with each other.
- The organization was simple to generate and had better local control but the data of an organization is dispersed throughout the functional subsystem.

Example


Orders Department
Accounting Department
Payroll Department
Program A
Program B
Program C
Program A
Program B
Program A
Program B
Order Filling System
Invoicing System
Payroll System
Customer Master File
Inventory Master File
Back Order File
Inventory Pricing File
Customer Master File
Employee Master File

Example

- Three of the computer application based on the file processing approach is shown here
- The systems are order filling, invoicing and payroll
- Order filling system has 3 files:

-Customer master file

-Inventory master file

-Back order

- There is duplication of some files used by the 3 applications which is typical of file processing system
- With the traditional file system, each applications has its own private files and users have little opportunity to share the data outside their own applications.

Example

- In accounting department, have the access to the invoicing system and its files but they probably do not have the access to the order filling system or to the payroll system and their files. So there is:

- duplication of data

-limited data sharing

-lengthy development times

-excessive program maintenance

Characteristics of File Processing System

- It is group of files storing data of an organization.
- Each file is independent from one another.
- Each file is called flat file.
- Each file contained and processed information for one specific function like accounting or inventory.
- Files are designed by using application programs written in programming languages such as COBOL, C, C++, etc….

Advantages of File Processing System

- Cost friendly
- Easy to use
- High scalability

Advantages of File Processing System

1. **Data Security:** Traditional file organization has security advantages over electronic filing.
2. **Complexity:** Traditional file systems are less complex than electronic systems, which can make it easier for untrained people to access and manipulate data.
3. **Access Time:** Traditional file systems have provide enhanced data retrieval techniques to retrieve data stored in files in easy and efficient way.
4. **Simple to Use:** The traditional filing system is very simple to set up and use.
5. **Minimal Investment:** If you're using a traditional file system, you also don't have to make an investment in a software database that may include an upfront purchase and ongoing licensing fees.

Disadvantages of File Processing System

- Data Redundancy
- Data Inconsistency
- Difficulty in Accessing Data
- Limited Data Sharing
- Integrity Problems
- Atomicity Problems
- Concurrent Access Anomalies
- Security Problems

Disadvantages of File Processing System

Data Redundancy

Due to the manual storing of data, the same data used to be present in multiple locations using the space in each of the Hard Disk, below are some following reasons for which Data Redundancy can be a major issue

Data inconsistency

Data redundancy leads to data inconsistency, lets take the same example, a student is enrolled for two courses and we have student address stored twice, now lets say student requests to change his address, if the address is changed at one place and not on all the records then this can lead to data inconsistency.

Disadvantages of File Processing System

Atomicity

Atomicity of a transaction refers to "All or nothing", which means either all the operations in a transaction executes or none.

- For example: Let's say Steve transfers Rs100 to Negan's account. This transaction consists multiple operations such as debit Rs 100 from Steve's account, credit Rs 100 to Negan's account. Like any other device, a computer system can fail lets say it fails after first operation then in that case Steve's account would have been debited by Rs 100 but the amount was not credited to Negan's account, in such case the rollback of operation should occur to maintain the atomicity of transaction. It is difficult to achieve atomicity in file processing systems.

Disadvantages of File Processing System

Data Security

Data should be secured from unauthorized access, for example a student in a college should not be able to see the payroll details of the teachers, such kind of security constraints are difficult to apply in file processing systems.

Data Integrity

The data values stored in the database must satisfy certain types of consistency constraints.

Example: bank account balance should never fall below zero, The maximum marks of the student can never be more than 100.

Disadvantages of File Processing System

Concurrent Access Anomalies

- For the sake of overall performance of the system and faster response, many systems allow multiple users to update the data simultaneously.
- But if data is getting updates by two users at the same time, there is possibility of inconsistency of data.
- To guard against this possibility, the system must maintain some form of supervision.
- In file processing system supervision is difficult to provide because data may be accessed by many different application programs that are coordinated.

Disadvantages of File Processing System

Difficulty in accessing data

- To carry out any new task a new program has to be written.
- For example, an application program may be available to generate a list of all students, but if we are required to generate a list of students who stay in a particular location, we need to write a new program as this is not anticipated.
- Accessing data is not convenient and efficient in file processing system.

Disadvantages of File Processing System

Limited Data Sharing

There is limited data sharing possibilities with the traditional file system. Each application has its own private files and users have little choice to share the data outside their own applications

Why Databases?

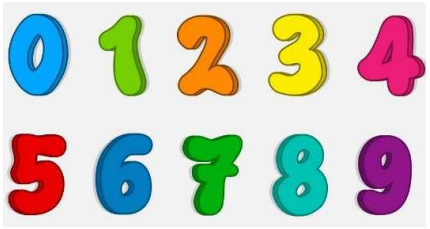
- Scientists (biologists) have to manage huge quantities of data
  - Results of experiments
  - References to relevant publications
- Those data need to be stored in a consistent way, shared and analyzed
  - Which are the experiments on a cellular biology done at my lab in 2009?
  - Which are the publications of my group?
- Databases are a possible solution to this issue

Basic Definitions

- **Data:**
  - Known facts that can be recorded and have an implicit meaning.







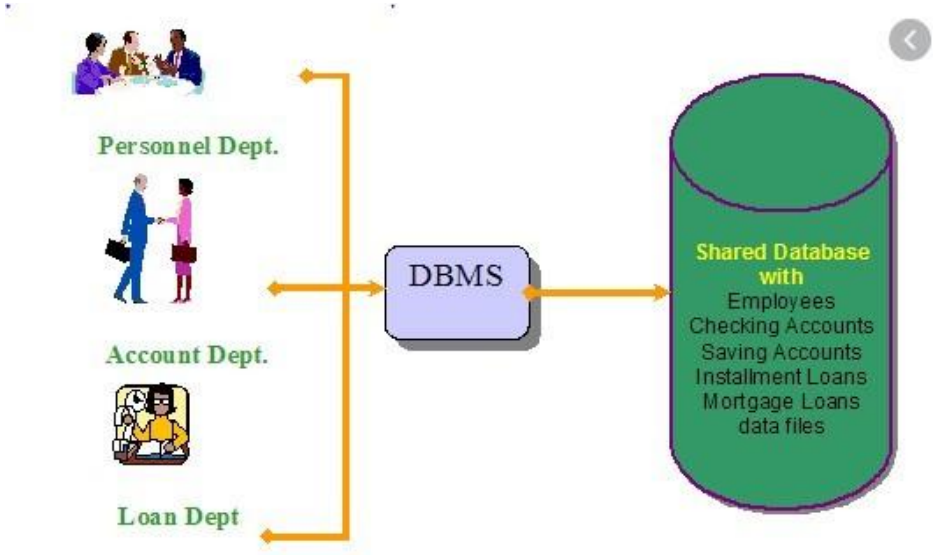




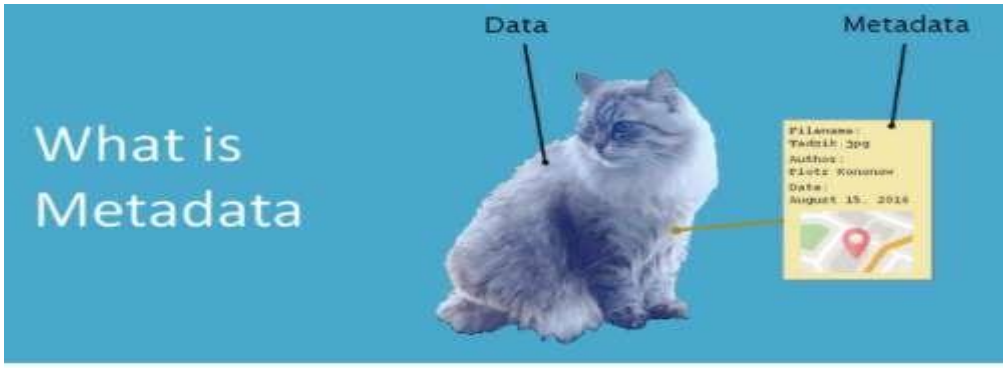


- **Database:**
  - A database is an organized collection of data, so that it can be easily accessed and managed.
- **Database Management System (DBMS):**
  - Collection of inter-related data and set of programs to access those data.
  - Helps to store and retrieve database information in a convenient and efficient manner.
- **Database System:**
  - The DBMS software together with the data itself. Sometimes, the applications are also included.
  - DB + DBMS = DBS

Database System



- **Metadata:**
  - Data about data



- **Data dictionary:**
  - A Data Dictionary is a collection of names, definitions, and attributes about data elements that are being used or captured in a database.
  - It is made up of two words, data which means collected information through multiple sources, and dictionary meaning the place where all this information is made available.
  - Data Dictionary provides consistency by reducing data redundancy in the collection and use of data across various members of a team.

- **Metadata:**

Examples

- The table of content and the index in a book may be treated metadata for the book
- A table (which is a record) has a name (e.g. table titles) and there are column names of the tables that may be treated as meta data.
- The figure also have titles or names.

- **Data dictionary:**

Attribute Name	Data Type	Max Field Size	Description	isRequired
Employee ID	Integer	10	A unique ID for each Employee	Yes
Name	Text	25	Name of the Employee	Yes
Date of Birth	DateTime	10	Date of Birth of the Employee	Yes
Mobile Number	Integer	10	Contact Number of the Employee	Yes

Field Name	Datatype	Field Length	Constraint	Description
Student_ID	Number	5	Primary Key	Student id
Student_Name	Varchar	20	Not Null	Name of the student
Student_Address	Varchar	30	Not Null	Address of the student
Student_City	Varchar	20	Not Null	City of the student

Simplified database system environment

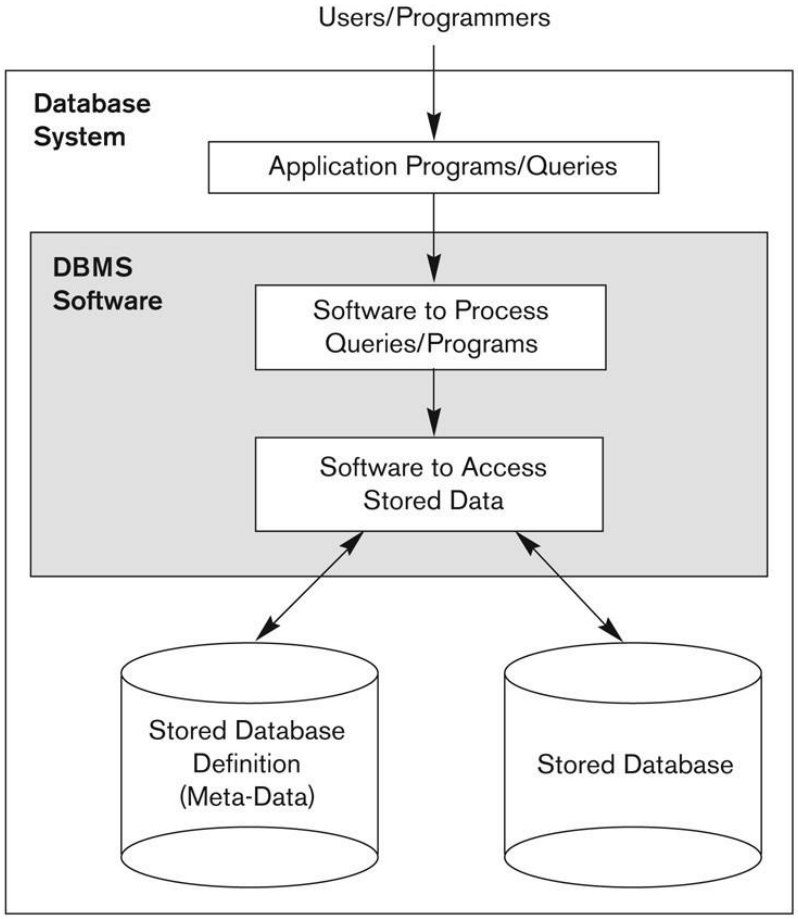


Figure 1.1
A simplified database system environment.

Simplified database system environment

A database environment is a collective system of components that comprise and regulates the group of data, management, and use of data, which consist of software, hardware, people, techniques of handling database, and the data also.

Here, the hardware in a database environment means the computers and computer peripherals that are being used to manage a database, and the software means the whole thing right from the operating system (OS) to the application programs that include database management software like **M.S. Access or SQL Server**.
Again the people in a database environment include those people who administrate and use the system. The techniques are the rules, concepts, and instructions given to both the people and the software along with the data with the group of facts and information positioned within the database environment.

Functions of DBMS

- ***Define*** **a particular database in terms of its data types, structures, and constraints**
- ***Construct*** **or Load the initial database contents on a secondary storage medium**
- ***Manipulating*** **the database**
- **Protection or Security measures to prevent unauthorized access**
- ***Processing*** **and** ***Sharing*** **by a set of concurrent users and application programs – yet, keeping all data valid and consistent**
- **Presentation and Visualization of data**
- **Maintaining the database and associated programs over the lifetime of the database application**

Example of a simple database

COURSE

Course_name	Course_number	Credit_hours	Department
Intro to Computer Science	CS1310	4	CS
Data Structures	CS3320	4	CS
Discrete Mathematics	MATH2410	3	MATH
Database	CS3380	3	CS

SECTION

Section_identifier	Course_number	Semester	Year	Instructor
85	MATH2410	Fall	04	King
92	CS1310	Fall	04	Anderson
102	CS3320	Spring	05	Knuth
112	MATH2410	Fall	05	Chang
119	CS1310	Fall	05	Anderson
135	CS3380	Fall	05	Stone

GRADE_REPORT

Student_number	Section_identifier	Grade
17	112	B
17	119	C
8	85	A
8	92	A
8	102	B
8	135	A

PREREQUISITE

Course_number	Prerequisite_number
CS3380	CS3320
CS3380	MATH2410
CS3320	CS1310

Figure 1.2
A database that stores student and course information.

Example of a simplified database catalog

RELATIONS

Relation_name	No_of_columns
STUDENT	4
COURSE	4
SECTION	5
GRADE_REPORT	3
PREREQUISITE	2

COLUMNS

Column_name	Data_type	Belongs_to_relation
Name	Character (30)	STUDENT
Student_number	Character (4)	STUDENT
Class	Integer (1)	STUDENT
Major	Major_type	STUDENT
Course_name	Character (10)	COURSE
Course_number	XXXXNNNN	COURSE
….	….	…..
….	….	…..
….	….	…..
Prerequisite_number	XXXXNNNN	PREREQUISITE

Note: Major_type is defined as an enumerared type with all known majors. XXXXNNNN is used to define a type with four alpha characters followed by four digits

Figure 1.3
An example of a database catalog for the database in Figure 1.2.

Advantages of Database System

- Controls data redundancy
- Program-data independence
- Data abstraction (DBMS provide data abstraction which hides the data storage and management details from the users.This allows the users to focus on the data rather than the details of how the data is stored and managed)
- Data sharing and concurrent access
- Data consistency (Data Inconsistency exists when different versions of the same data Appear in different Places.)
- Multiple user interface
- Enforcing security and integrity (deals with the accurate and consistent data)
- Providing back up and recovery

Advantages of Database System

- Controls database redundancy: It can control data redundancy because it stores all the data in one single database file and that recorded data is placed in the database.
- Data sharing: In DBMS, the authorized users of an organization can share the data among multiple users.
- Easily Maintenance: It can be easily maintainable due to the centralized nature of the database system.
- Reduce time: It reduces development time and maintenance need.
- Backup: It provides backup and recovery subsystems which create automatic backup of data from hardware and software failures and restores the data if required.
- Multiple user interface: It provides different types of user interfaces like graphical user interfaces, application program interfaces

Users of DBMS

It is broadly categorised into

1. Application Programmer
2. End User
3. Database Administrator

Users of DBMS

Application Programmer

- Write application programs that used in the database
- Application programmers also referred as system analysts or simply a software engineers those are the back-end programmers who writes the code for the application programs.
- They are the computer professionals.
- These program could be written in high level language such as Java, .net, php, etc
- The application programmer are the users who are responsible for developing the application programs or user interface.
- Application development is done by specialized professionals. These DBMS users, code the application and its interface after system and requirements study of the client or organization.
- The application programs are written in a programming language or by using a development tool. The application programmers use the database commands provided within the programming language or development tool.

Users of DBMS

End User

- It interacts with the system either by involving an application programs or by writing their queries in a database query language to perform all the basic operations like insertion, deletion, updation, retrieval etc., on the data.
- End users are final users.
- These are the people for whom the DBA, Application Developers and Database Designers are working together.
- They create easily accessible, secure and consistent database for end users.
- End users can be the people who have little knowledge of computer systems and can access data through a interactive and user-friendly application.
- They can also be expert professionals who access data by querying the database directly though query interface of a DBMS.
- Another category of end users can be those who need to use data stored in a database into high-end applications Knowledge base, Decision Systems, Intelligent Systems, Data Analytics or storage and management of Big Data.

Users of DBMS

Database Administrator

- Centralised control of the database is executed by a person or a group of persons under the supervision of a high level administrator. This person or group is referred as DBA.
- An administrator has full control of the database
- An administrator defines the logical and physical schemas and manages all three levels of the database.
- They design the overall structure of the database including layouts, functioning and procedures
- They are responsible for routine maintenance, back up and recovery of the database.
- They perform admin related activities like time to time updating, insert new required value/functions, modify the existing etc
- They control all these operations like: Security, integrity, redundancy, concurrency, hardware and software management
- They always update the database in terms of technology, function, aim and requirements to make the database ready for future scope.

Types of Users in DBMS

Naive users −The naive users need not be aware of the presence of the database system.They are end users of the database who works through a menu driven application programs, where the type and range of response is always indicated to the user (online library system, ATMs)

Online users − Online users may communicate with databases directly through an online terminal or indirectly through user interface and application programs.

Sophisticated Users −They are those users who interact with the system without writing the program instead they form their request in database query language.They are the SQL programmers, who are going to deal directly with the database.They write queries to delete or select or insert and update the database. (engineers, scientists, business analysts)

Specialized Users − Specialized users who write specialized database applications that do not fit into the fractional database processing framework. (bibliographic, Multimedia, Numeric)

Guest Users- A guest user is a user who has limited access to the database, typically for read-only purposes.This type of user may be used by external organizations or individuals who need access to a subset of the data in the database

Casual/Temporary Users- These type of users communicate with the database for a little period of time.

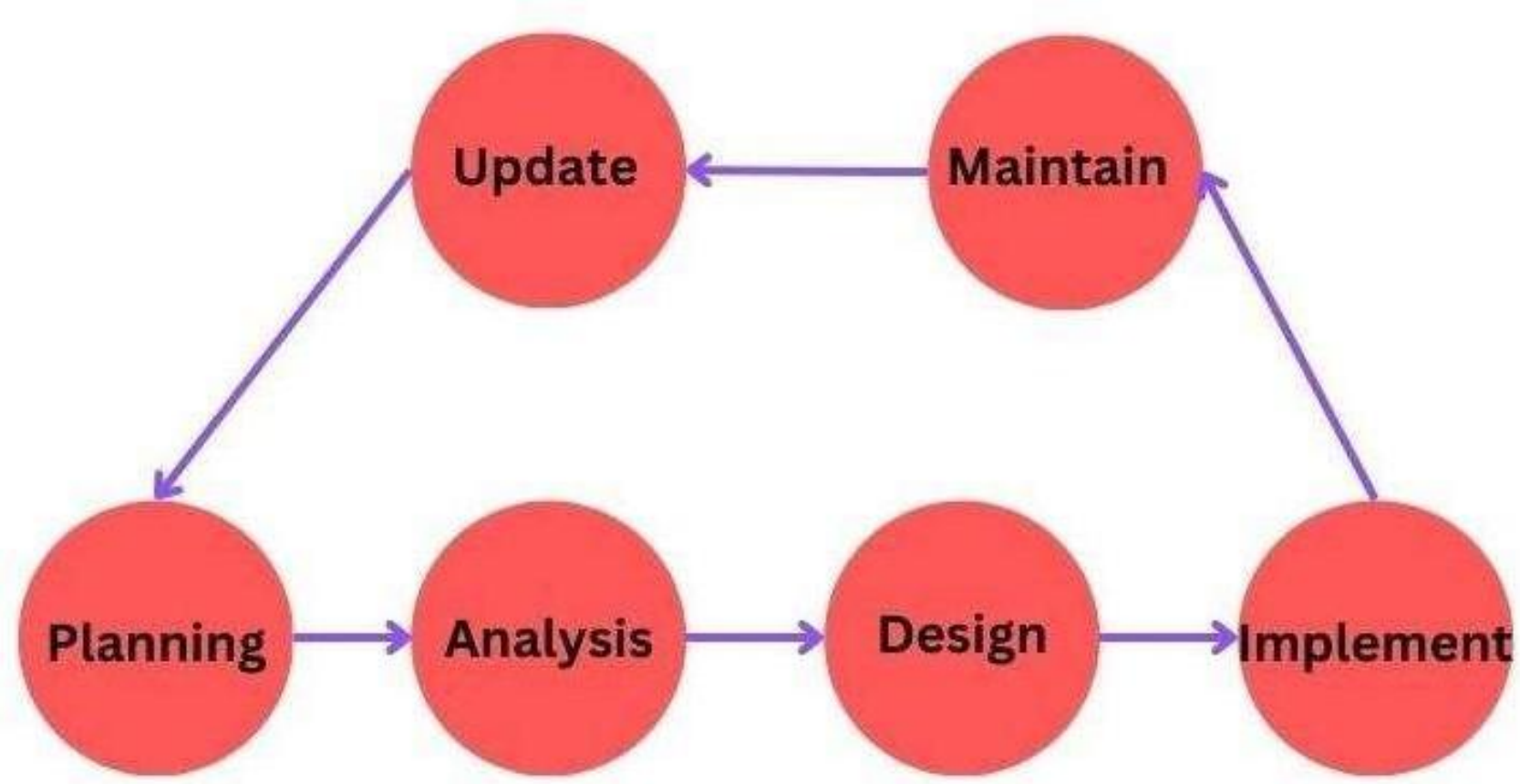
Functions of DBA

- Installing and configuring database management systems (DBMS)
- Designing and implementing database structures
- Importing and exporting data to and from databases
- Setting up user accounts and managing access to the database
- Monitoring and optimizing database performance
- Ensuring data security and integrity through backups, testing, and recovery processes
- Troubleshooting and resolving issues with the database
- Staying up-to-date with new technologies and best practices in database management.
- Installing and upgrading the database management system (DBMS) related software.
- Creating and maintaining databases, including backup and recovery procedures.

Functions of DBA

- Monitoring and optimizing database performance, including troubleshooting any issues that may arise.
- Setting up user accounts and managing access to the database.
- Ensuring data integrity and security through the use of backup and recovery procedures, as well as implementing security measures such as firewalls and data encryption.
- Developing and enforcing database policies and procedures.
- Providing support and training to users.
- Coordinating with developers to design and optimize database structures and schemes.
- Performing data migration between different DBMS platforms or between different versions of the same DBMS.
- Analyzing and optimizing the database for improved performance and scalability.

DBA Life Cycle



Database Administrator

Planning

Planning is the first step. In this stage, we try to understand what the stakeholders require and how we can use the database to support business needs. The DBA collaborates with other stakeholders to find out what data is needed. Then they plan how to design and set up the system and set goals for the database.

Analysis

During the analysis stage, the focus is on the organization's data. The goal is to determine how to organize the data to meet the system's new requirements. This involves reviewing existing databases, data dictionaries, and models.

Design

The design stage creates a model for the organization's data needs. This includes defining the tables, fields, relationships, dictionaries and Data Flow Diagram.

Database Administrator

Implementation

In the implementation stage, we put the database system into action based on the design plan from the planning stage. This means creating tables, importing data, and setting up user accounts. User accounts are set with various hierarchies and access permissions.

Maintenance

The system needs maintenance to work well. This means checking and maintaining it to avoid problems. This includes performing regular backups and monitoring performance. Also, the DBA optimizes the queries to improve efficiency.

Update

Over time, the needs of the organization could change. So, accordingly DBA would need to make changes to the database system. This could involve making changes to the data model. For example, adding new fields or updating systems to support new business processes.

Database Administrator

There are various roles and responsibilities of a Database Administrator:

- Managing the relationship with the database vendors and service providers.
- Conducting regular audits of the database systems. It is done to ensure the proper functioning and security of the database.
- Providing technical support to the end-users and other IT staff regarding database issues.
- Developing and implementing policies and maintaining them. It is done to ensure that databases are managed effectively.
- Working with people who need data to create a database that fits their needs. Developing database solutions to meet the needs of the stakeholders.
- Identifying and troubleshooting database issues and resolving them in a timely manner.
- Creating and maintaining data dictionaries and data flow diagrams.
- Ensuring that databases are optimized for efficient data storage and retrieval.

Functions of DBA

- **Identifying User Requirements:**

Database Administrator identifies the current user environment. He closely studies the current system and its outputs. He also interviews different users to know their current and future requirements.

- **Developing & Maintaining Data Dictionary:**

Data dictionary is a very useful collection of data about database. The DBA (Database Administrator) stores data item names, source, meanings and usage in data dictionary. The data dictionary is revised regularly to update it as the project continues.

- **Designing Logical Model:**

After analyzing the user environment, the DBA (Database Administrator) develops a logical data model for the organization. This data model consists of all entities, attributes and relationships.

Functions of DBA

- **Choosing a DBMS(Database Management System):**

The DBA (Database Administrator) chooses a particular database management system on the basis of logical data model.The selected DBMS should satisfy all requirements and constraints identified in the logical data model.

- **Developing Physical Model:**

The DBA (Database Administrator) creates the exact layout of data according the facilities of selected of DBMS and available resource of software and hardware.

- **Creating & loading Database:**

After developing physical model, the DBA (Database Administrator) creates the structure of database by using DBMS. He also loads the data into the database.

Functions of DBA

- **Developing User Views:**

Any project is basically developed to serve the user. DBA (Database Administrator) should guide the user about the functionality of database. If any request of the user is not satisfied, DBA (Database Administrator) should justify it.

- **Writing & Maintaining Documentation:**

The documentation is very important when the project is reviewed or revised, it is normally maintained by data dictionary system of DBMS.The Database Administrator should ensure the proper documentation of the project.

Functions of DBA

- **Helping Database Users:**

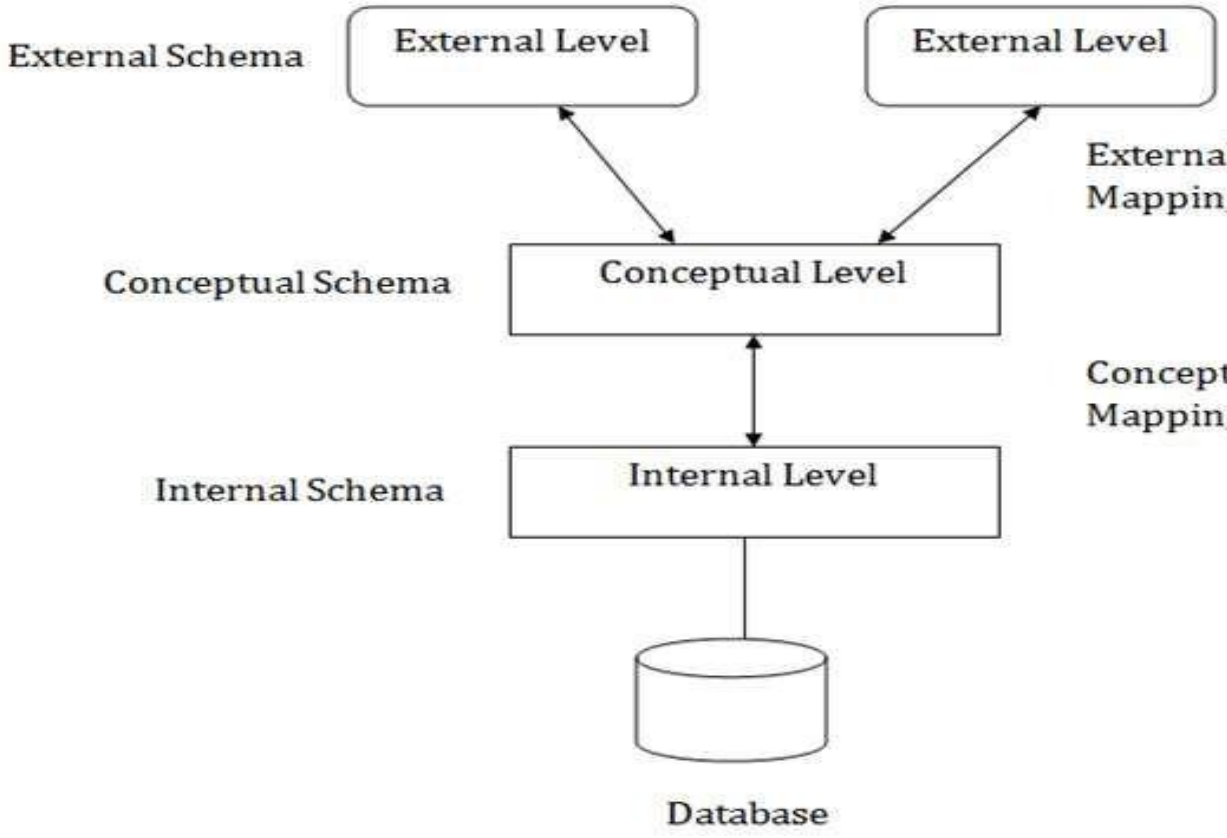
The Database Administrator helps the database users by:

- making sure that the data they require is available
- assisting them on using correctly the system.
- Defining Backup and Recovery Procedures:

If any portion of database is damaged by human error or by hardware, it should be restored as soon as possible. It important to backup the information of the database on a backup server so that it may be recovered in case of emergency.

Three Schema Architecture of DBMS

The three-schema architecture is as follows:



Individual Users view
How user is viewing the data

What data stored

How data is stored and where it is stored or storage view

Three Schema Architecture of DBMS

- The architecture is called as three level architecture as it is divided basically into 3 blocks or levels:

  -External level

  - Conceptual level

  -Internal level

- The view (subset of database) at each of these levels are being described by a schema (plan or outline)

Three Schema Architecture of DBMS

- **External Level**
- This is the highest level which is closest to the user.
- Each individual user is not concerned with the entire databases as only the part i.e. relevant to him/her is visible.
- External view is described by means of schema called as External Schema
- External Schema consists of definition of logical records and relationships in external view

Three Schema Architecture of DBMS

- **Conceptual Level**
- This level comes in between external and internal level
- Conceptual level is used by DBA
- At this level of database all the database entities, relationships among them can be viewed.
- Conceptual level is described by means of a schema is called as conceptual schema
- It describes the structure of database for community of users
- It hides the details of storage structure, concentrates on describing entities, data types, relationships and constraints of the restrictions.

Three Schema Architecture of DBMS

- **Internal Level**
- This is the lowest level of the database
- It describes the physical storage structure of database
- It indicates how the data will be stored and describes the data structure and access methods to be used by the database.

Objectives of Using Architecture

1. Mapping
2. Data Independence

Mapping:

The process of transforming request and response in between the three levels is known as mapping

- Mapping is of two types:

i. External/Conceptual Mapping

It defines the correspondence between the external view and conceptual view

ii. Conceptual/Internal Mapping

It is the correspondence between the conceptual view to internal view or and stored database

Objectives of Using Architecture

Data Independence:

- Data Independence can be defined as the capacity to change the schema at one level of database system without having to change the schema at next higher level.
- Data Independence are categorised into two:

i. Physical data independence

It indicates that the physical storage device which are used to storing the data can be changed without change in conceptual and external schema

Example- To change the location of database from C drive to D drive

ii. Logical data independence

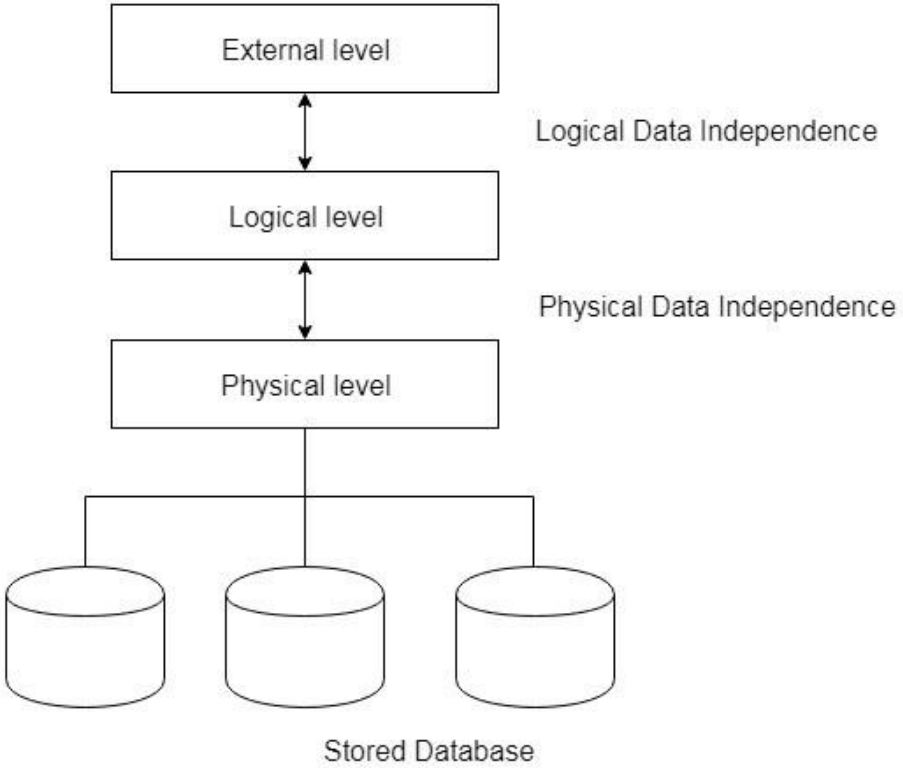
The conceptual schema can be change without affecting the external schema

Conceptual schema can be change in order to expand database or to reduce the database or to change the constraints.

Example-To add/modify/delete a new attribute into a relation

Data Independence

- **Data independence** refers characteristic of being able to modify the schema at one level of the database system without altering the schema at the next higher level.



DBMS facilities or Database languages

1. Data definition language (DDL)

   create, alter, drop, truncate, rename

2. Data Manipulation language (DML)

   inserting, deleting, modifying, retrieving, select, update, merge

3. Data control language (DCL)

   Commit, rollback, savepoint, transaction

DBMS facilities or Database languages

1. Data definition language (DDL)- statements used to define the database structure or schema.

- CREATE- to create objects in the database
- ALTER- alters the structure of the database
- DROP-delete objects from the database
- TRUNCATE- remove all records from a table, including all spaces allocated for the records are removed
- COMMENT- add comments to the data dictionary
- RENAME- rename an object

DBMS facilities or Database languages

2. Data Manipulation language (DML)- statements are used for managing data within schema objects

- SELECT- retrieve data from the database
- INSERT- insert data into a table
- UPDATE-It is used to update existing data within a table
- LOCK TABLE- control concurrency

DBMS facilities or Database languages

3. Data Control Language (DCL)

- GRANT- gives user's access privileges to database
- REVOKE-withdraw access privileges given with the GRANT command

4.Transaction Control Language (TCL)- statements are used to manage the change made by DML statements

- COMMIT- save work done
- SAVEPOINT- identify a point in a transaction to which you can later roll back
- ROLLBACK-restore database to original since the last COMMIT

Data Models

- A data model is a collection of concepts that can be used to describe the structure of a database.
- It defines the data, relationship between the data and constraints on the data.
- It has 3 main components:

i. A set of rules according to which database can be constructed
ii. A set of integrity rules
iii. Types of operations that can be performed on the data

- Data models can be classified into the following types:

iv. Object based logical model
v. Record based logical model

Data Models

Object based logical model

E-R Model is a widely known Object based logical model. They are used to describe the data and at the conceptual and view levels

Record based logical models

(Network model, hierarchical model, relational model)

They are used to describe data at the conceptual and view levels. They are used both to specify the overall logical structure of the database and to provide a higher level description of implementation.

-Database is structured in fixed format records of several types

-Each record type defines a fixed number of fields or attributes.

-Each field is usually of a fixed length.

Data Models

Network Model

The Network model replaces the hierarchical tree with a graph thus allowing more general connections among the nodes.

The main difference of the network model from the hierarchical model, **is its ability to handle many to many (N:N) relations.** In other words, **it allows a record to have more than one parent.** Suppose an employee works for two departments.

The strict hierarchical arrangement is not possible here and the tree becomes a more generalized graph – a network.

The network model was evolved to specifically handle non-hierarchical relationships.

A network structure thus allows 1:1 (one: one), 1: M (one: many), M: M (many: many) relationships among entities.

In network database terminology, a relationship is a set.

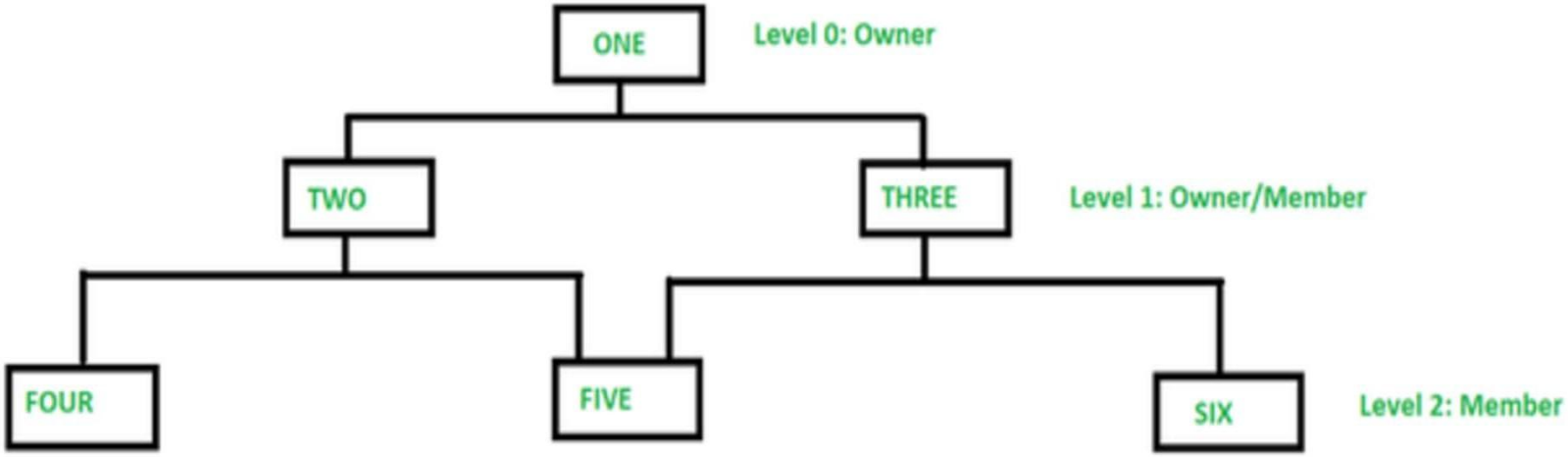
Data Models

Network Model

- This model can consist of multiple parent segments and these segments are grouped as levels but there exists a logical association between the segments belonging to any level.
- Mostly, there exists a many-to-many logical association between any of the two segments.
- It can have M: N relations i.e, many-to-many which allows a record to have more than one parent segment.

Data Models

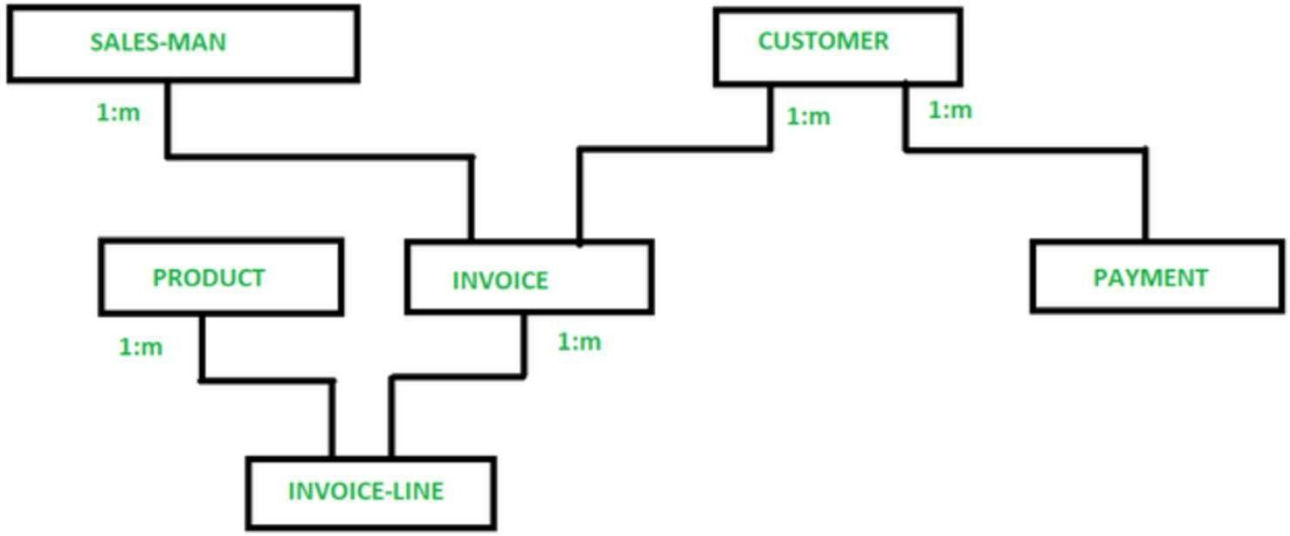
Network Model



- In the above figure, member TWO has only one owner ‘ONE’ whereas member FIVE has two owners i.e, TWO and THREE. Here, each link between the two record types represents 1 : M relationship between them.
- Therefore, it allows 1: 1, 1 : M, M : N relationships among the given entities which helps in avoiding data redundancy problems as it supports multiple paths to the same record.
- **Example :** Network model for a Finance Department.

Data Models

Network Model



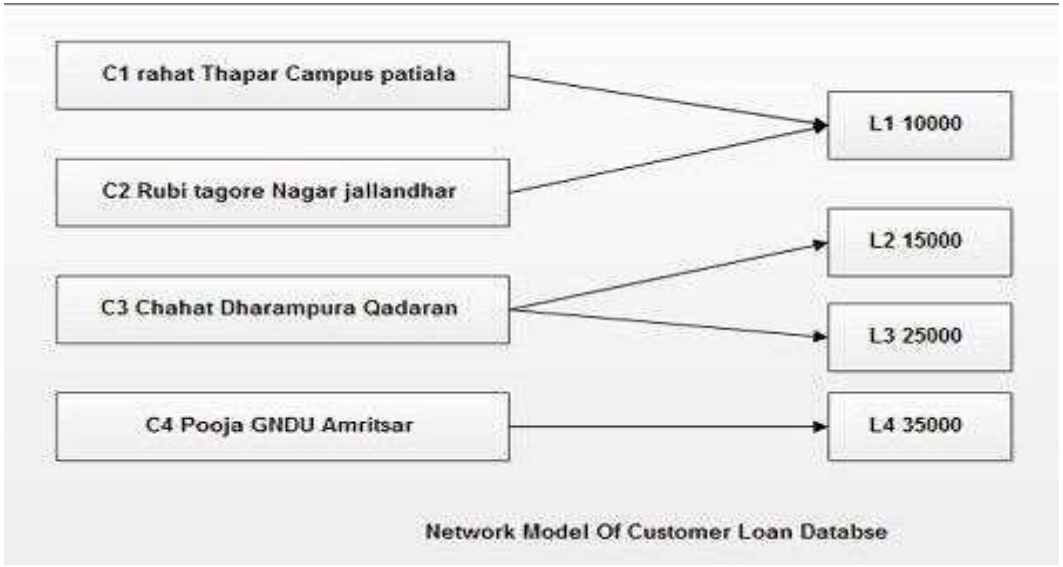
- So, In a network model, a one-to-many (1: N) relationship has a link between two record types. Now, in the above figure, SALES-MAN, CUSTOMER, PRODUCT, INVOICE, PAYMENT, INVOICE-LINE are the types of records for the sales of a company.
- Now, as you can see in the given figure, INVOICE-LINE is owned by PRODUCT & INVOICE. INVOICE has also two owners SALES-MAN & CUSTOMER.

Data Models

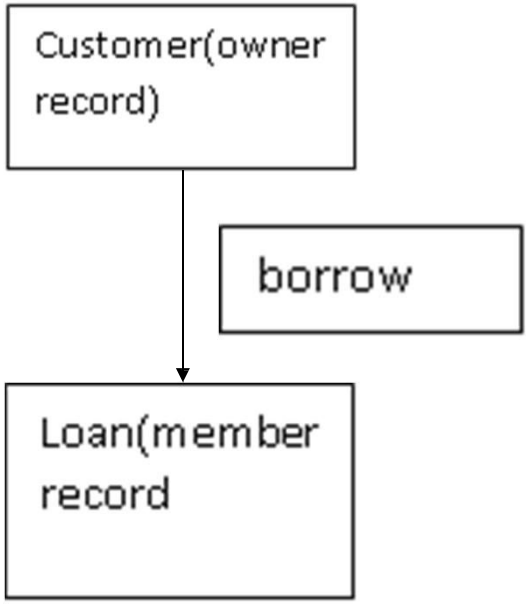
Each set is made up of at least two types of records: an owner record (equivalent to parent in the hierarchical model) and a member record (similar to the child record in the hierarchical model).

The database of Customer-Loan, which we discussed earlier for hierarchical model, is now represented for Network model as shown.

It can easily depict that now the information about the joint loan L1 appears single time, but in case of hierarchical model it appears for two times

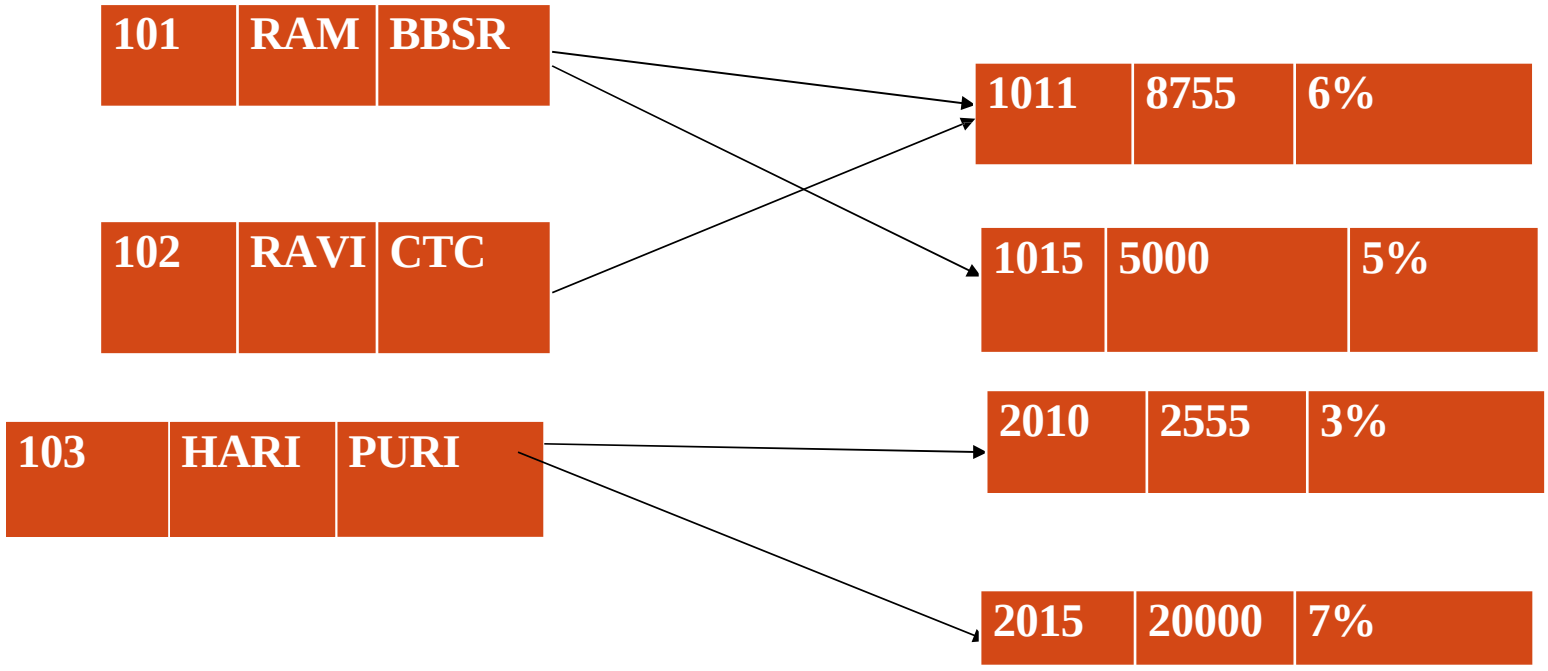


Network Model Of Customer Loan Databse



Data Models

SCHEMA WITH NETWORK MODEL REPRESENTATION



Data Models

Advantages of Network Model

- This model is very simple and easy to design like the hierarchical data model.
- This model is capable of handling multiple types of relationships which can help in modeling real-life applications, for example, 1: 1, 1: M, M: N relationships.
- In this model, we can access the data easily, and also there is a chance that the application can access the owner's and the member's records within a set.
- This network does not allow a member to exist without an owner which leads to the concept of Data integrity.

Data Models

Thus, it reduces the redundancy and is better as compared to hierarchical model.

Disadvantages of Network Model

Even though the **network database model was significantly better than the hierarchical database model, it also had many drawbacks**. Some of them are:

System complexity: All the records are maintained using **pointers** and hence the whole database structure becomes very complex.

Operational Anomalies: As discussed earlier, network model's insertion, deletion and updating operations of any record require large number of **pointer adjustments**, which makes its implementation very complex and complicated.

Absence of structural independence: If changes are made to the database structure then all the application programs need to be modified before they can access data. Thus, even though the network database model succeeds in achieving data independence, it still fails to achieve structural independence. (A change in structure demands a change in the application as well, which leads to lack of structural independence)

Data Models

Disadvantages of Network Model

- The schema or the structure of this database is very complex in nature as all the records are maintained by the use of pointers.
- There's an existence of operational anomalies as there is a use of pointers for navigation which further leads to complex implementation.
- The design or the structure of this model is not user-friendly.
- This model does not have any scope of automated query optimization.
- This model fails in achieving structural independence even though the network database model is capable of achieving data independence.
- The system may become progressively complex as the number of relationships to be managed increases. To effectively work with the model, a user needs to have an in-depth understanding of it.
- Modifications such as updates, deletions, or insertions are quite challenging to implement.

Data Models

Hierarchical Model

In **hierarchical model**, data is organized into a tree like structure with each record is having one parent record and many children. (PARENT CHILD RELATIONSHIP)

A hierarchical model represents the data in a tree-like structure in which there is a single parent for each record.

These types of models are designed basically for the early mainframe database management systems, like the Information Management System (IMS) by IBM.

This model structure allows the **one-to-one and a one-to-many** relationship between two/ various types of data.

Note: Hierarchical models are rarely used now.

Data Models

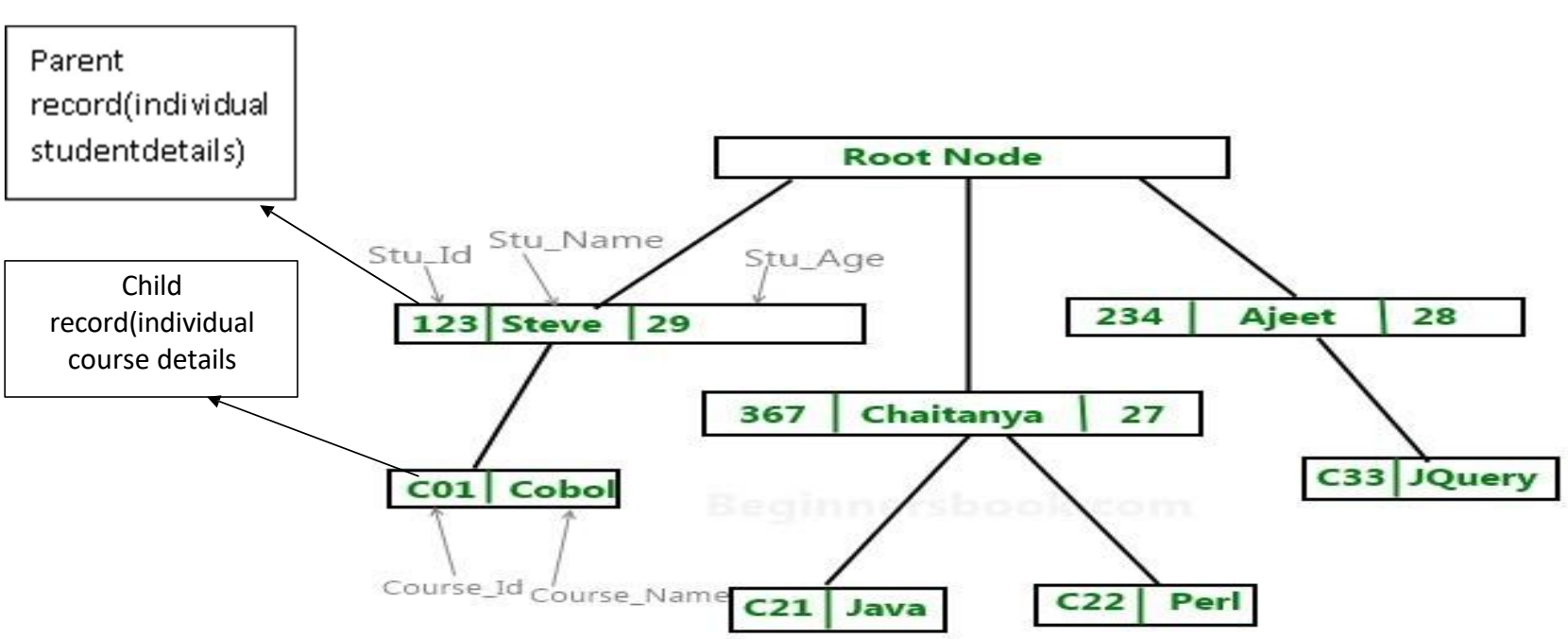
Hierarchical Model

- It is based on tree structure.
- It consists of a collection of records that are connected to each other by links.
- So, Hierarchical model is a collection of routed trees and the relationship that exists in the hierarchical model is one to many and one to one.

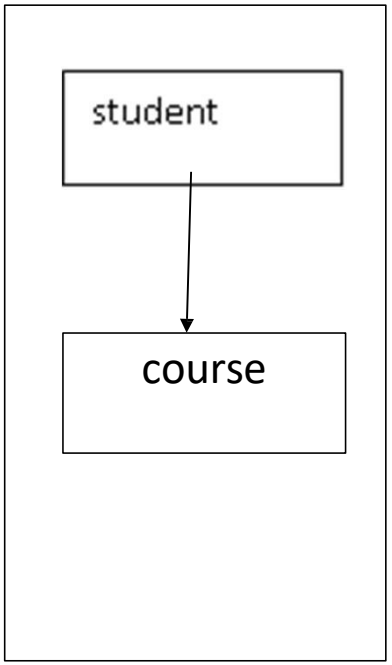
Data Models

Sample Hierarchical Model Diagram:

Lets say we have few students and few courses and a course can be assigned to a single student only, however a student take any number of courses so this relationship becomes one to many.

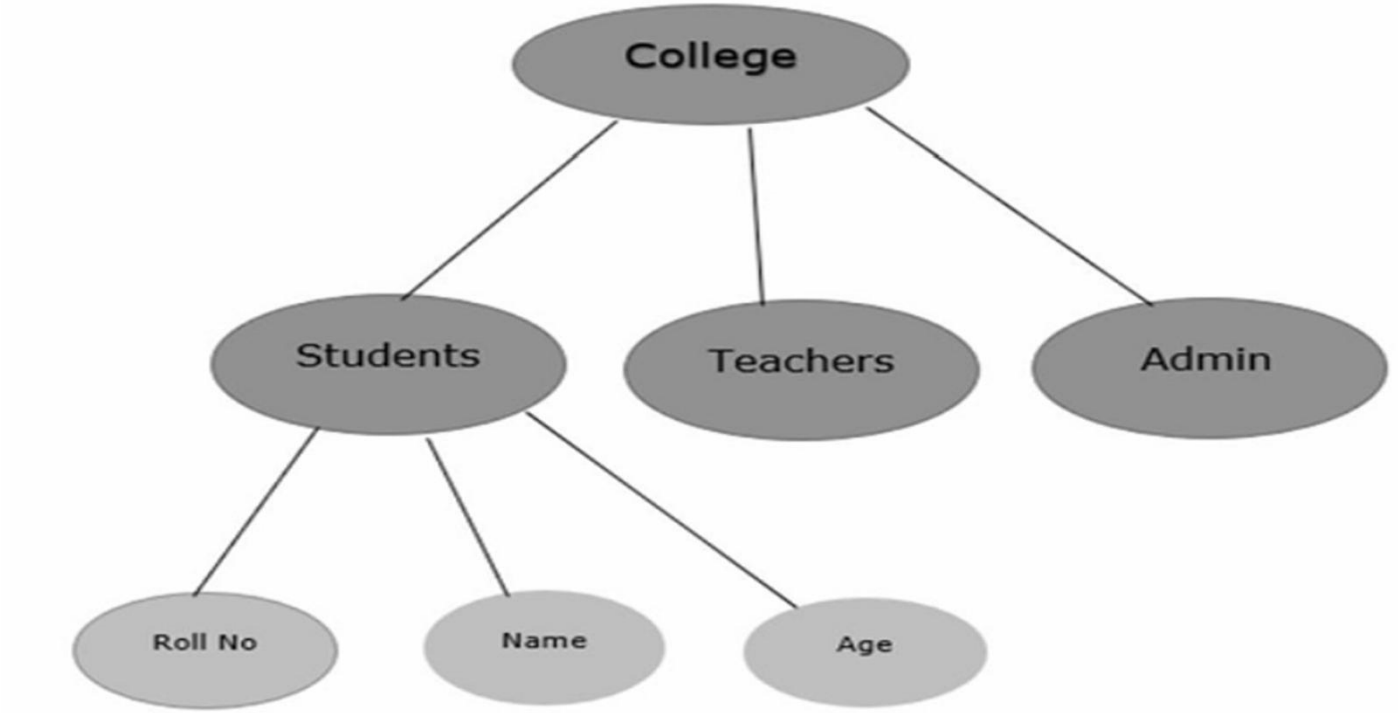


Hierarchy given is:



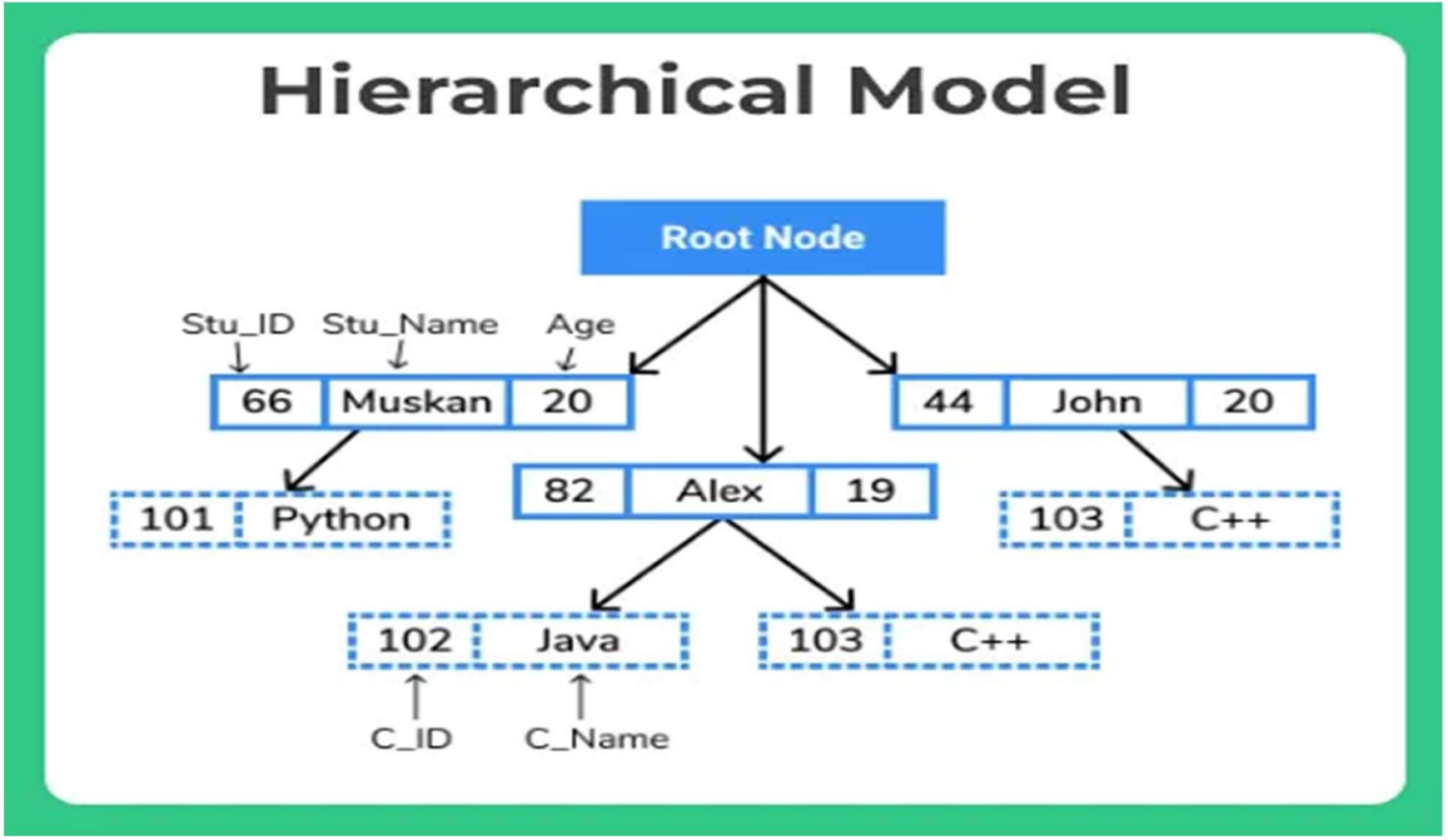
Data Models

HIERARCHICAL MODEL

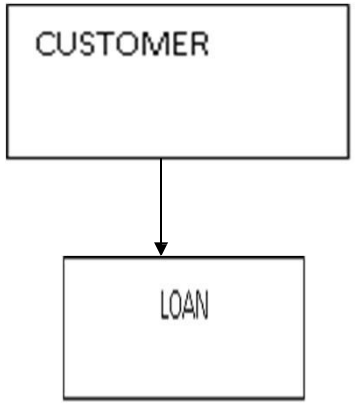


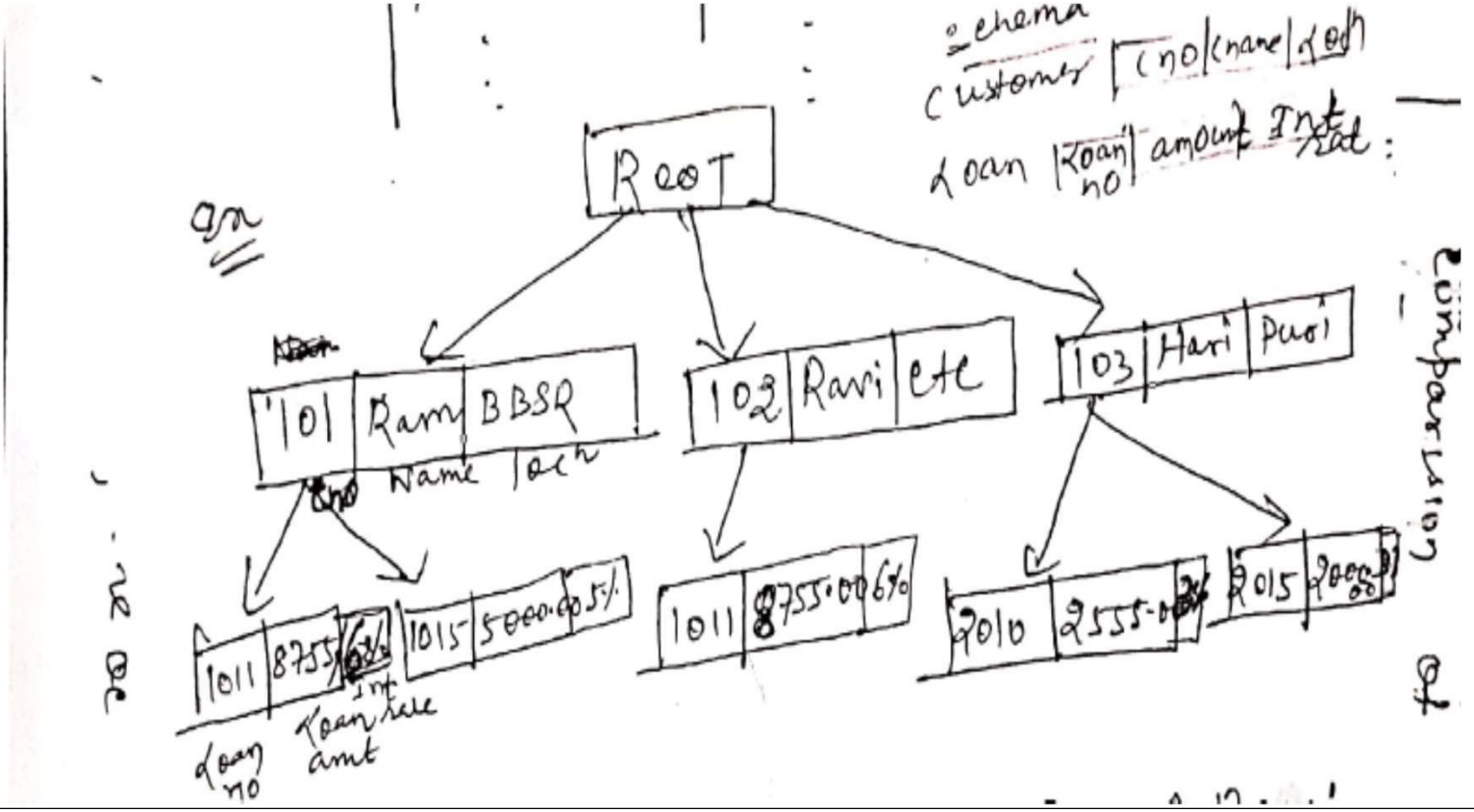
Data Models

HIERARCHICAL MODEL



Data Models


CUSTOMER
LOAN


ROOT
101 Ram BBSR
102 Ravi etc
103 Hari Puri
1011 8755
1015 5000
1011 8755
2010 2555
2015

Data Models

ANOMALIES IN HIERRCHICAL MODEL (such inconsistencies may arise when have a particular record stored in multiple locations and not all of the copies are updated)

There are four basic operations Insert, Update, Delete and Retrieve that can be performed on each model.

Insert Operation: We can say that insert anomaly exists only for those children, which has no corresponding parents.(insert anomaly exists only for those children, which has no corresponding parents)

Update Operation: We can say that update anomalies only exist for children not for parent because children may have multiple entries in the database like for 1011 loan record if I want to change interest rate it has to be changed 2 times.

Delete Operation: In hierarchical model, deletion of parent causes the deletion of child records also and if the child occurrence is the only occurrence in the whole database, then the information of child records will also lost with the deletion of parent.

Data Models

HIERRCHICAL MODEL

Advantages of the hierarchical model :

- As the database is based on this architecture the relationships between various layers are logically simple so, it has a very simple hierarchical database structure.
- It has data sharing as all data are held in a common database data and therefore sharing of data becomes practical.
- It offers data security and this model was the first database model that offered data security.
- There's also data integrity as it is based on the parent-child relationship and also there's always a link between the parents and the child segments.

Data Models

HIERRCHICAL MODEL

Advantages

1. **Fast and efficient data retrieval:** One of the key advantages of hierarchical databases is that they allow for fast and efficient data retrieval, since data is organized in a predictable and structured way.
2. **Easy to add/delete information:** In this type of database, you can easily add or delete the information.
3. **Predictable data structure:** The hierarchical database model has a predictable and structured data structure, which makes it easier to understand and navigate.
4. **Efficient storage of data:** Since hierarchical databases store data in a parent-child relationship, they can be more efficient in terms of storage compared to other database models.
5. **Good performance:** Hierarchical databases can have good performance in terms of data retrieval and querying, especially for large datasets.

Data Models

HIERRCHICAL MODEL

Disadvantages of the hierarchical model :

- Even though this model is conceptually simple and easy to design at the same time it is quite complex to implement.
- This model also lacks flexibility as the changes in the new tables or segments often yield very complex system management tasks. Here, a deletion of one segment can lead to the involuntary deletion of all segments under it.
- It has no standards as the implementation of this model does not provide any specific standard.
- It is also limited as many of the common relationships do not conform to the 1 to N format as required by the hierarchical model.

Data Models

HIERRCHICAL MODEL

Disadvantages of the hierarchical model :

1. **Limited flexibility:** One of the major disadvantages of hierarchical databases is that they are not as flexible as other types of databases, and are not well suited for handling complex data relationships or changes in data structures.
2. **Difficult to maintain and update:** Hierarchical databases can be difficult to maintain and update, since changes to the data structure can impact the entire database.
3. **Limited support for data manipulation:** Hierarchical databases do not support complex data manipulation operations, such as data aggregation or data mining.
4. **Limited interoperability:** Hierarchical databases are not compatible with other database models, which can limit their interoperability with other systems and applications.
5. **Lack of standardization:** Since hierarchical databases are not as widely used as other database models, there is a lack of standardization and best practices for their use and management.

Data Models

Relational Model:

Relational model stores data in the form of tables. This concept purposed by Dr. E.F. Codd, a researcher of IBM in the year 1960s. The relational model consists of three major components:

1. The set of relations and set of domains that defines the way data can be represented (data structure).

2. Integrity rules that define the procedure to protect the data (data integrity).

3. The operations that can be performed on data (data manipulation).

A relational model database is defined as a database that allows you to group its data items into one or more independent tables that can be related to one another by using fields common to each related table.

Data Models

Characteristics of Relational Database

Relational database systems have the following characteristics:

- The whole data is represented as an orderly arrangement of data into rows and columns, called a relation or table.
- That is, at any given row/column position in the relation there is one and only one value.

Dr. Codd, when formulating the relational model, chose the term “relation” on behalf of “table”.

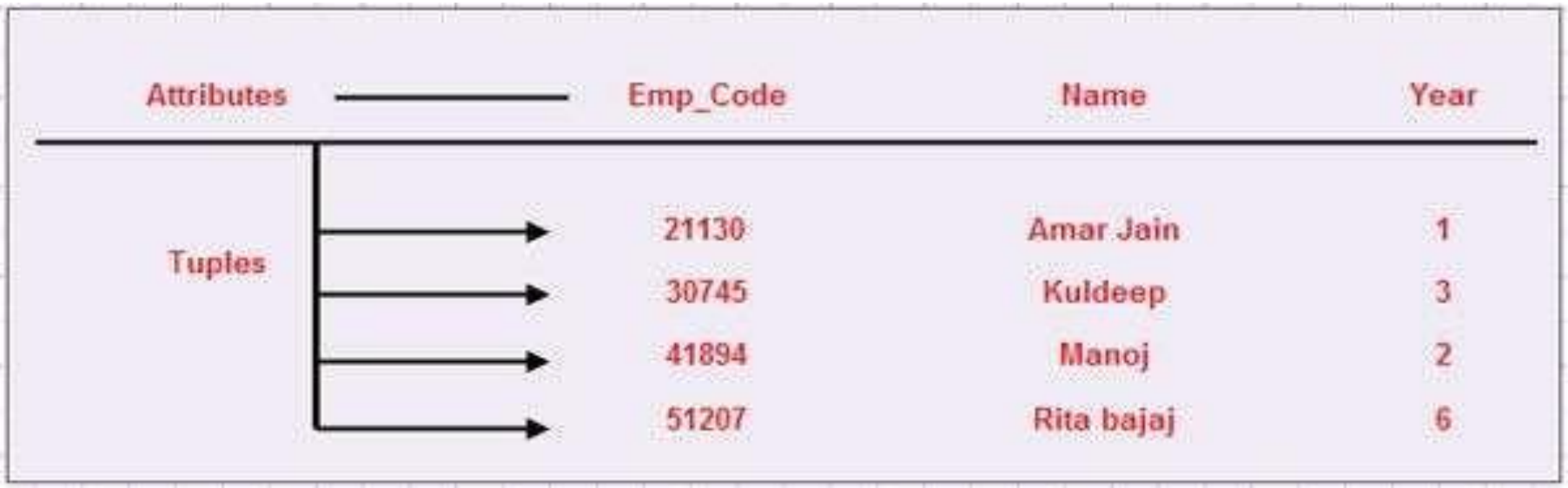
Relational Model satisfies the Codd’s 12 rules whereas network and hierarchical model does not satisfies.

The DBMS that supports Relational model is known as RDBMS(Relational Database mgt. system)

Data Models

Basic Terminology used in Relational Model

The figure shows a relation with the Formal names of the basic components marked the entire structure is, as we have said, a relation.



Data Models

Tuples of a Relation

Each row of data is a tuple.

Cardinality of a relation: The number of tuples in a relation determines its cardinality. In this case, the relation has a cardinality of 4.

Degree of a relation: Each column in the tuple is called an attribute. The number of attributes in a relation determines its degree. The relation in figure has a degree of 3.

Domains: A domain definition specifies the kind of data represented by the attribute.

More- particularly, a domain is the set of all possible values that an attribute may validly contain. Domains are often confused with data types, but this is inaccurate. Data type is a physical concept while domain is a logical one. “Number” is a data type and “Age” is a domain. To give another example “Street Name” and “Surname” might both be represented as text fields, but they are obviously different kinds of text fields; they belong to different domains.

For example, Person Age could be defined as “an integer in the range 0 to 120” whereas Exhibit Age (age of any object for exhibition) might simply by “an integer equal to or greater than 0.”

Data Models

Body of a Relation: The body of the relation consists of an unordered set of zero or more tuples. The relation is unordered. Record numbers do not apply to relations. A relation with no tuples still qualifies as a relation. Therefore, for a table to qualify as a relation each record must be uniquely identifiable and the table must contain no duplicate records.

Keys of a Relation

It is a set of one or more columns whose combined values are *unique* among all occurrences in a given table. A key is the relational means of specifying uniqueness. Some different types of keys are:

Primary key is an attribute or a set of attributes of a relation which posses the properties of uniqueness and irreducibility.

Foreign key is the attributes of a table, which refers to the primary key of some another table. Foreign key permit only those values, which appears in the primary key of the table to which it refers or may be null (Unknown).

Data Models

Example of Relational Model :

It can easily understood that, this model is very simple and has no redundancy. The total database is divided in to two tables.

Customer table contains the information about the customers with CNO as the primary key.

The Cutomer_Loan table stores the information about CNO, LNO and AMOUNT. It has the primary key combination of CNO and LNO.

Here, CNO also acts as the foreign key and refers to CNO of Customer table.

It means, only those customer number are allowed in transaction table Cutomer_Loan that have their entry in the master Customer table.

Data Models

Customer Table

CNO	NAME	ADDRESS	CITY
C1	Rahal	Thapar campus	Patiala
C2	Ruhi	Tagore Nagar	Jalandhar
C3	Chachat	Dharampura	Qadian
C4	Pooja	GNDU	Amritsar

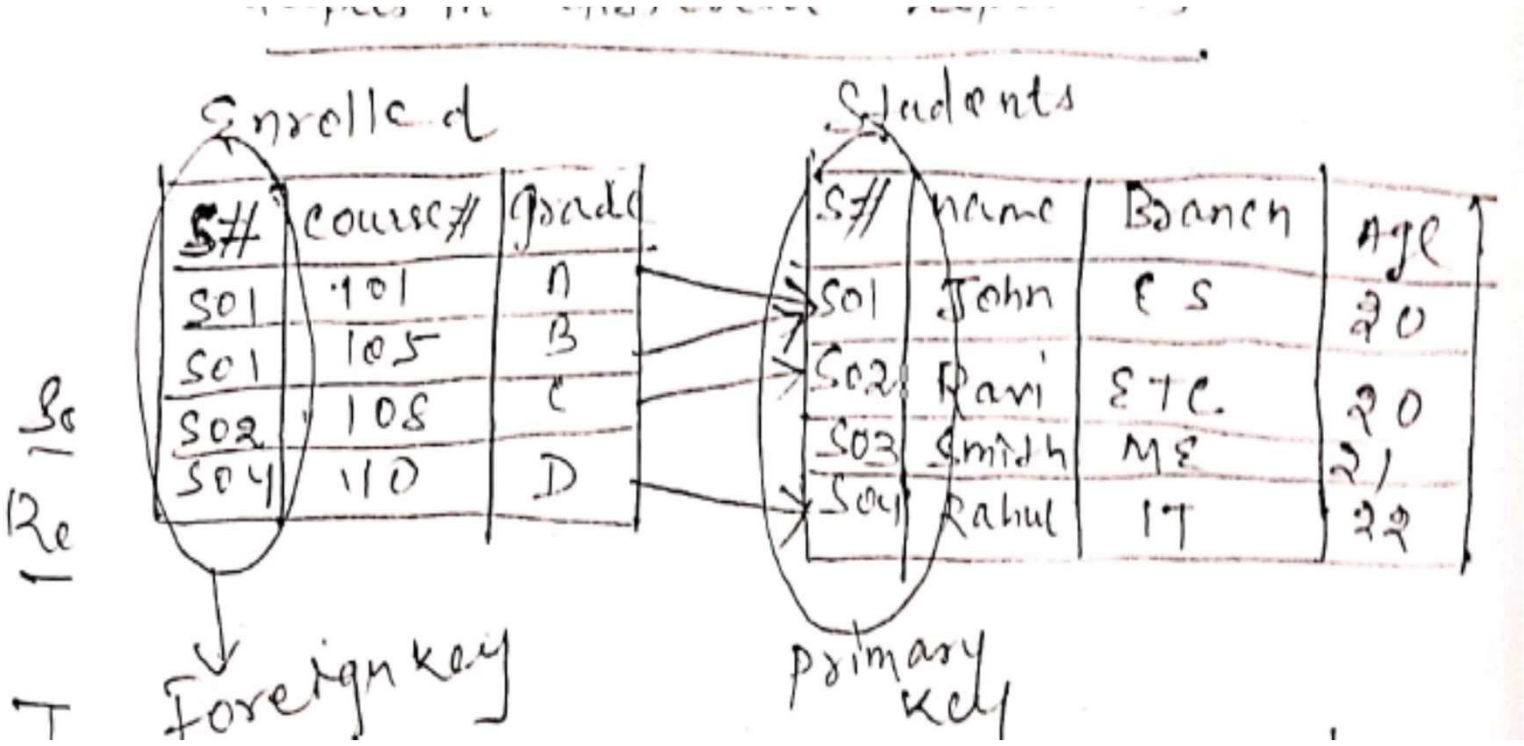
Customer_Loan Table

CNO	LNO	AMOUNT
C1	L1	10000
C2	L1	10000
C3	L2	15000
C3	L3	25000
C4	L4	35000

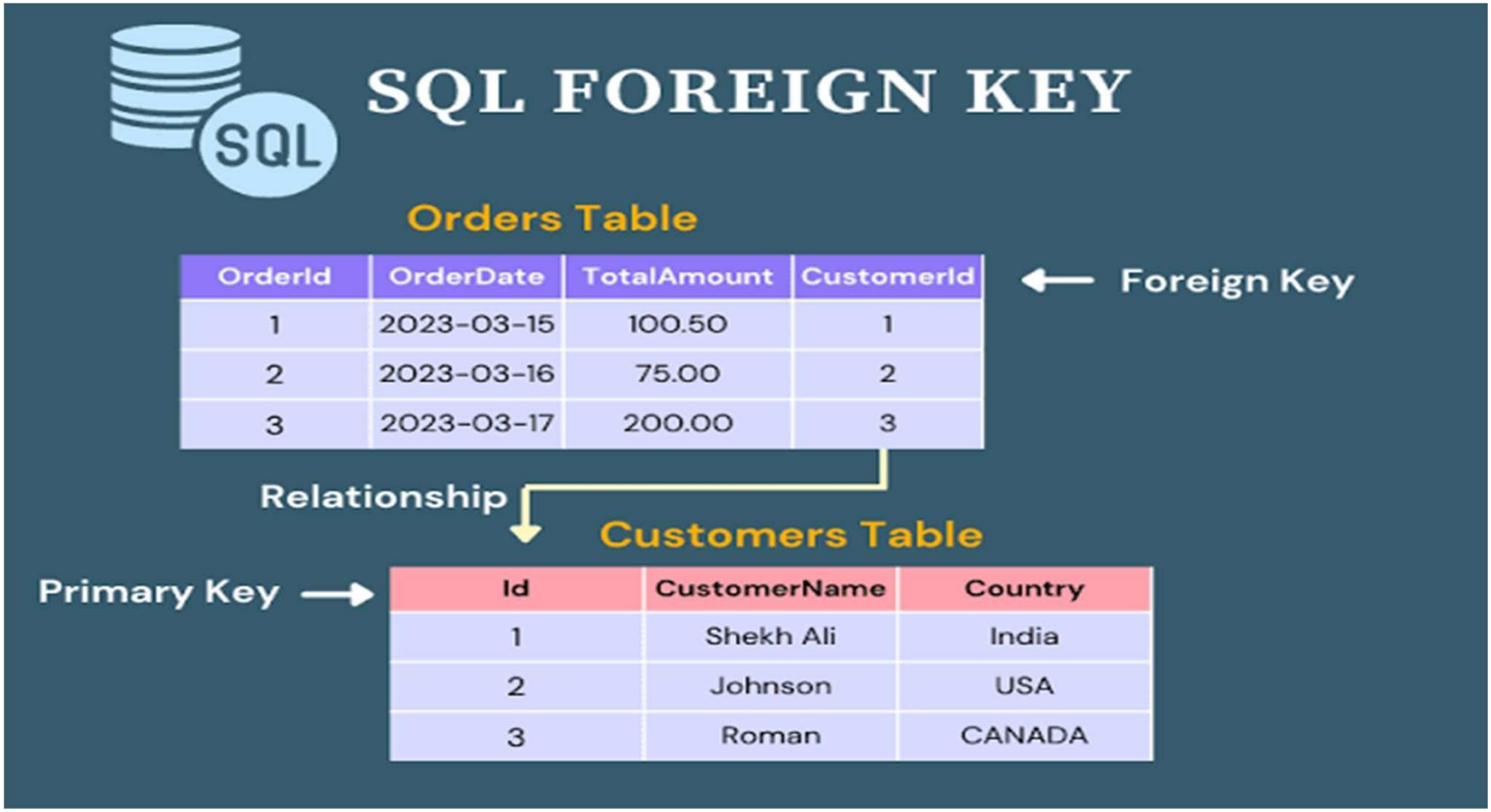
Relationat Model of Custmor-Loan database

Data Models

Example 2 of Relational Model



Data Models


SQL FOREIGN KEY
Orders Table
OrderId | OrderDate | TotalAmount | CustomerId
1 | 2023-03-15 | 100.50 | 1
2 | 2023-03-16 | 75.00 | 2
3 | 2023-03-17 | 200.00 | 3
Foreign Key
Relationship
Customers Table
Primary Key
Id | CustomerName | Country
1 | Shekh Ali | India
2 | Johnson | USA
3 | Roman | CANADA

Data Models

Collection of records are called record types in hierarchical and network model (i.e equivalent to a table)

SCHEMA-1.CUSTOMER(CNO,CNAME,CUST _LOCN)

2.LOAN(L_NO,LOAN_AMOUNT,INTEREST_RATE)

3.CUST_LOAN(CNO, L_NO)(WHICH CUSTOMER HAS TAKEN WHICH LOAN)

RELATIONAL MODEL REPRESENTATION of customer loan schema

CUSTOMER TABLE/CUSTOMER RELATION

CNO	CNAME	CUST_LOCATION
101	RAM	BBSR
102	RAVI	CTC
103	HARI	PURI

Data Models

LOAN

LOAN_NO	L_AMOUNT	INTREST_RATE
1011	8755	6%
1015	5000	5%
2010	2555	3%
2015	20000	7%

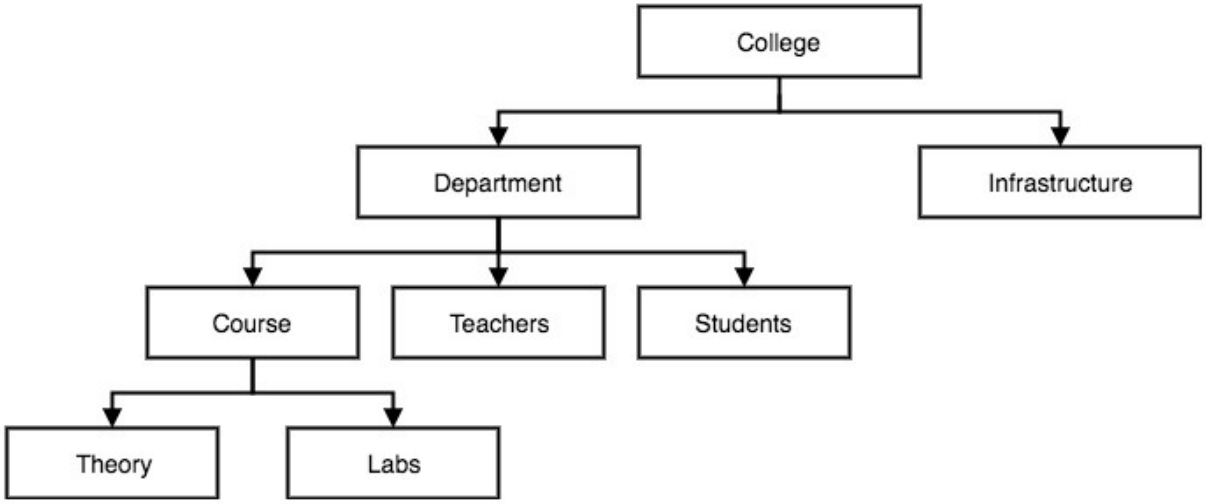
Data Models

CUST_LOAN(Which customer has taken which loan)

CNO	LOAN_NO
101	1011
102	1011
103	2010
103	2015
101	1015

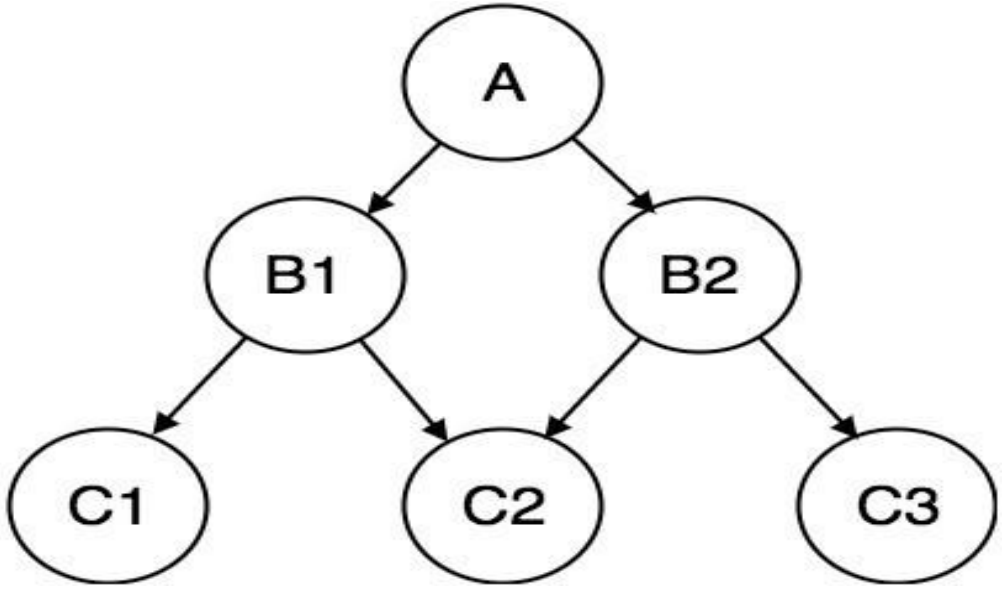
Data Models

Hierarchical Data Model: (tree like structure)

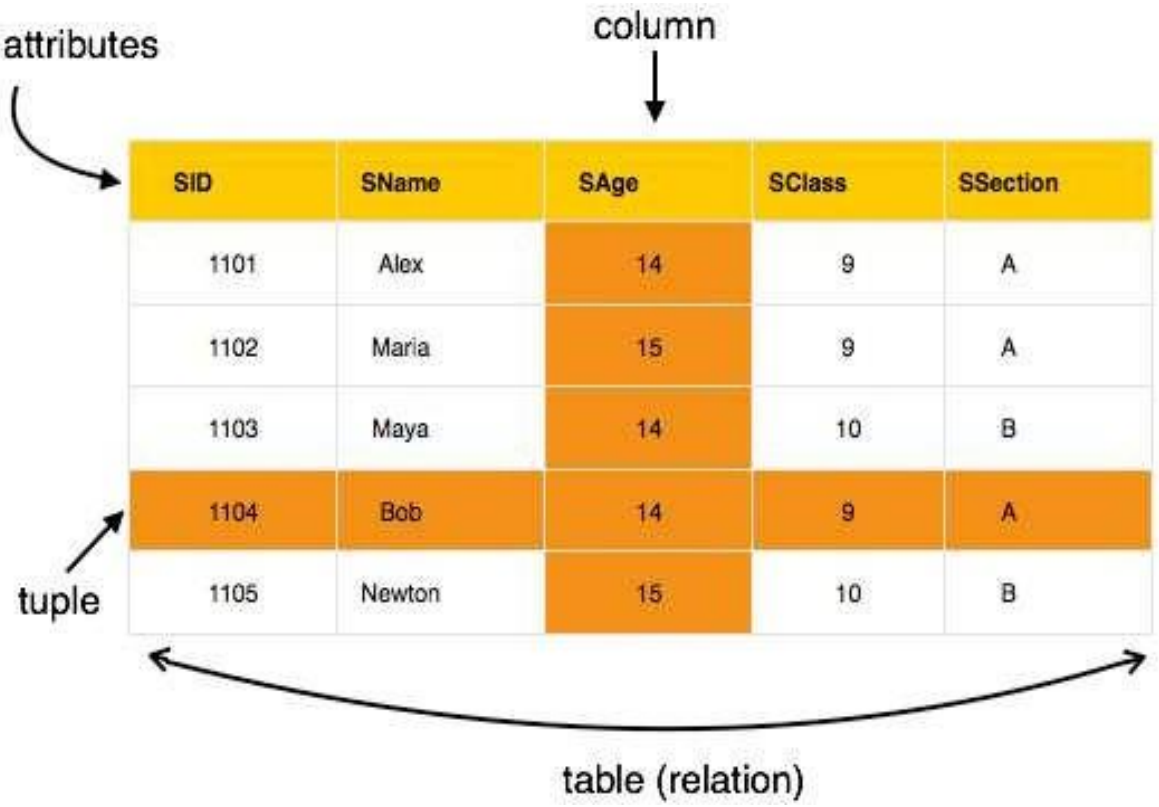


- **Network Model:**

allows each record to have multiple parent and multiple child records, which when visualized, form a web-like structure of networked records. In contrast, a hierarchical model data member can only have a single parent record but can have many child records.



- **Relational Model:**



Key

- Key of a Relation:
  - Each row has a value of a data item (or set of items) that uniquely identifies that row in the table
    - Called the ***key***
  - In the STUDENT table, SSN is the key
  - Sometimes row-ids or sequential numbers are assigned as keys to identify the rows in a table
    - Called ***artificial key*** **or** ***surrogate key***

Schema

- The **Schema** (or description) of a Relation:
  - Denoted by R(A1, A2, ...An)
  - R is the **name** of the relation
  - The **attributes** of the relation are A1, A2, …, An
- Example:

  CUSTOMER (Cust-id, Cust-name, Address, Phone#)
  - CUSTOMER is the relation name
  - Defined over the four attributes: Cust-id, Cust-name, Address, Phone#
- Each attribute has a **domain** or a set of valid values.
  - For example, the domain of Cust-id is 6 digit numbers.

Definition Summary

Informal Terms		Formal Terms
Table		Relation
Column Header		Attribute
All possible Column Values		Domain
Row		Tuple
Table Definition		Schema of a Relation

Data Model

It is a collection of concepts that can be used to describe the structure of a database

Categories

1. **Relational Model**
2. **Network Model**
3. **Object Oriented Model**

Data modelling Objects

These are the basic building blocks of the modelling which are categorised into three types:

1. **Entity**
2. **Attributes**
3. **Relationships**

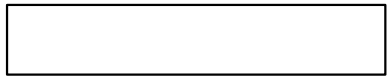
Entity

It is a real world object or concept

OR

Entity is a thing or object in real world with an independent existence

Example- student, employee, college, bank, house, school

It is represented by 

Data Model

Attributes

The property that characterized an entity set is referred to as Attributes

Example- If student is an entity then,

Regd. No, roll no, section, branch, contact no, etc can be the Attributes

- If Employee is an entity set then,

Employee name, employee Id, age, salary, date of joining, contact no, date of birth can be the attributes

Attributes can be categorized into three types:

1. **Simple and composite attributes**
2. **Single and multivalued attributes**
3. **Stored and derived attributes**

Data Model

1. Simple and Composite Attributes

Simple Attributes

The attributes which cannot be simplify further

Example- Regd. No, loan no, course code, course id

Composite Attributes

The attributes which can be further divided into smaller sub parts

Example- Address, name

2. Single and Multi valued Attributes

Single value

It can hold single value of an entity

Example- emp id, loan no, regd no, age

Multi value

It can hold multiple value of an entity

Example- phone no, degree of a person

3. Stored and derived Attributes

The attribute which is derived from other attributes is derived attributes and from whom it is derived is known as stored attributes

Example- DOB ← Age (Marks ← Percentage)

(Stored attributes) (Derived attributes) (Stored attributes) (Derived attributes)

Data Model

NULL Value

NULL values are mainly used to represent the missing information or when the attribute value is unknown

Entity Set

The collection of all entities of a particular entity type in the database at any point of time is called as entity set

An entity types defines a collection or set of entities that have the same attributes

Entity Type Name: Student

Attribute: Name, Roll no, Branch

Entity	**Student 1**
	Abhay, 1, CSE
Entity	**Student 2**
	Rahul, 2, CSE
Entity	**Student 3**
	Diksha, 3, CSE

Entity Set

Data Model

Key attribute of an entity type

An entity type usually has an attribute whose values are distinct for each individual entity in the entity set.

-Such an attribute is called as an key attribute and its values can be used to identify each entity uniquely.

Example- Regd. No of a student

Account number of a person or customer

Employee Id of an employee

Value set (Domain) of attributes

Each simple attribute of an entity type is associated with a value set or (Domain of values) which specifies the set of values that may be assigned to that attributes for each individual entity.

Example- If the salary allowed for employees is in between 20,000 to 1 lakh.Then we can specify the value set of the salary attributes to be the set of integers between 20,000 to 1 lakh.

Data Model

Schema

Student

Regd No	Name	Branch	Section	Mark	Phone no

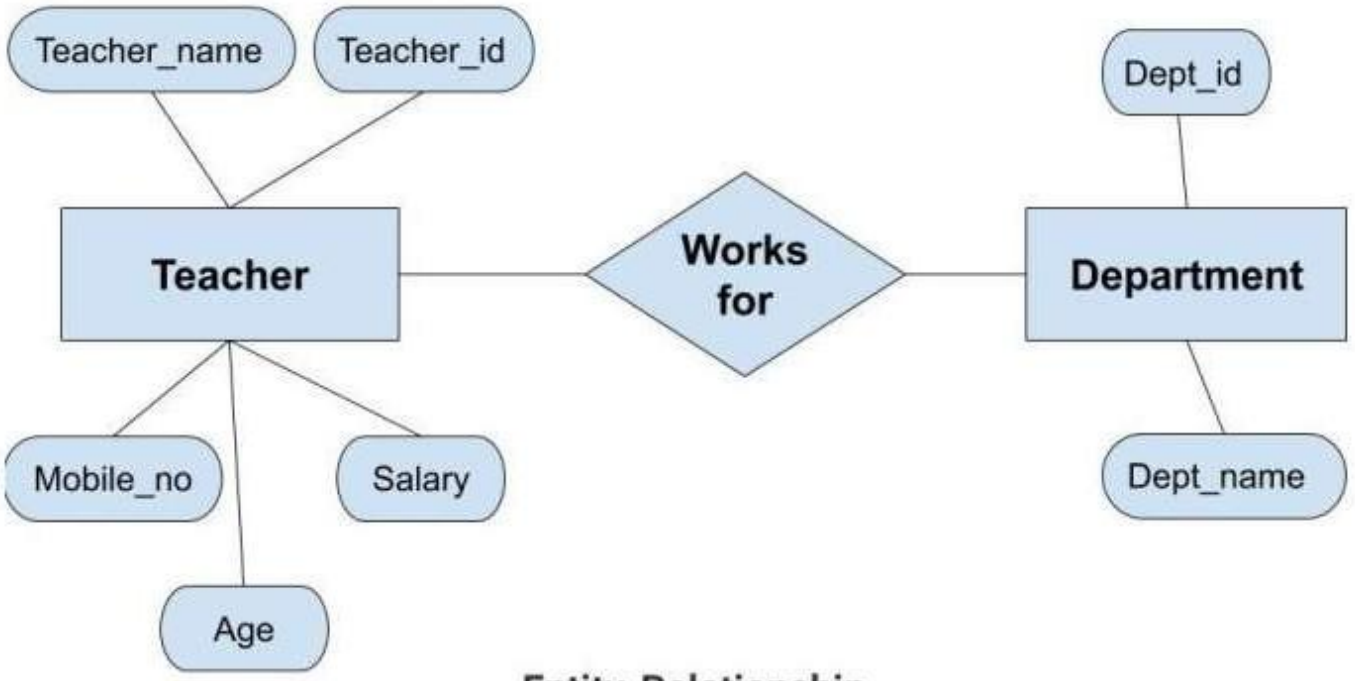
Employee

Emp name	Emp id	age	salary	Date of joining	Dept

Subject

Subject name	Subject code	credit	department	semester

- **ER- Model:**

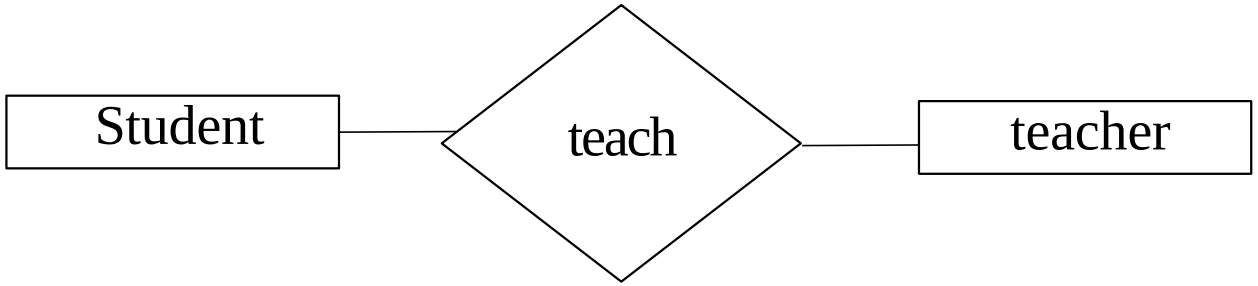


Entity-Relationship Model

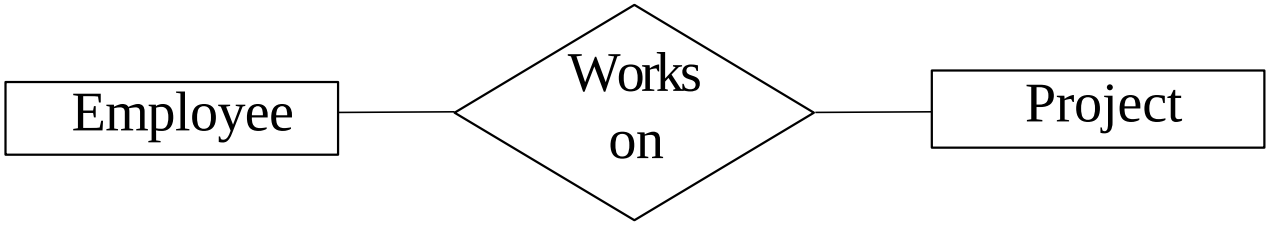
Entity Relationship Model

- Provides the graphical or conceptual view of the database design
- Describes data as a set of entities, attributes, and relationships
- Defines the structure of a database with the help of a diagram called as ER-Diagram

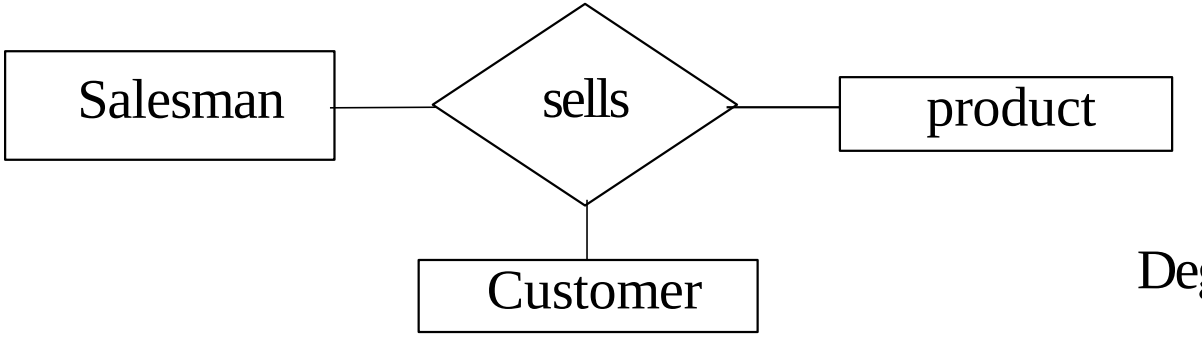
Relationship Degree



Degree-2 (Binary Relationship)



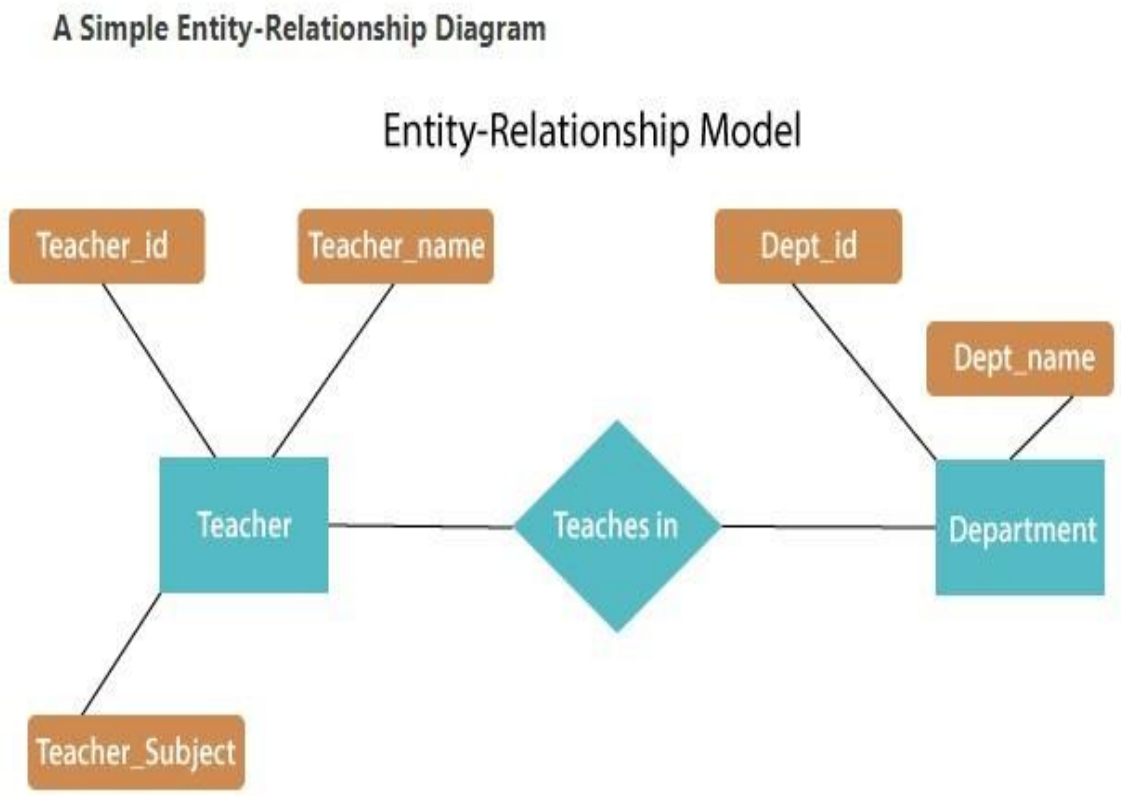
Degree-2 (Binary Relationship)



Degree-3 (Ternary Relationship)

The relationship with <u>degree 2</u> is known as <u>Binary Relationship</u> and the relationship with <u>degree 3</u> is known as <u>Ternary relationship</u>

ER-Diagram



Main Components:

- **Entities**
- **Attributes**
- **Relationship**

Entity Relationship Diagram (ER Diagram Model)

The database structure employing the ER model it shown pictorially or diagrammatically by using entity relationship diagram i.e., ER diagram or model.

In order to draw the ER model the following things are to be identified:

-Identify objects

-Remove duplicate entities

-List out the attributes of each entity

-Mark the key attribute

-Define the relationships

-Describe the cardinality ratio of the relationship

Symbols for Representation



Represents Entity



Represents Attribute



Represents Relationship

Links Attribute(s) to entity set(s) or
Entity set(s) to Relationship set(s)



Represents Multivalued Attributes



Represents Derived Attributes



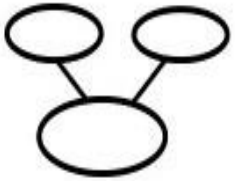
Represents Total Participation of Entity



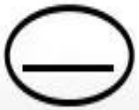
Represents Weak Entity



Represents Weak Relationships



Represents Composite Attributes



Represents Key Attributes / Single Valued
Attributes

Strong Entity Type

The type of entity that have the key attributes of their own is known as strong entity type

-Strong entity always has a primary key.

-It is represented by a single rectangle

-Two strong entity's relationship is represented by a single diamond

-Strong entity's have either total participation or not

Weak Entity Type

Entity type that do not have key attribute of their own.

OR

They do not have sufficient attributes to form a key are known as weak entity type

-Weak entity does not have primary key

-Weak entity is represented by a double rectangle

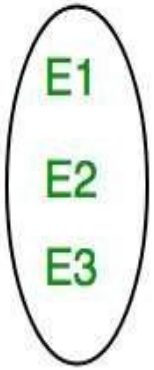
-While the relation between one strong and one weak entity is represented by a double diamond

-While weak entity always has a total participation

Entity

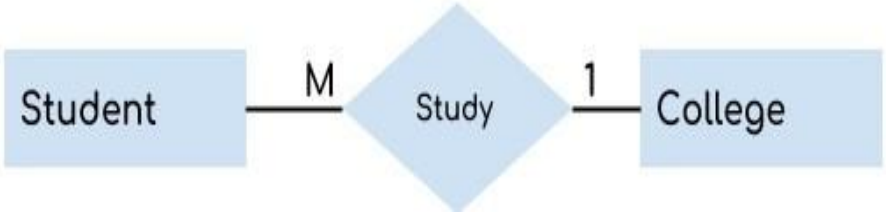


Entity Type

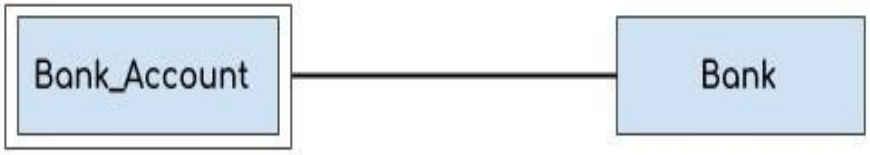


Entity Set

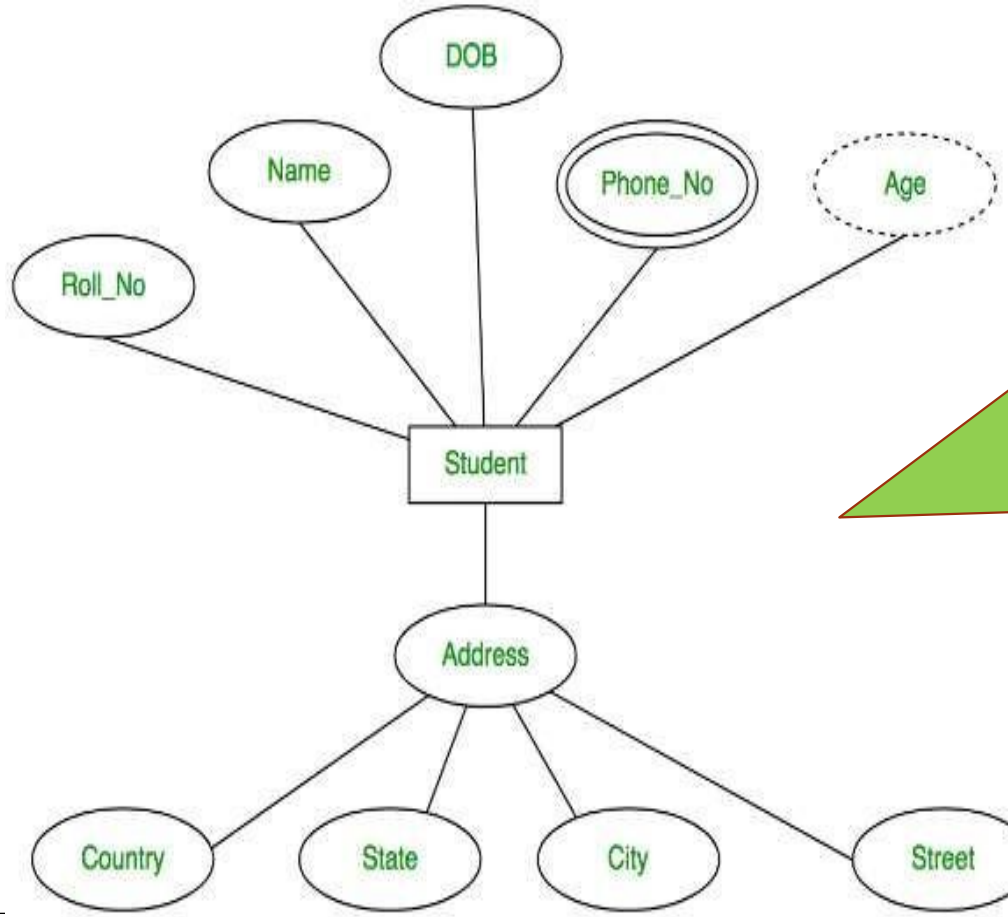
Strong Entity



Weak Entity



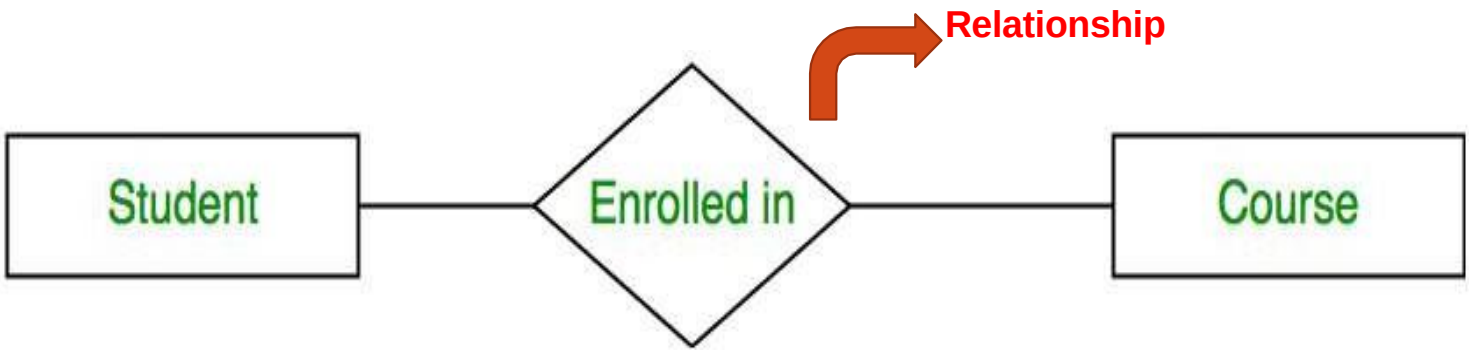
Attribute



Types of attributes:

Key
Simple
Composite
Single valued
Multi-valued
Derived

Relationship



Degree of Relationship:

The number of participating entities in a relationship defines the degree of the relationship.

- **Binary = degree 2**
- **Ternary = degree 3**
- **n-ary = degree**

Mapping Cardinalities

Cardinality defines the number of entities in one entity set, which can be associated with the number of entities of other set via relationship set.

It is the specification of the number of occurrences of one entity type that can be related to the number of occurrences of another entity type.

It specifies the maximum number of relationship instances that an entity can participate

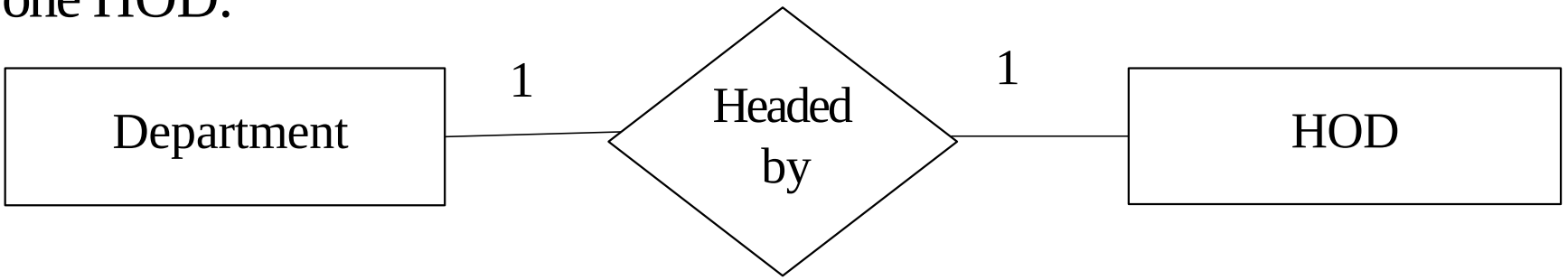
Four types:

- One-to-one
- One-to-many
- Many-to-one
- Many-to-many

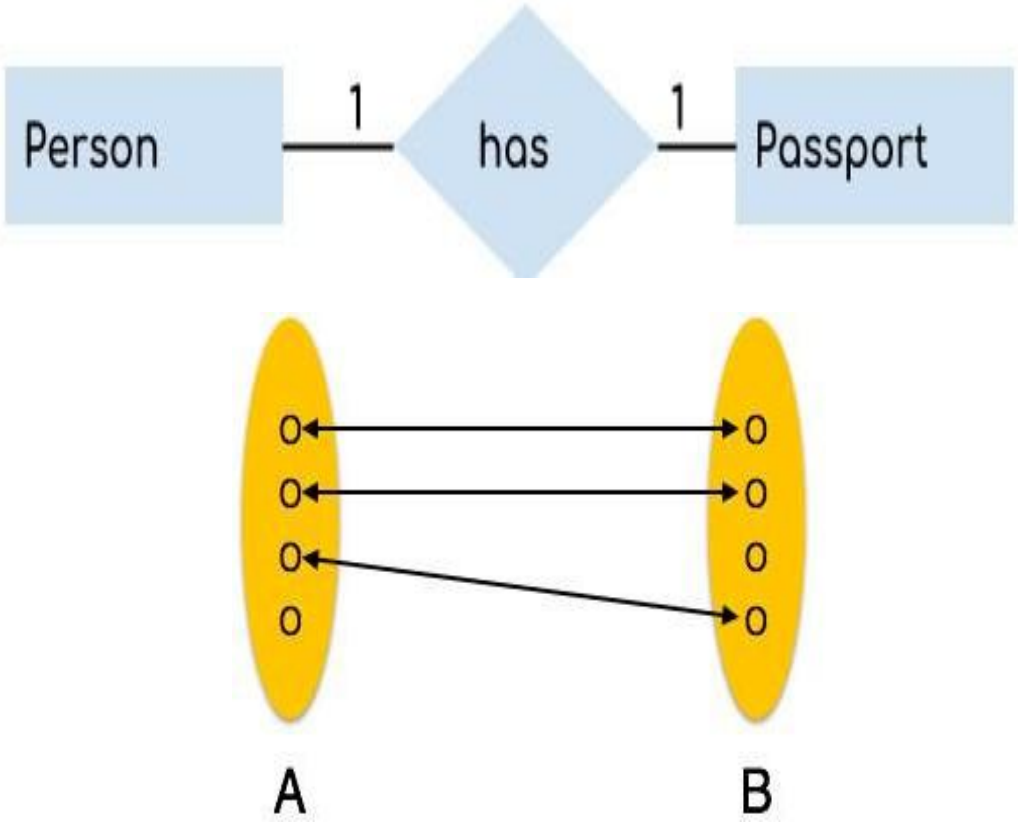
One - to - one

An occurrence of entity A can relate to one and only one occurrence of entity B and an occurrence of entity B relate to only one occurrence of entity A

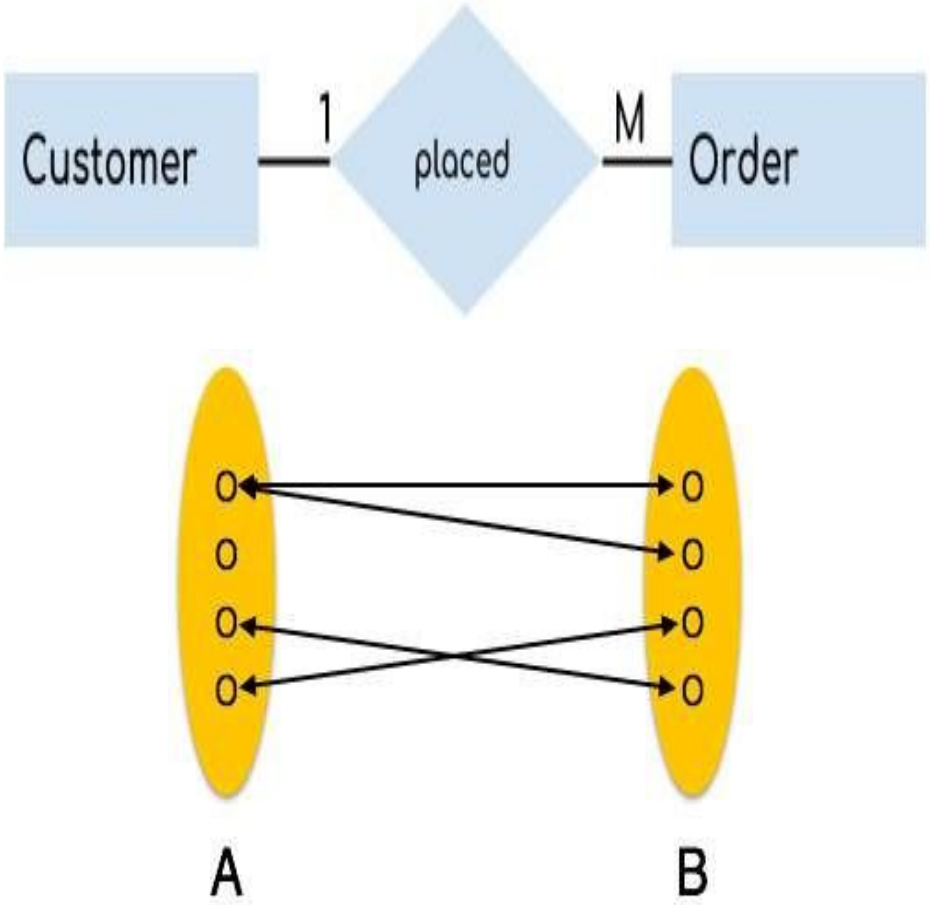
'Headed by' relationship is 1:1 relationship. As it represents that at any point of time one HOD can manage only one Department and a Department has only one HOD.



One-to-One Relationship



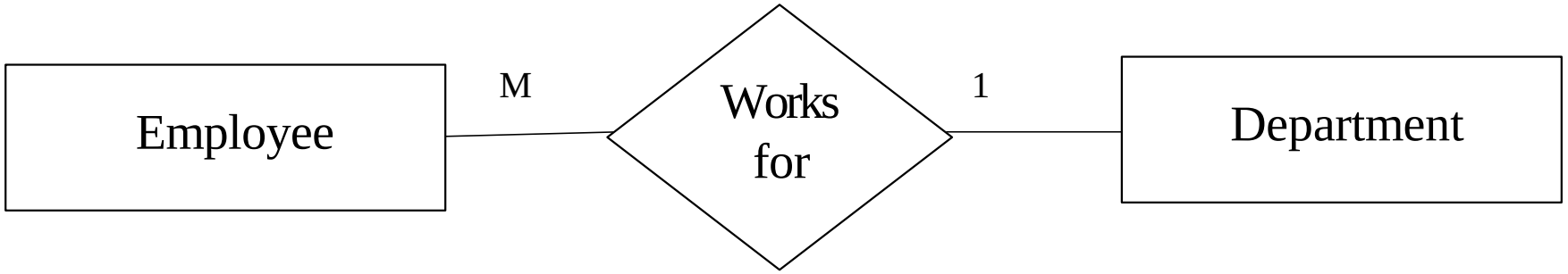
One-to-Many Relationship



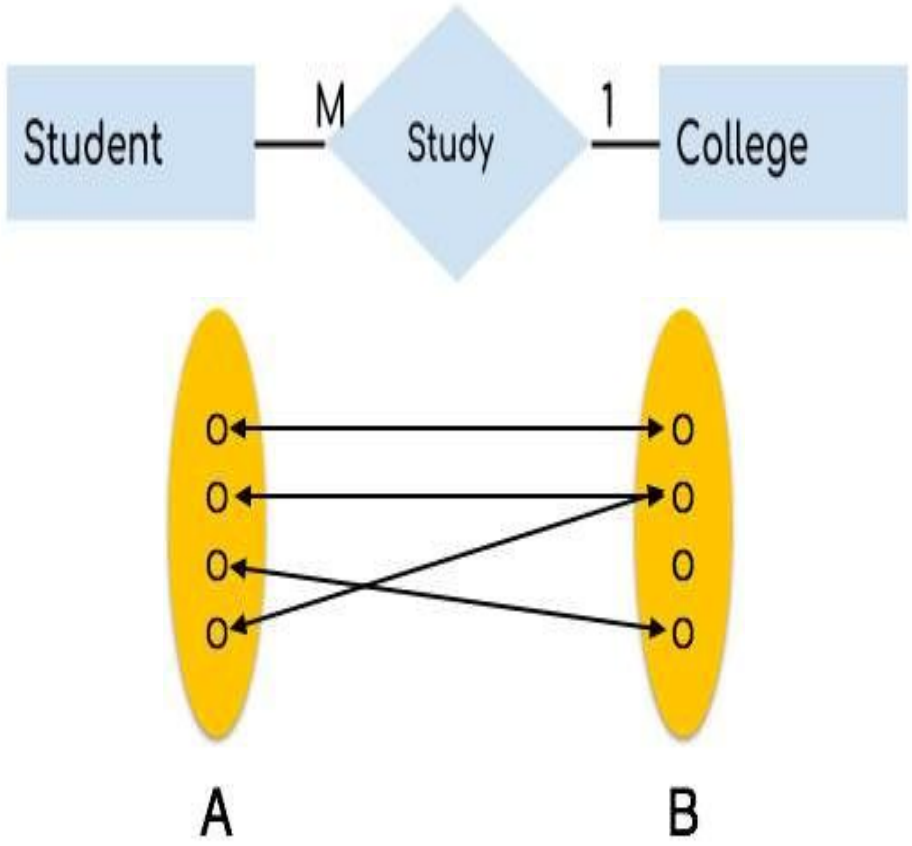
Many- to- One

An occurrence of entity A can relate to one occurrence of entity B but one occurrence of entity B can relate to many occurrences of entity B.

- The relationship 'works for' between employee and department is M:1 relationship i.e., each department can be related to any number of employees, but an employee can be related to only one of the department.

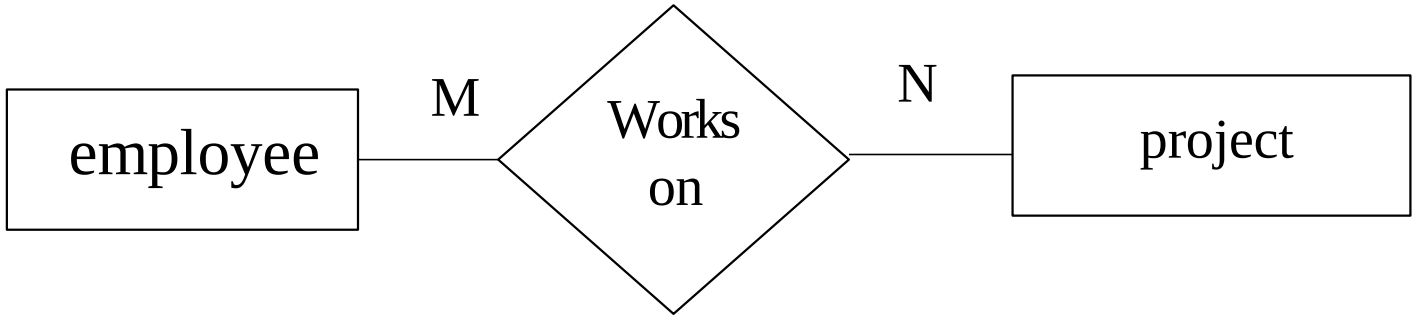


Many-to-One Relationship

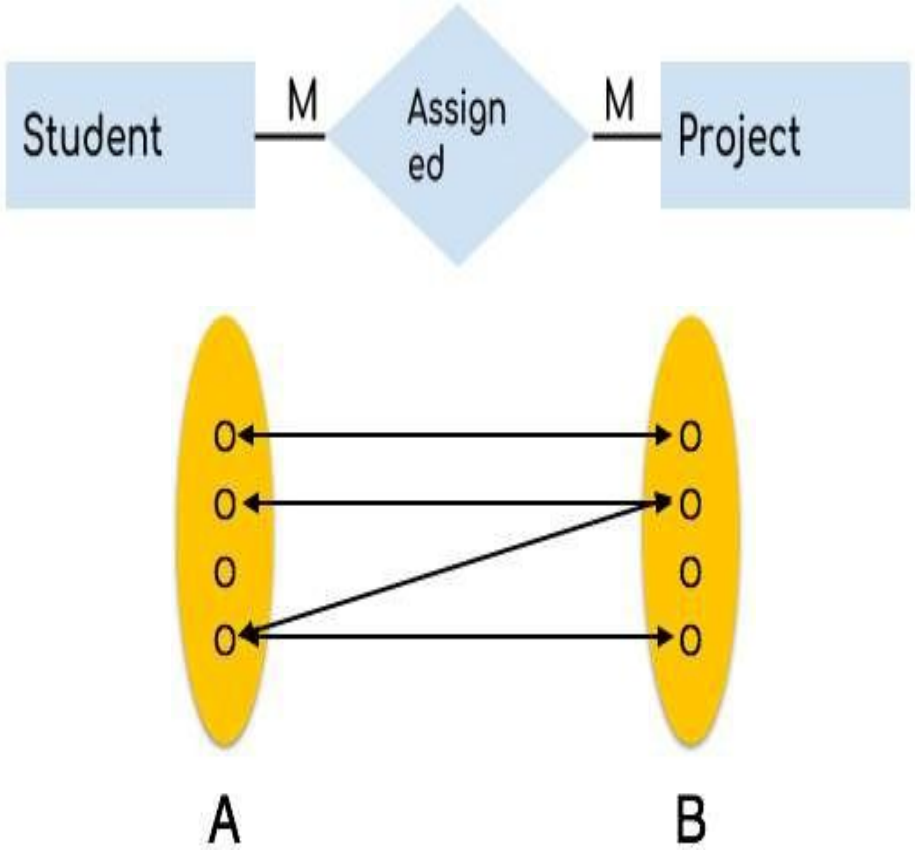


Many-to-Many

An occurrence of entity A can relate to one or more occurrence of entity B.While an occurrence of entity B can relate to one or more occurrences of entity A.



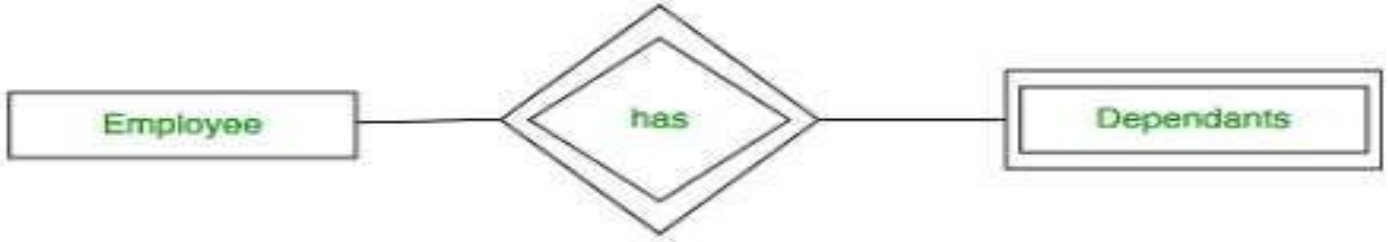
Many-to-Many Relationship



Identifying Relationship

The relationship associating the weak entity type with the identifying entity type is called as Identifying Relationship of weak entity type.

- A weak entity type always have a total participation with respect to its identifying relationship because a weak entity can not be identified without an owner entity.



Participation Constraints

- Specifies the number of instances of an entity that are participating in the relationship type

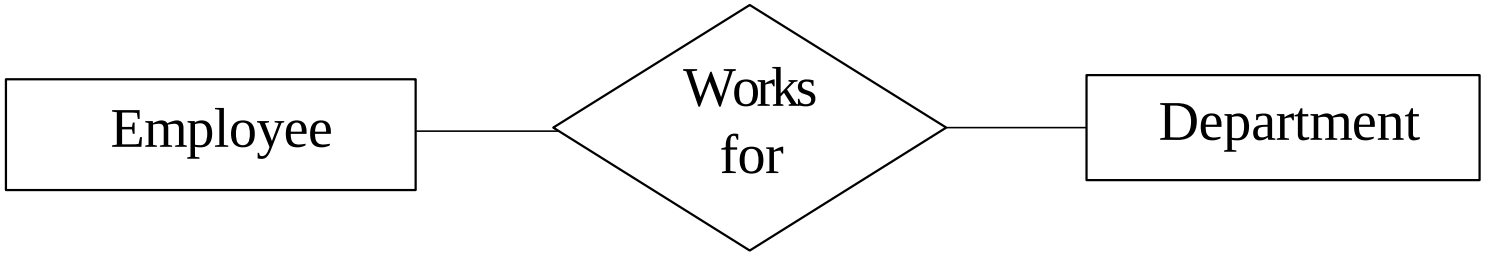
- There are two types of Participation constraint:

  - Total participation
  - Partial participation

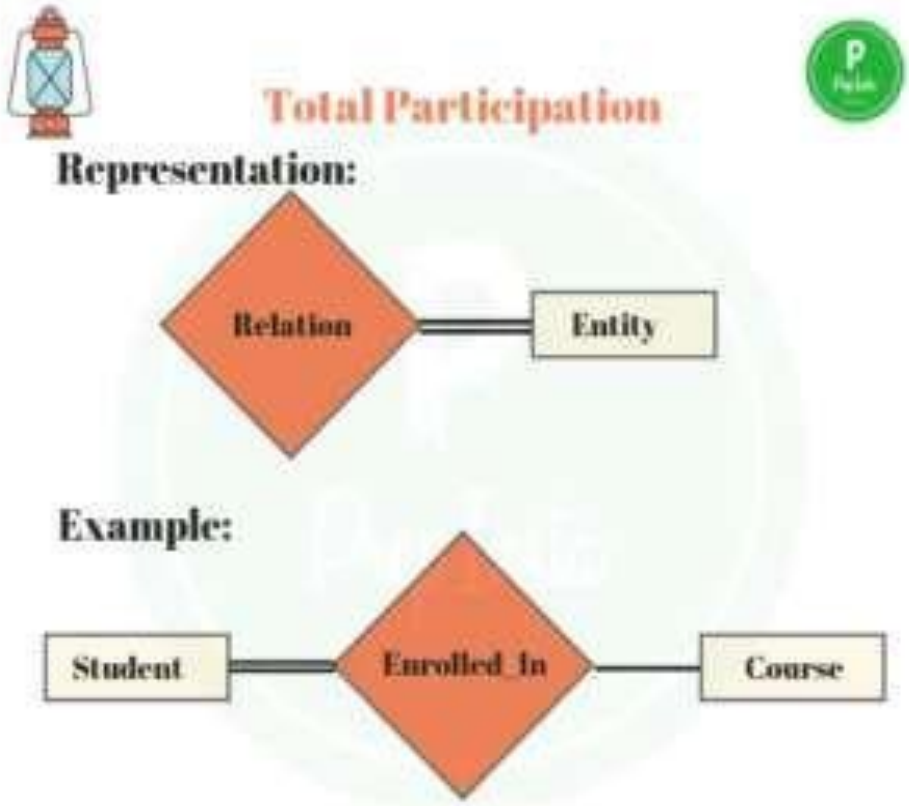
Total Participation Constraint

The participation of entity set E in a relationship set R is said to be total if every entity in E participates in at least one relationship in R

- The participation of the entity employee in 'Works for' relationship is total as every employee works for a particular department.



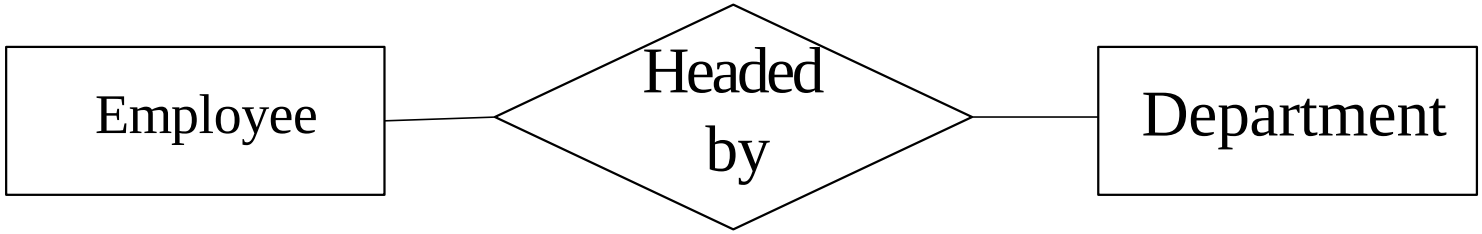
Total Participation



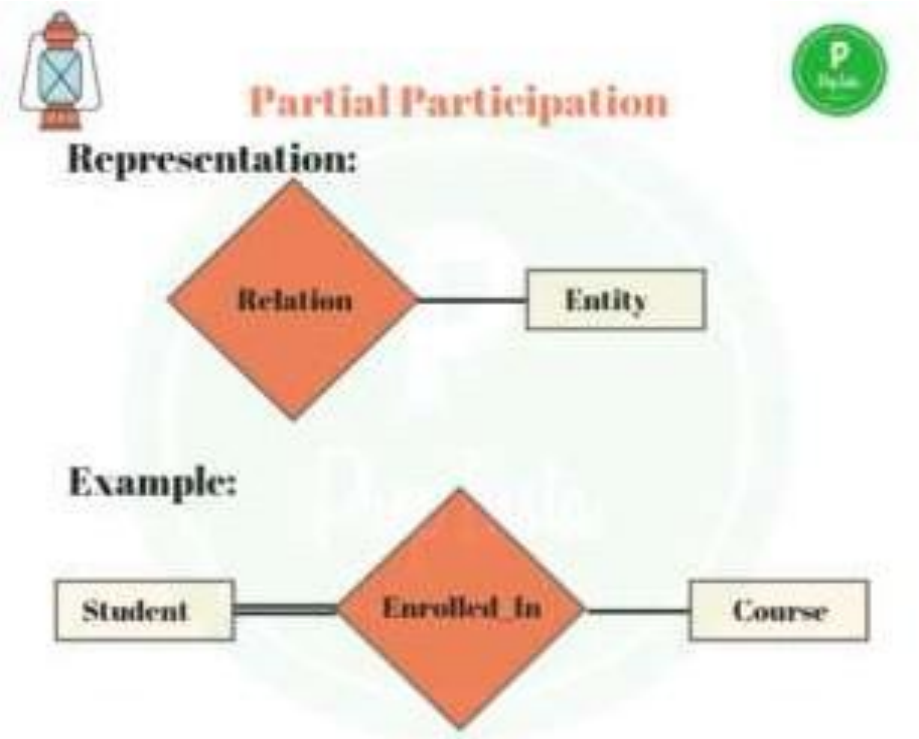
Partial Participation Constraint

The participation of entity set E in a relationship set R is said to be partial if only some entities in E participate in the relationship R

- The participation of the entity employee in ‘headed by’ relationship is partial as we do not expect that every employee is do not headed by the department.



Partial Participation



EER Features:

- EER is a high-level data model that incorporates the extensions to the original ER model.

- In addition to ER model concepts EER includes:

❖ Subclasses and Super classes.
❖ Specialization and Generalization.
❖ Category or union type.
❖ Aggregation.
❖ Attribute or relationship inheritances

Enhanced ER (EER) model

- Created to design more accurate database schemas
- Reflect the data properties and constraints more precisely
- More complex requirements than traditional application

Enhanced ER (EER) model

- EER model includes all modeling concepts of the ER model
- In addition, EER includes:
- Subclasses and superclasses
- Specialization and generalization
- Category or union type
- Attribute and relationship inheritance

- These concepts are used to create EE-R diagrams.
- EER diagrams are basically a more expansive version of ER diagrams.
- EER models are helpful tools for designing databases with high level models.
- With their enhanced features, you can plan databases more thoroughly by searching into the properties and constraints with greater precision.

When to use ER diagrams vs. EER diagrams

- An ER diagram gives you the visual outlook of your database. It details the relationships and attributes of its entities, covering the way for smooth databases development in the steps ahead.
- EER diagrams, on the other hand, are perfect for taking more detailed look at your information.When your database contains a large amount of data, its best to turn to an enhanced version to more deeply understand your model
- Both are useful, and it is mostly depends on the size and detail of your data.The more complicated the data, the more likely you will need to use an EER diagram to make sure you are properly organizing every relationship.

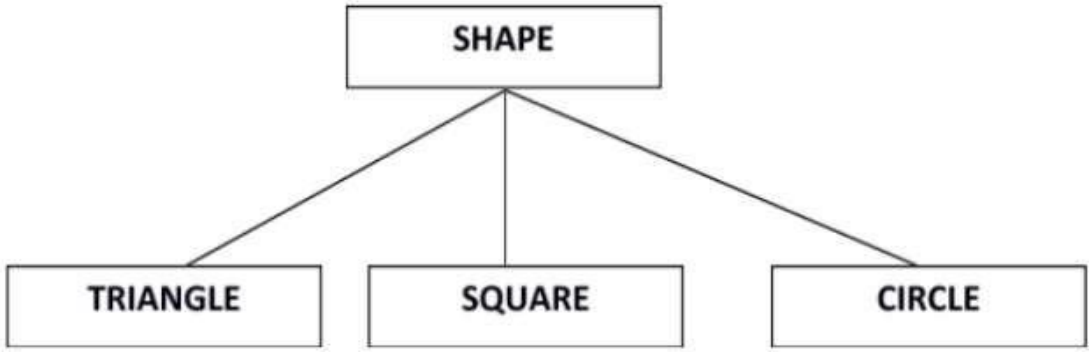
EER Features:

- **Features of EER Model**
- EER creates a design more accurate to database schemas.
- It reflects the data properties and constraints more precisely.
- It includes all modeling concepts of the ER model.
- Diagrammatic technique helps for displaying the EER schema.
- It includes the concept of specialization and generalization.
- It is used to represent a collection of objects that is union of objects of different of different entity types.

Subclasses and Super class

- Super class is an entity that can be divided into further subtype.
- Super class – An entity type that represents a general concept at a high level
- Sub class- An entity type that represents a specific concept at lower level

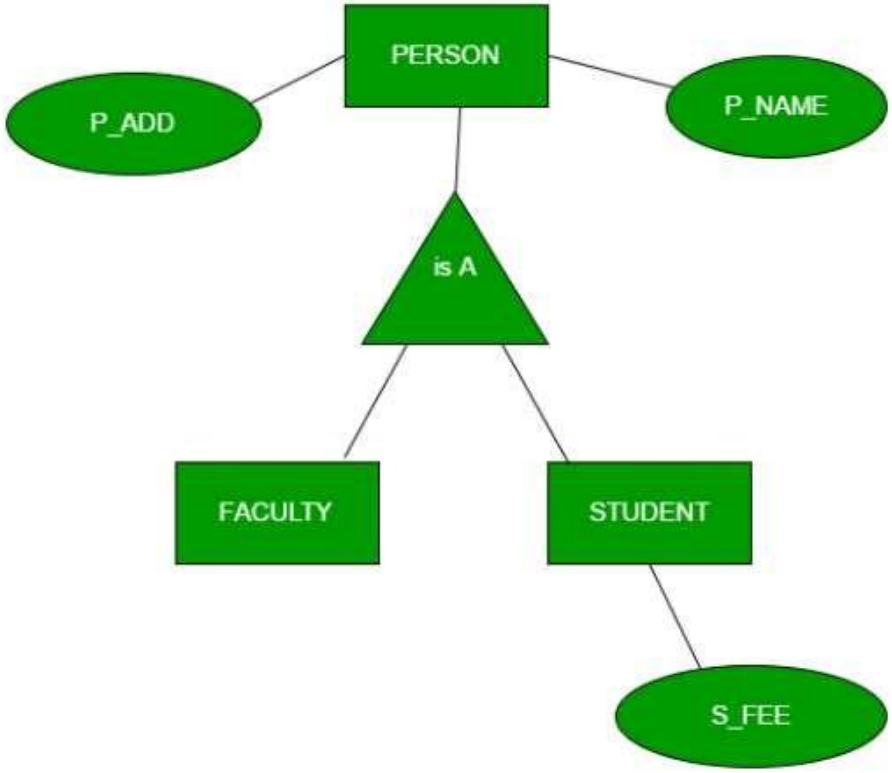
- For **example** – consider Shape super class.



Super class shape has sub groups: Triangle, Square and Circle.
Sub classes are the group of entities with some unique attributes. Sub class inherits the properties and attributes from super class.

Generalization

- Generalization is a process of generalizing an entity which contains generalized attributes or properties of generalized entities.

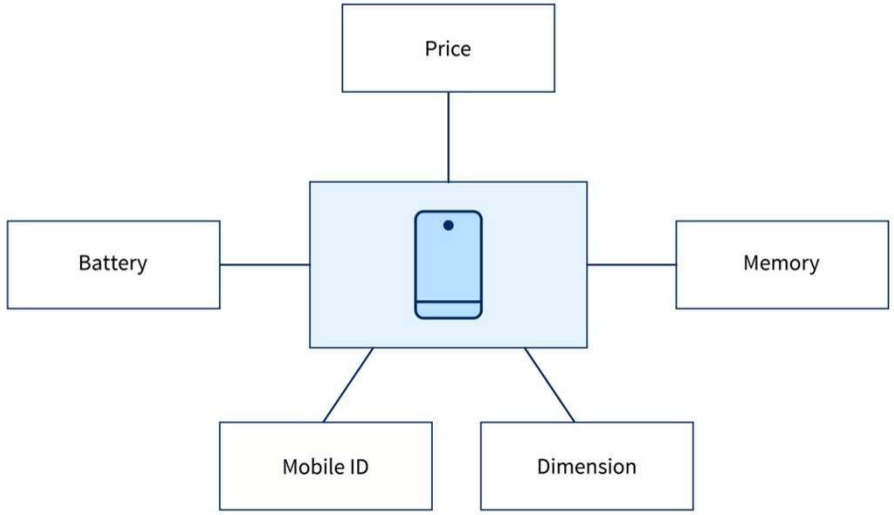


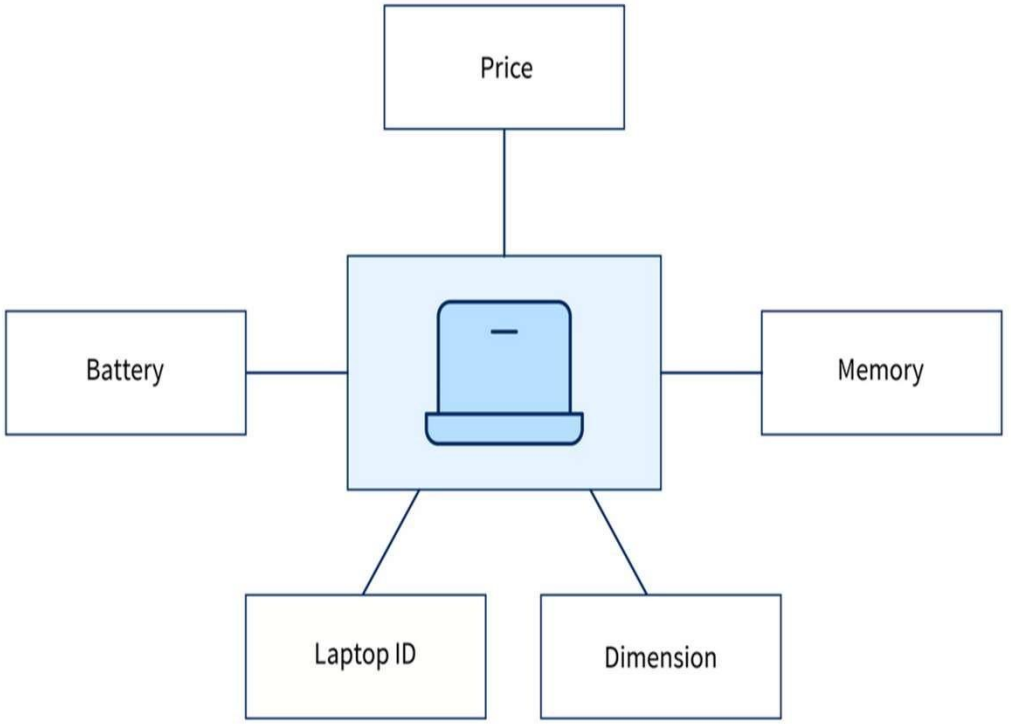
Generalization

- Generalization is simply gathering the common properties from entities and creating a generalized concept from those extracted data.
- Generalization helps in improving the flexibility, and reusability of the database.
- Generalization is important in the database because it helps to gather important information so that it becomes easier and faster for the user the analysis of data and it also helps in making decisions faster.

Generalization

Entities before Generalization

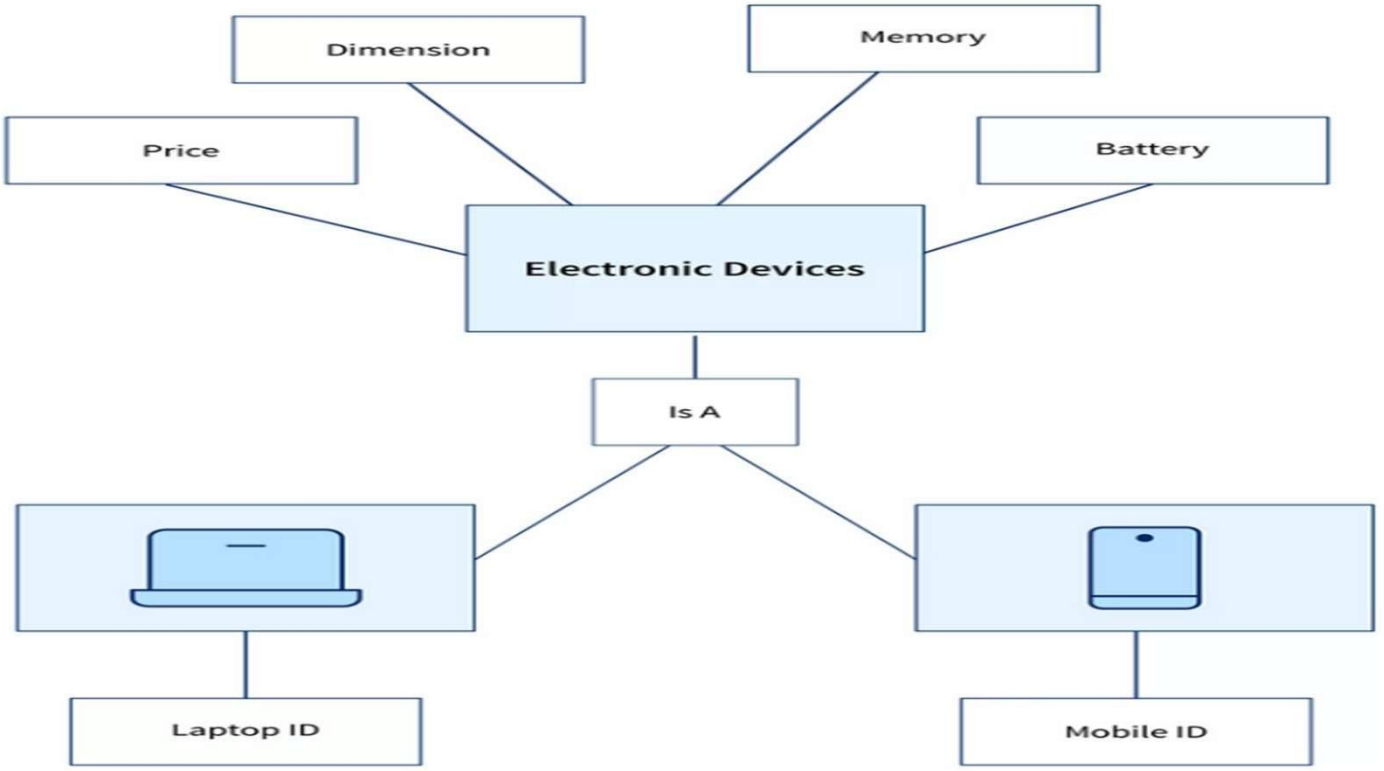




Generalization

ER Diagram after Generalization

We can see that for laptops and mobiles some attributes like price, memory, battery, and dimension are the same so we can apply generalization and can group them into an entity called Electronic Devices.



Generalization

- Generalization is the process of generalizing the entities which contain the properties of all the generalized entities.
- It is a bottom approach, in which two lower level entities combine to form a higher level entity.
- Generalization is the reverse process of Specialization.
- It defines a general entity type from a set of specialized entity type.
- It minimizes the difference between the entities by identifying the common features.

For example:

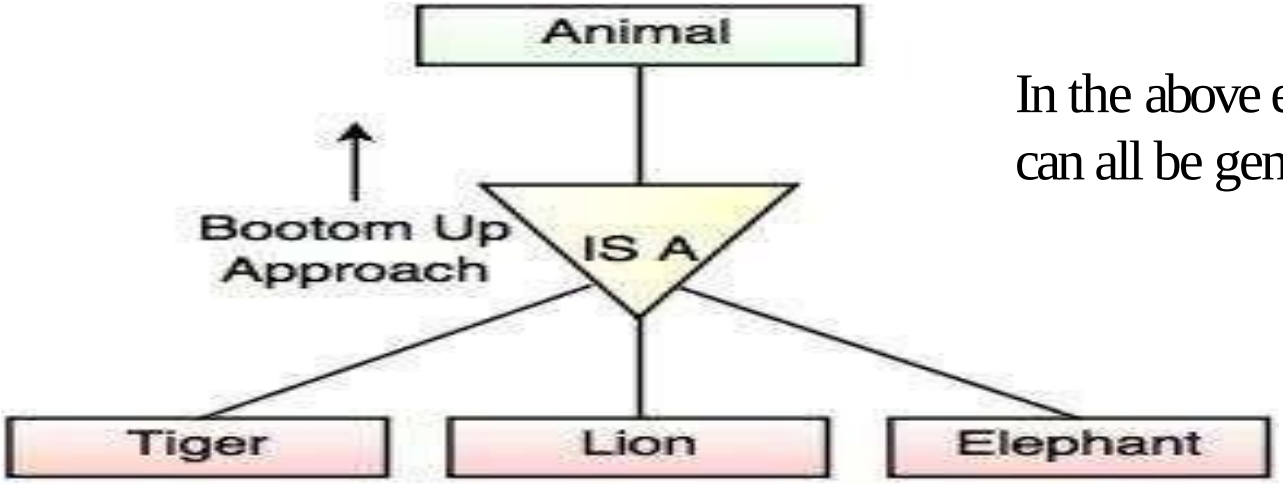
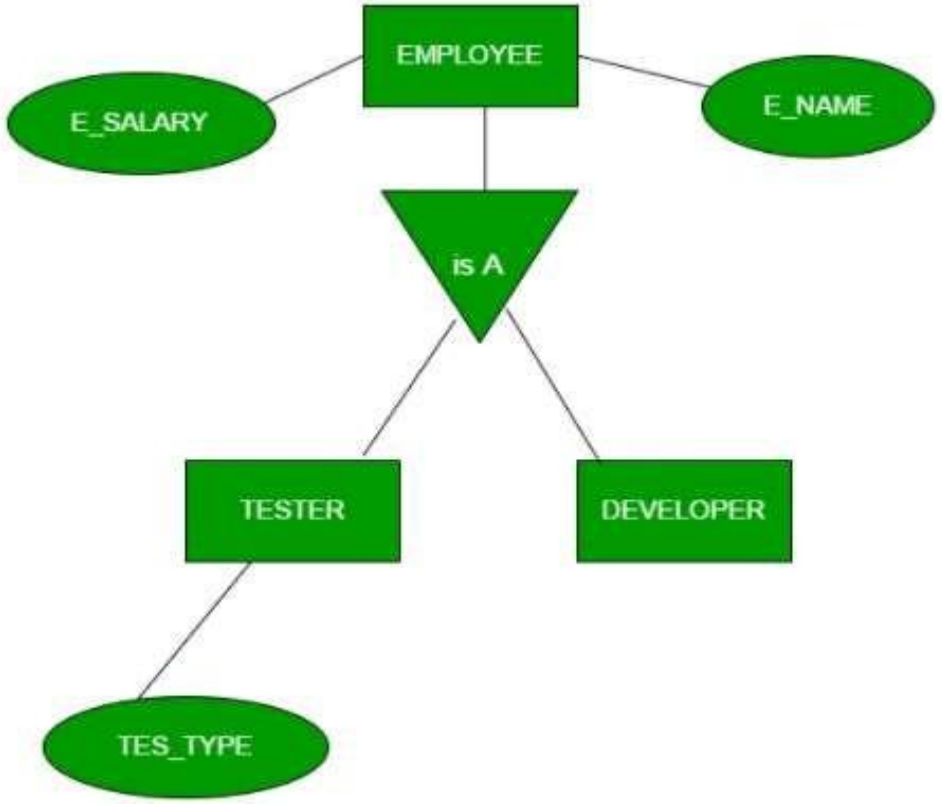


Fig. Generalization

In the above example, Tiger, Lion, Elephant can all be generalized as Animals.

Specialization

- Specialization is a process where a higher level entity is specialized into two or more lower level entities based on their characteristics.



Specialization

- Specialization is a process that defines a group entities which is divided into sub groups based on their characteristic.
- It is a top down approach, in which one higher entity can be broken down into two lower level entity.
- It maximizes the difference between the members of an entity by identifying the unique characteristic or attributes of each member.
- It defines one or more sub class for the super class and also forms the superclass/subclass relationship.

- **For example**

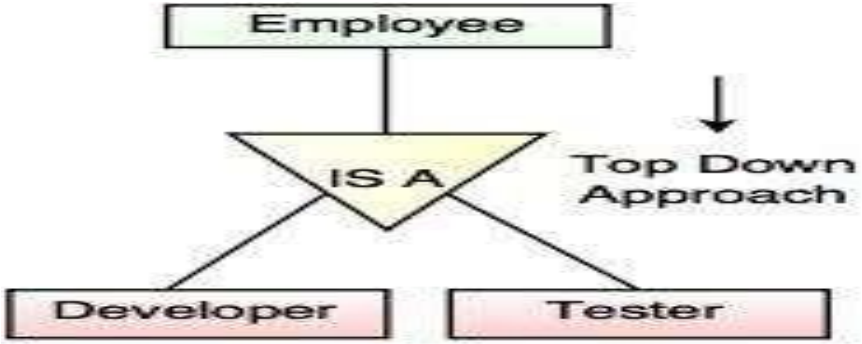


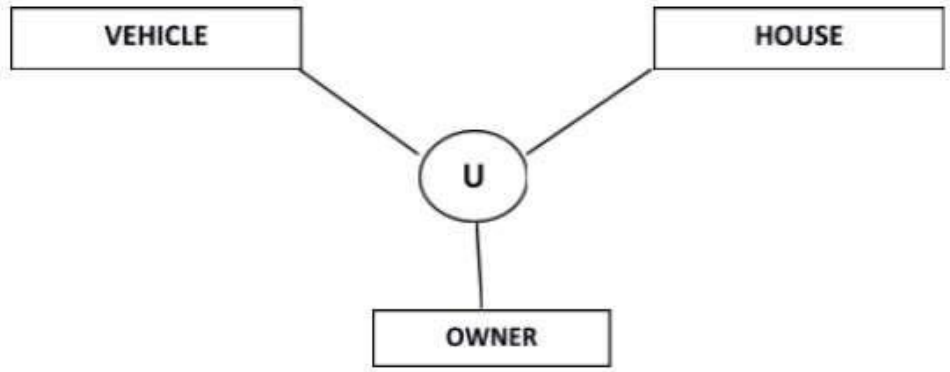
Fig. Specialization

In the above example, Employee can be specialized as Developer or Tester, based on what role they play in an Organization.

Category or Union

- Relationship of one super or sub class with more than one super class.

- Owner is the subset of two super class:Vehicle and House.

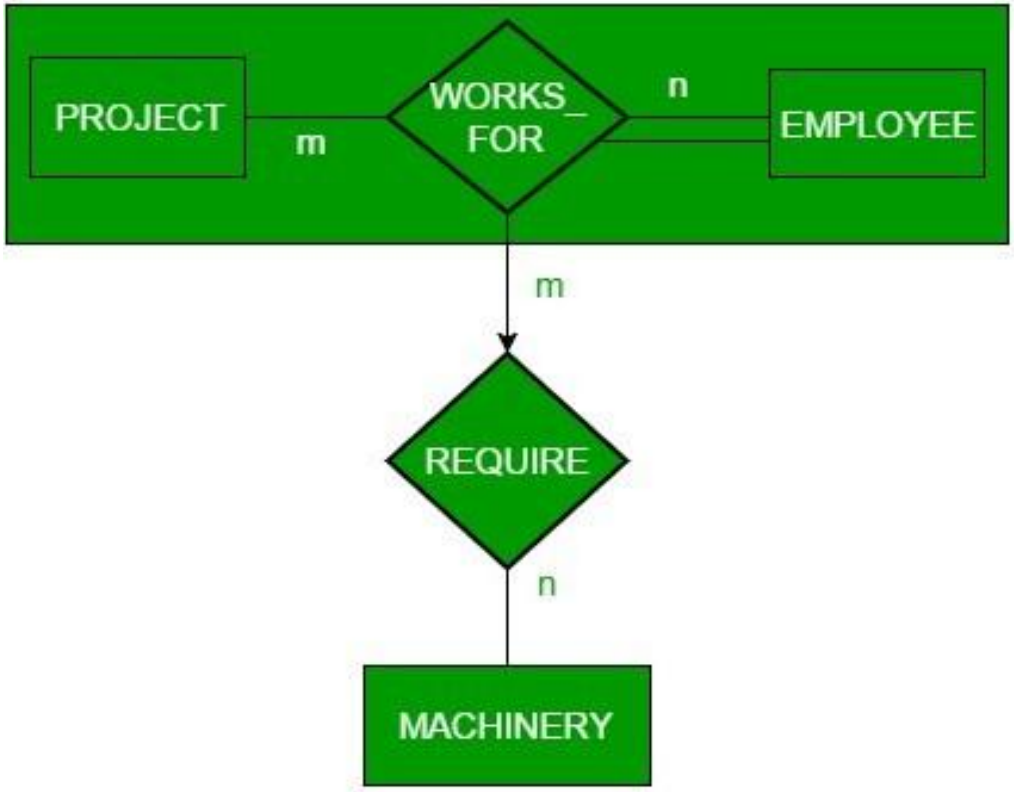


Aggregation

- Aggregation is a process that represent a relationship between a whole object and its component parts.
- It abstracts a relationship between objects and viewing the relationship as an object.
- It is a process when two entity is treated as a single entity.

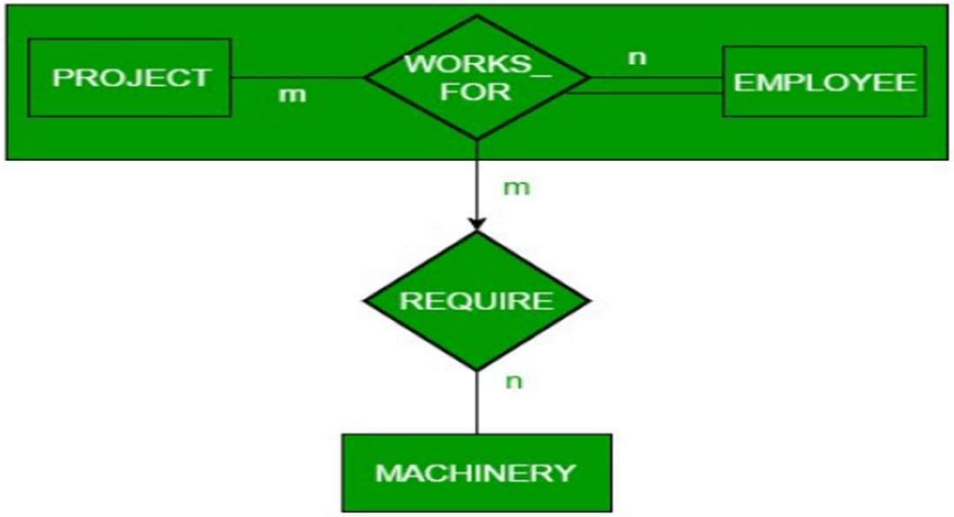
Aggregation

- Specialization is an abstraction where a relationship with its corresponding entities is aggregated into a higher level entity.



Aggregation

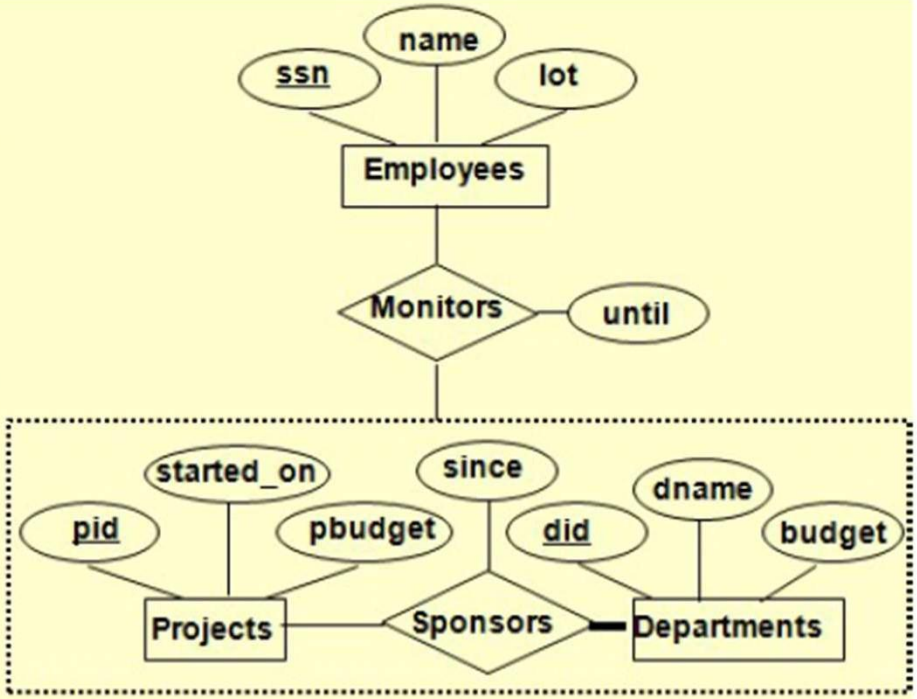
An ER diagram is not capable of representing relationship between an entity and a relationship which may be required in some scenarios. In those cases, a relationship with its corresponding entities is aggregated into a higher level entity. For Example, Employee working for a project may require some machinery. So, REQUIRE relationship is needed between relationship WORKS_FOR and entity MACHINERY. Using aggregation, WORKS_FOR relationship with its entities EMPLOYEE and PROJECT is aggregated into single entity and relationship REQUIRE is created between aggregated entityandMACHINERY. EXAMPLE:



Aggregation

Aggregation

- Used when we have to model a relationship involving (entitity sets and) a *relationship set*.



Aggregation allows us to treat a relationship set as an entity set for purposes of participation in (other) relationships.

Inheritance

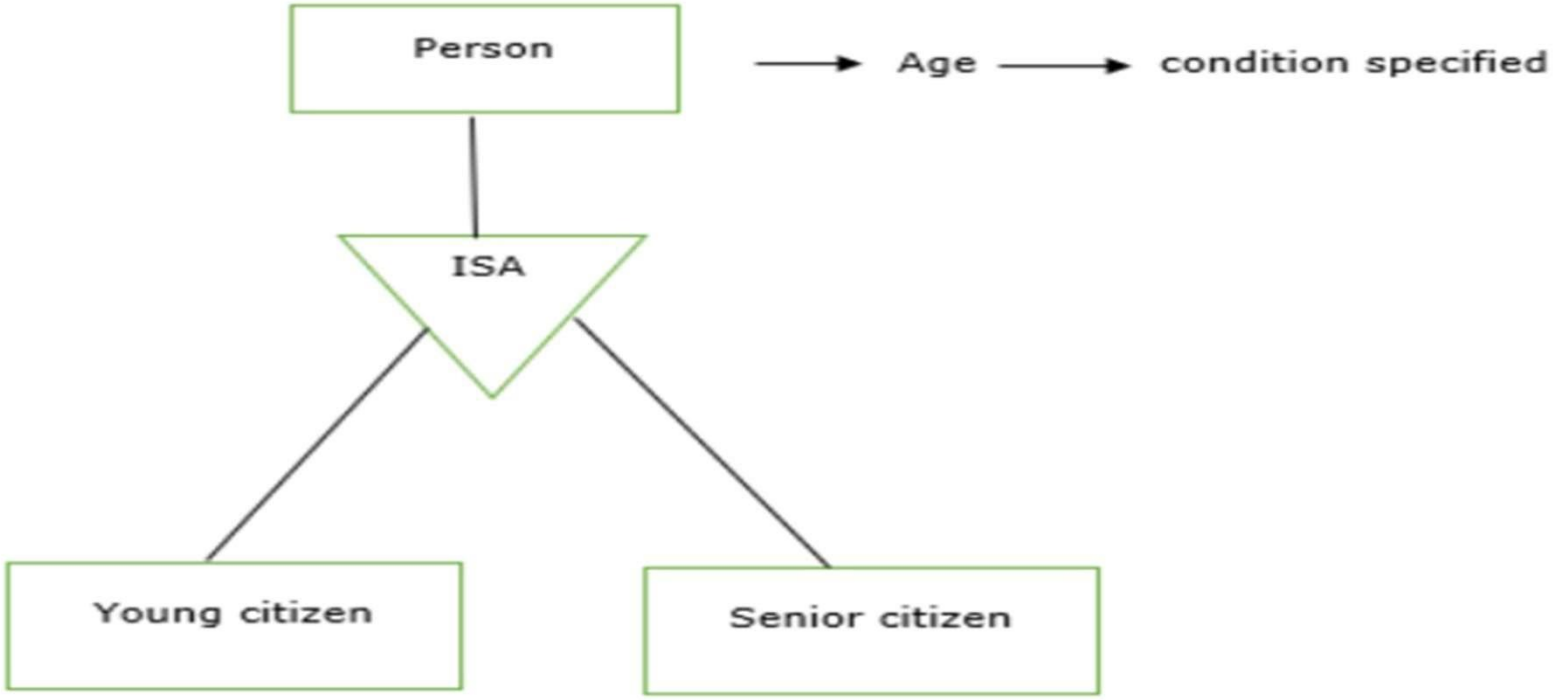
- Attributes of higher-level entity-sets are inherited by lower-level entity-sets
- Relationships involving higher-level entity-sets are also inherited by lower-level entity-sets!

Constraints on Generalization and Specialization

- Membership constraints
  - Condition-defined/attribute defined membership constraint
  - User-defined membership constraint
- Disjoint constraints
  - Disjoint constraint
  - Overlapping constraint
- Completeness constraints
  - Total completeness constraint
  - Partial completeness constraint

Constraints on Generalization and Specialization

- Membership constraints
  - Condition-defined/attribute defined membership constraint



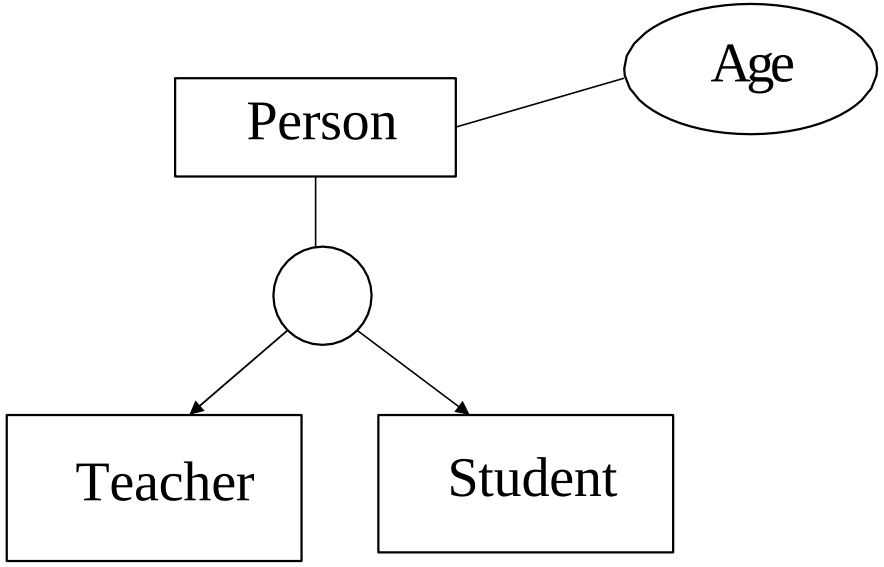
Constraints on Generalization and Specialization

Membership Constraints

i. Attribute defined

ii. User defined

Attribute Defined

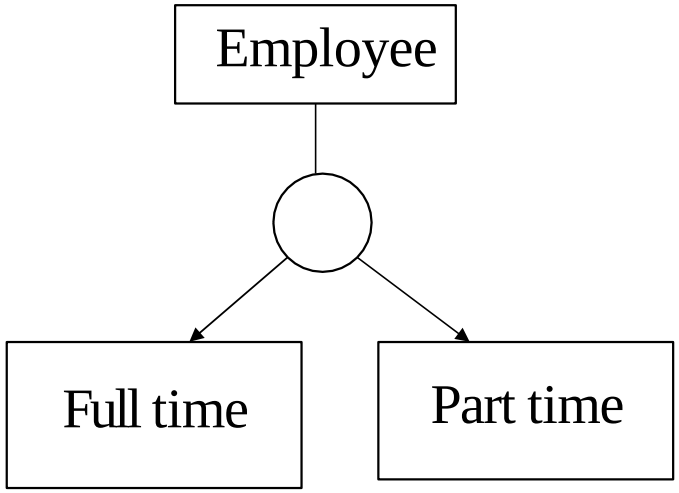


Age>18-teacher
Age<18-student
Which instance goes to teacher and student, decision will be taken by the attribute ‘age’

Constraints on Generalization and Specialization

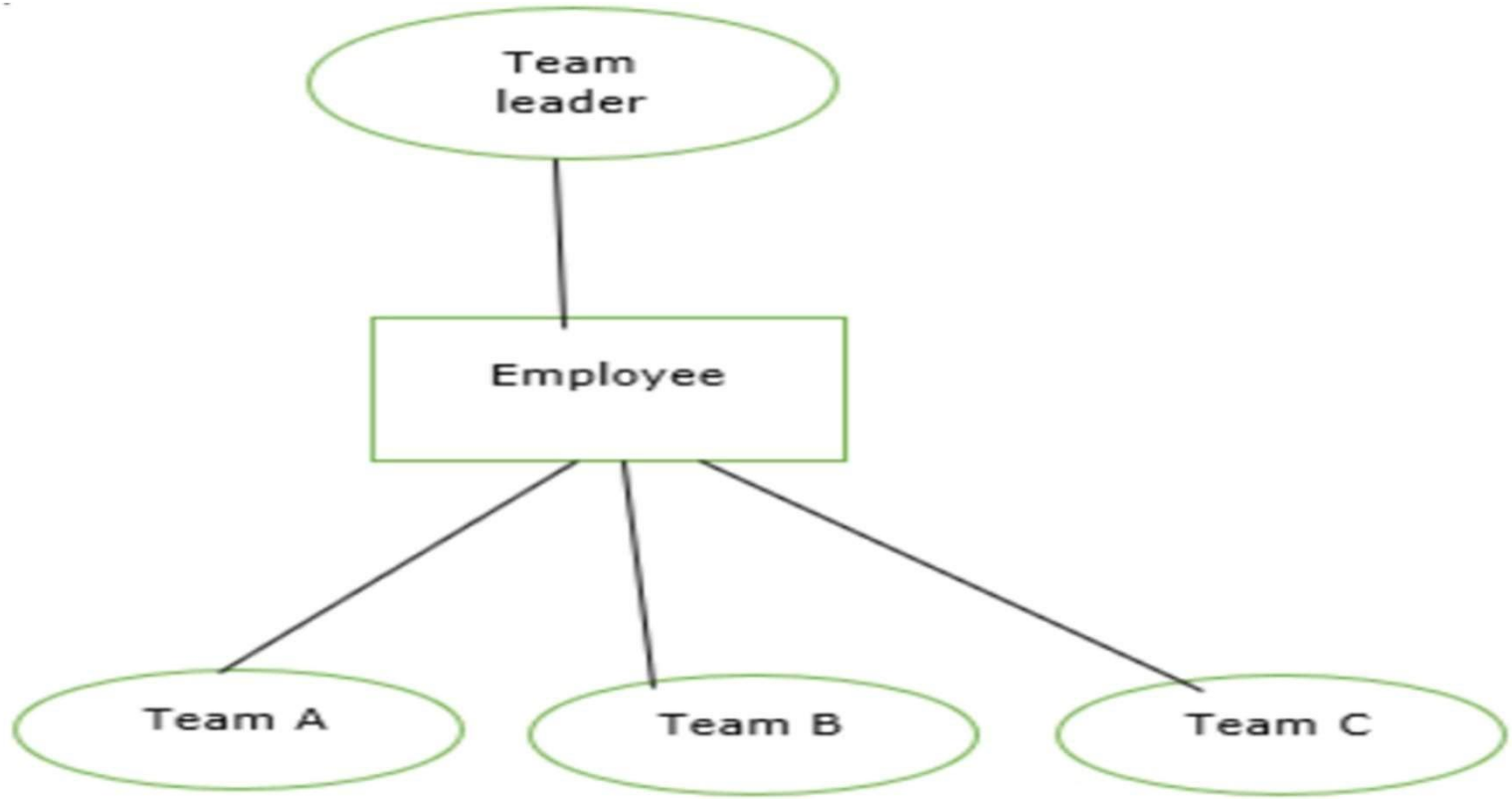
User defined (randomly)

When we will take decision, we don't have any attribute so we will take it as randomly



Constraints on Generalization and Specialization

- Membership constraints
  - User defined membership constraint (we will consider it as randomly)



Constraints on Generalization and Specialization

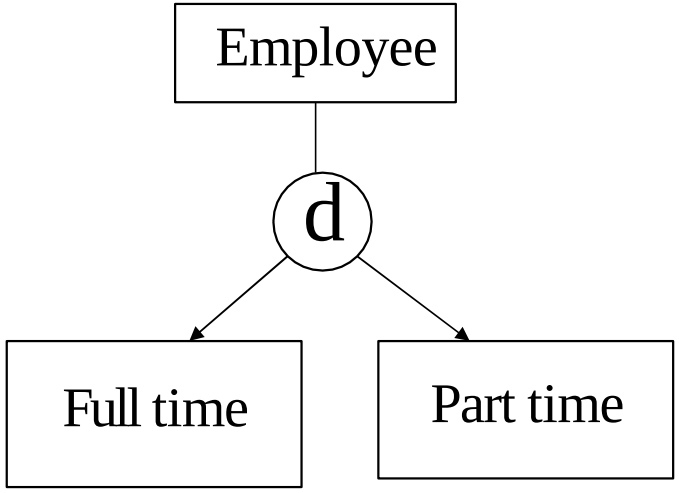
Disjoint Constraints

i. Disjoint
ii. Overlapping

Disjoint

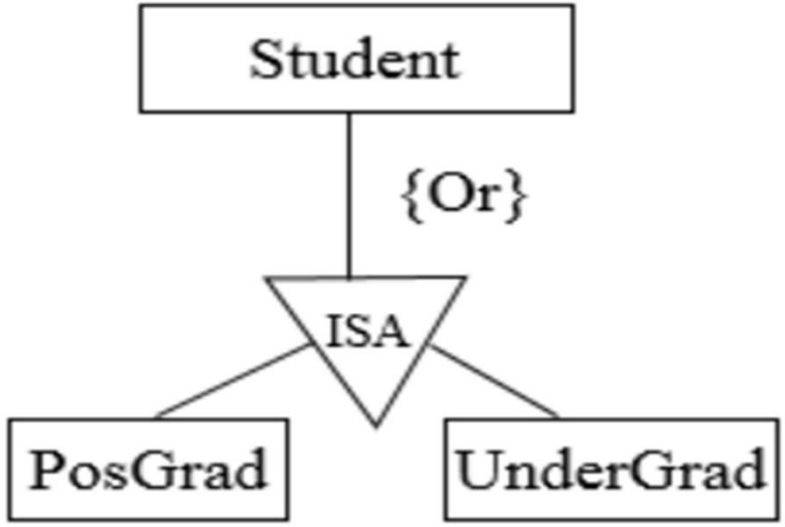
Employee has to work only either as FT or PT (both not possible)

Instance of superclass will belongs to the sub type of each of the sub classes or it can not belongs to multiple sub classes



Constraints on Generalization and Specialization

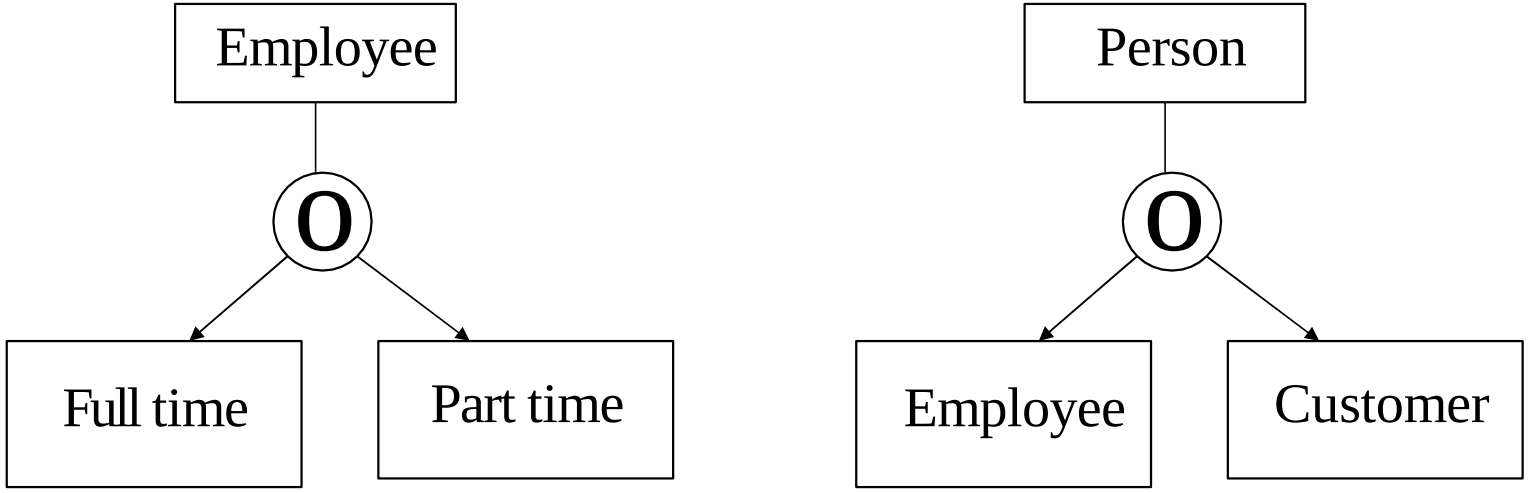
- Disjoint constraints
  - **Disjoint:** The disjoint constraint only applies when a superclass has more than one subclass. If the subclasses are disjoint, then an entity occurrence can be a member of only one of the subclasses, e.g. postgrads or undergrads, you cannot be both. **To represent a disjoint superclass/subclass relationship, Or is used. (Either PosGrad or UnderGrad but both not possible)**
  - An entity belong to no more than one lower level entity set



Constraints on Generalization and Specialization

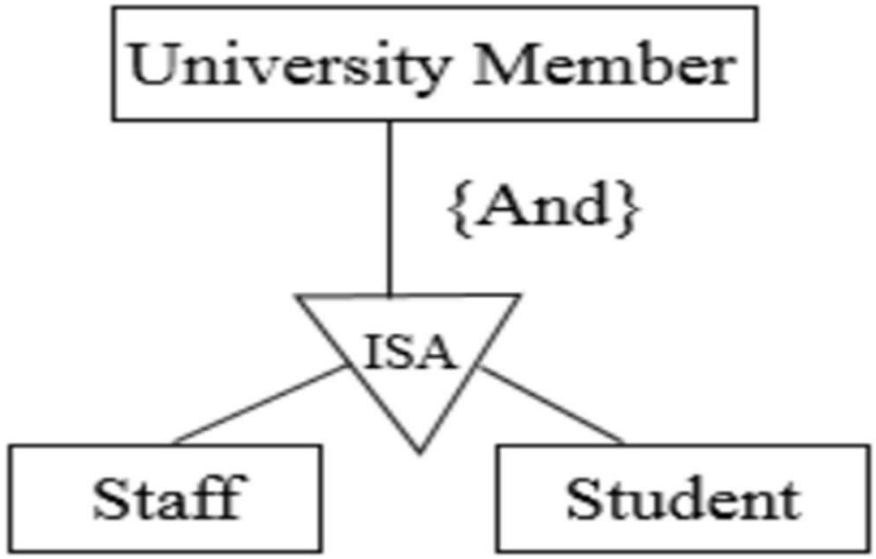
Overlapping

- Employee can be Full time and part time (both)
- Employee can work as both full time and part time
- Same entity may belong to more than one lower level entity set within a single generalization



Constraints on Generalization and Specialization

- Disjoint constraints
- **Overlapping:** This applies when an entity occurrence may be a member of more than one subclass, e.g. student and staff , some people are both. And is used to represent the overlapping specialization/generalization relationship in the ER diagram. (University member can work as staff as well as student). Example: **A person who knows Java and PHP can participate in both teams.**



Constraints on Generalization and Specialization

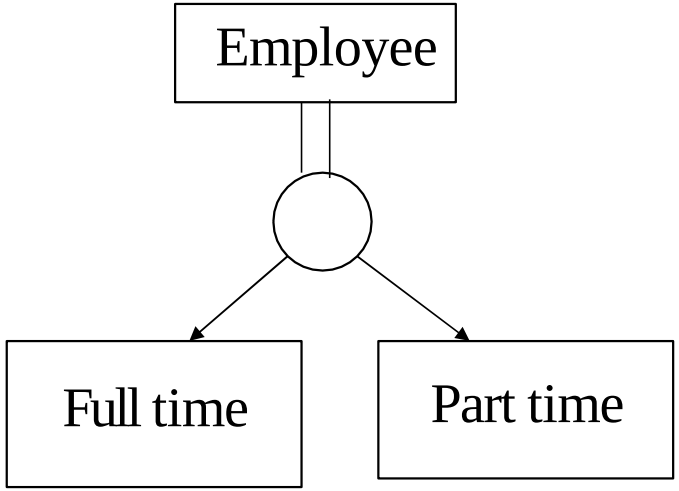
Completeness Constraint

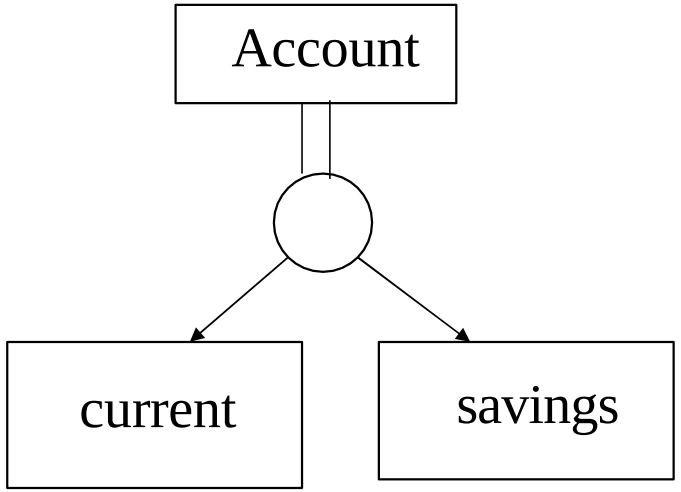
i. Total

ii. Partial

Total

- Every employee will be involving in either full time or part time. Neither of the employees will be there in full time or part time. Every employee should involve.
- Each higher level entity must belong to a lower level entity entity set

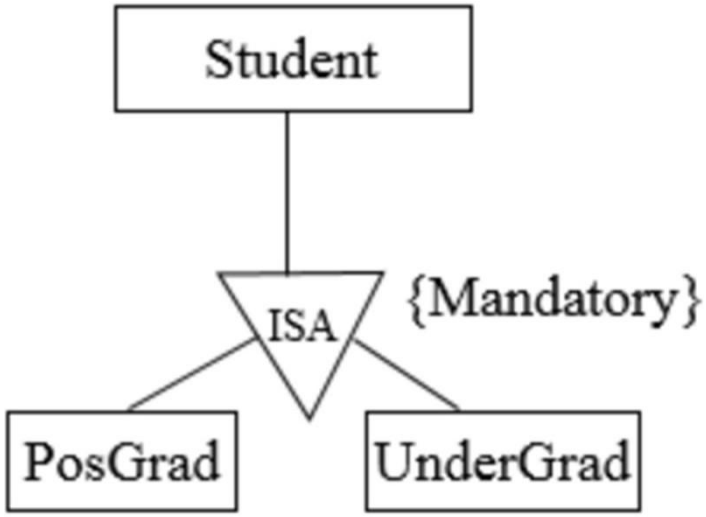




Constraints on Generalization and Specialization

- Completeness constraints

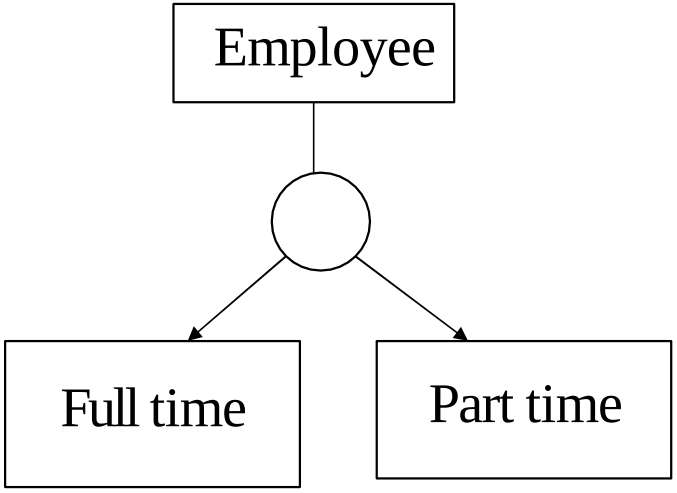
Total: Each superclass (higher-level entity) must belong to subclasses (lower-level entity sets), e.g. a student must be postgrad or undergrad. To represent completeness in the specialization/generalization relationship, the keyword Mandatory is used. (Every student will be either PosGrad or UnderGrad. Neither of the student will be there in PosGrad or UnderGrad. Every student should involve)



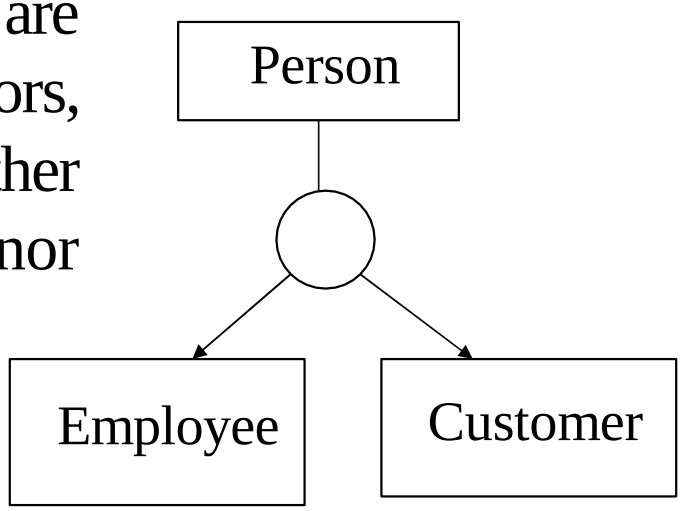
Constraints on Generalization and Specialization

Partial

- Some employee at University or company are neither full time nor part time.
- Some higher level entities may not belong to any lower level entity set. It is default constraint



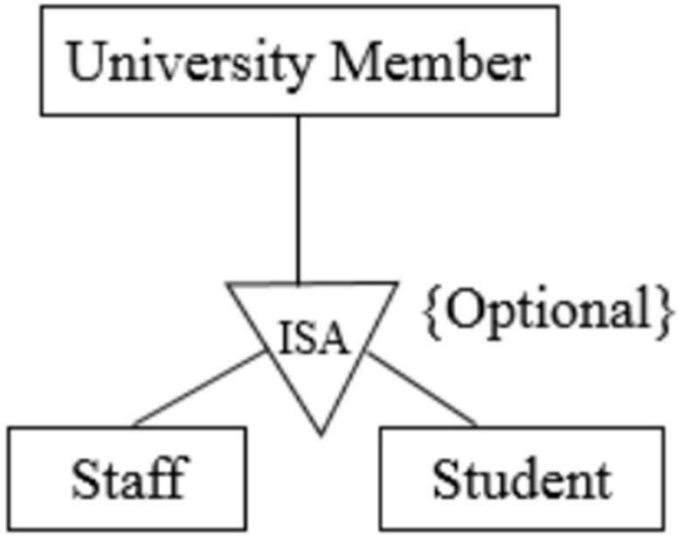
Some persons are there, they are visitors, they are neither employee nor customer



Constraints on Generalization and Specialization

- Completeness constraints

Partial: Some superclasses may not belong to subclasses (lower-level entity sets), e.g. some people at University are neither student nor staff. The keyword Optional is used to represent a partial specialization/generalization relationship. (university member can be staff and student)



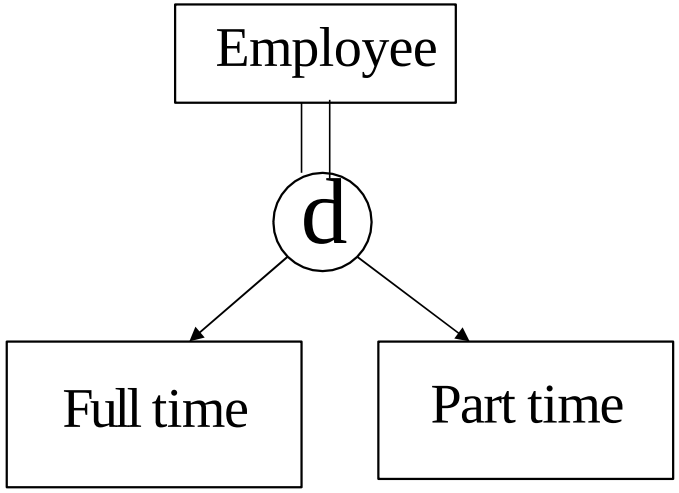
Constraints on Generalization and Specialization

- Hence, we have four types of specialization/generalization: –
- Disjoint, total
- Disjoint, partial
- Overlapping, total
- Overlapping, partial

Constraints on Generalization and Specialization

- Disjoint, total

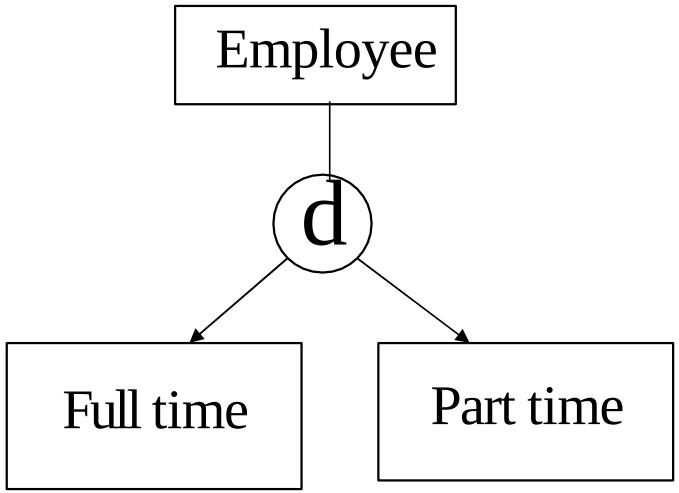
  All the instances coming, will map to one subclass only and will not be shared among other subclasses



Constraints on Generalization and Specialization

- Disjoint, partial

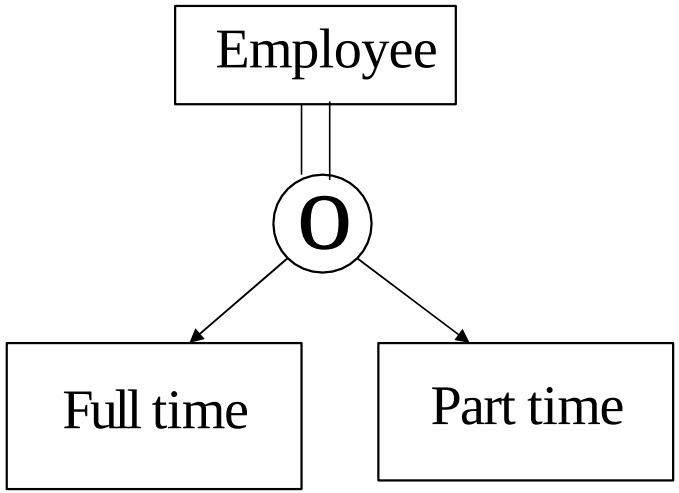
All instances coming, may stay with superclass or map to one of the sub classes only



Constraints on Generalization and Specialization

- Overlapping, total

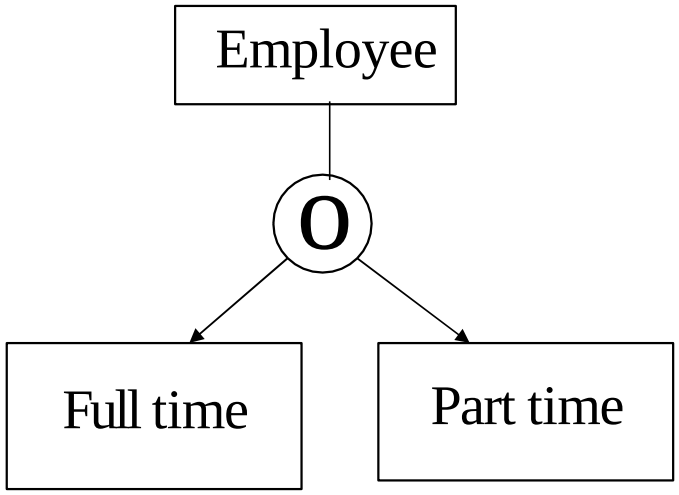
All instances coming, will map to multiple subclasses.



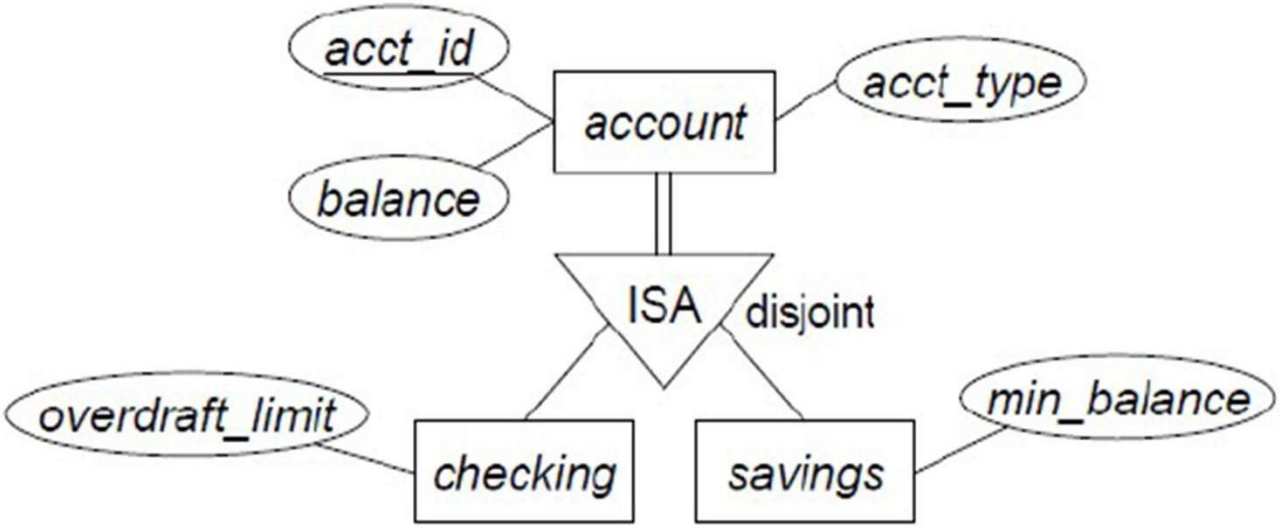
Constraints on Generalization and Specialization

- Overlapping, partial

All instances coming, may stay with the super class or map to multiple sub classes.



Schema Representation



- Schemas:

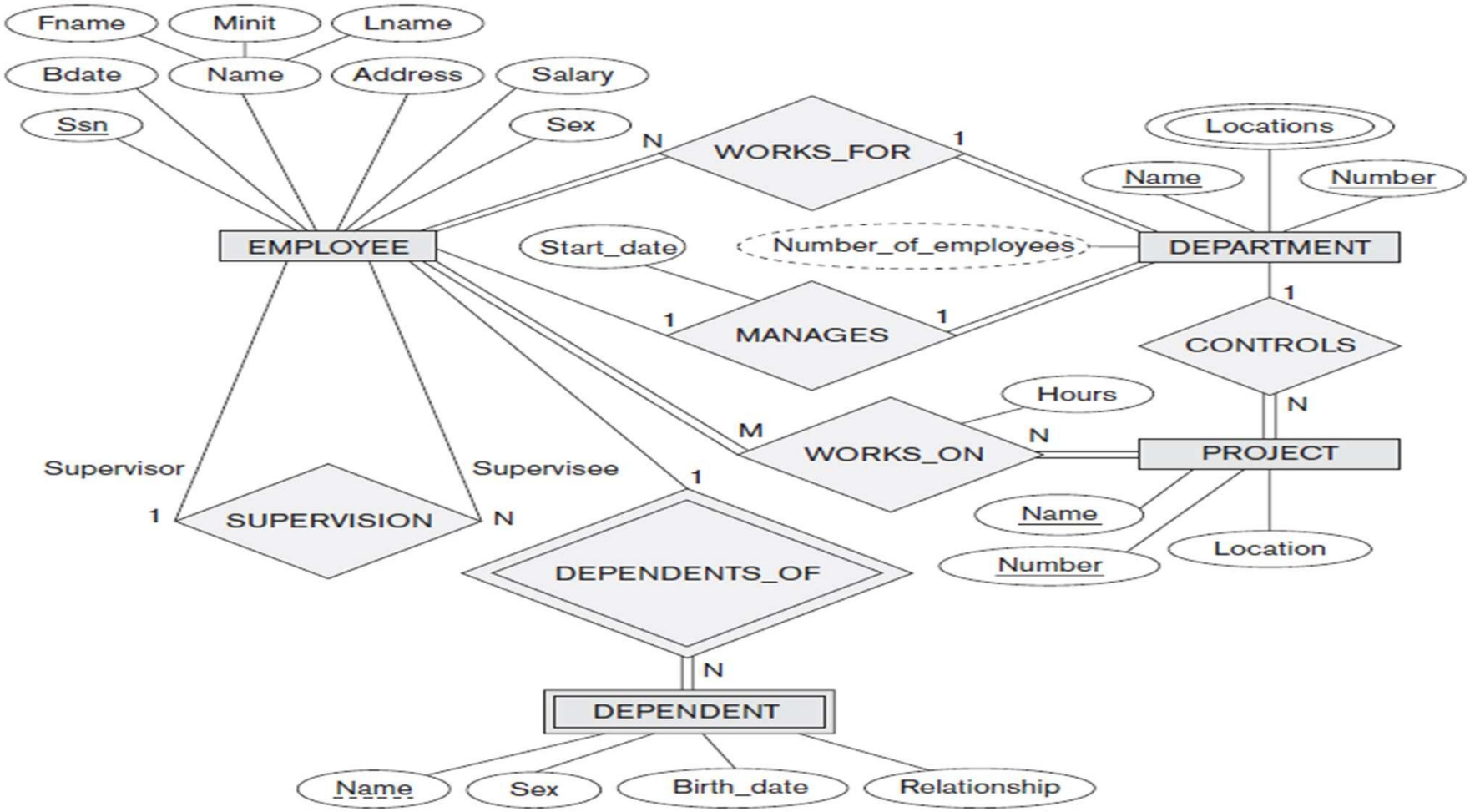
  checking(<u>*acct_id*</u>, *acct_type*, *balance*, *overdraft_limit*)

  savings(<u>*acct_id*</u>, *acct_type*, *balance*, *min_balance*)

Difference between Generalization and Specialization

GENERALIZATION	SPECIALIZATION
• Generalization works in Bottom-Up approach.	• Specialization works in top-down approach.
• In Generalization, size of schema gets reduced.	• In Specialization, size of schema gets increased.
• Generalization is normally applied to group of entities.	• We can apply Specialization to a single entity.
• Generalization can be defined as a process of creating groupings from various entity sets	• Specialization can be defined as process of creating subgrouping within an entity set
• In Generalization process, what actually happens is that it takes the union of two or more lower-level entity sets to produce a higher-level entity sets.	• Specialization is reverse of Generalization. Specialization is a process of taking a subset of a higher level entity set to form a lower-level entity set.
• Generalization process starts with the number of entity sets and it creates high-level entity with the help of some common features.	• Specialization process starts from a single entity set and it creates a different entity set by using some different features.
• In Generalization, the difference and similarities between lower entities are ignored to form a higher entity.	• In Specialization, a higher entity is split to form lower entities.
• There is no inheritance in Generalization.	• There is inheritance in Specialization.

ER Diagram of a Company Database



ER Diagram of a Company Database

- The company is organized into departments. Each department has a unique name, a unique number, and a particular employee who manages the department. We keep track of the start date when that employee began managing the department. A department may have several locations.
- A department controls a number of projects, each of which has a unique name, a unique number, and a single location. We store each employee's name, Social Security number, address, salary, sex (gender), and birth date. An employee is assigned to one department, but may work on several projects, which are not necessarily controlled by the same department. We keep track of the current number of hours per week that an employee works on each project. We also keep track of the direct supervisor of each employee (who is another employee).
- We want to keep track of the dependents of each employee for insurance purposes. We keep each dependent's first name, sex, birth date, and relationship to the employee.

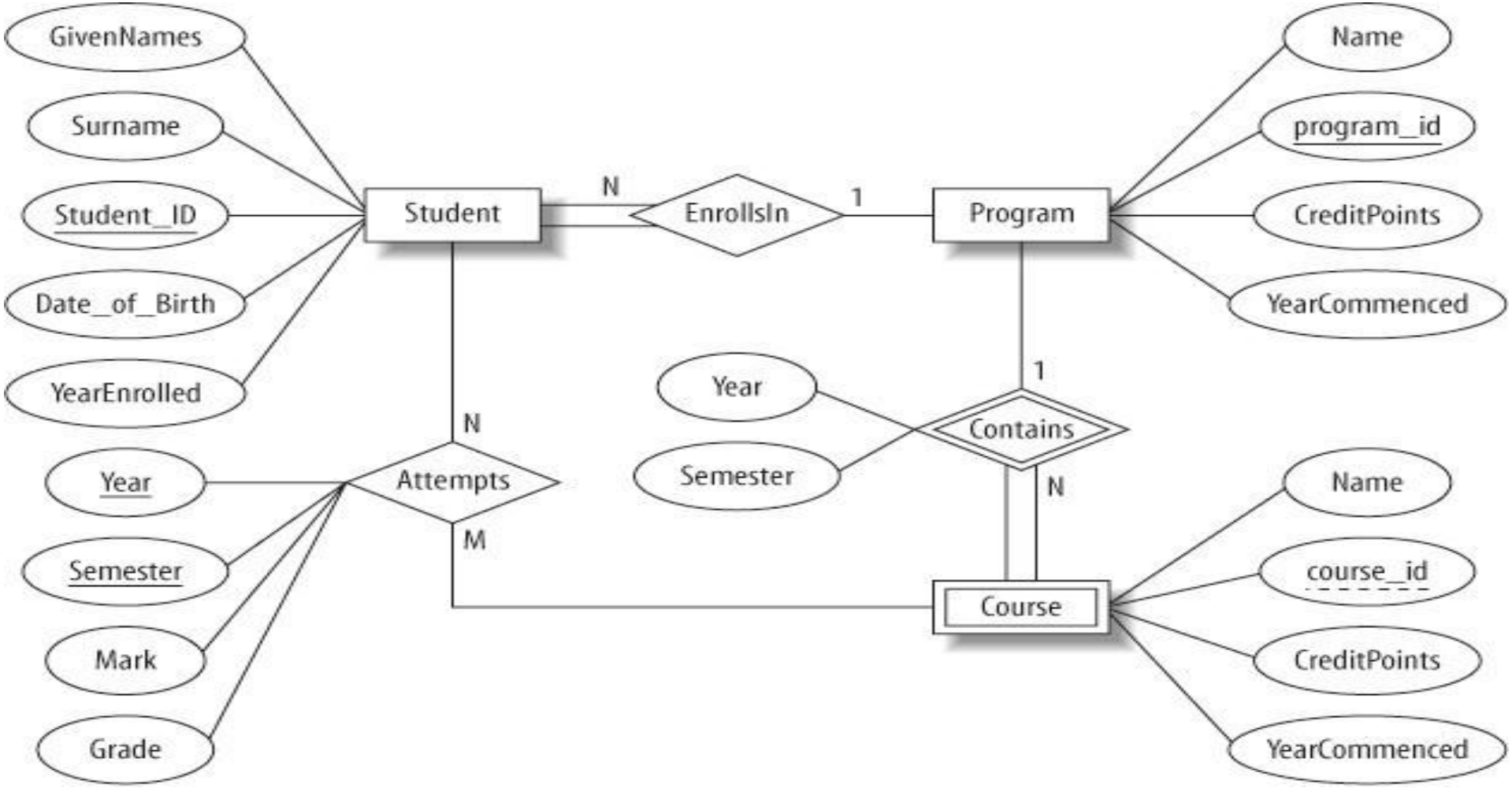
ER Diagram of a Company Database

- MANAGES, a 1:1 relationship type between EMPLOYEE and DEPARTMENT. EMPLOYEE participation is partial. DEPARTMENT participation is not clear from the requirements. We question the users, who say that a department must have a manager at all times, which implies total participation. The attribute Start_date is assigned to this relationship type.
- WORKS_FOR, a 1:N relationship type between DEPARTMENT and EMPLOYEE. Both participations are total.
- CONTROLS, a 1:N relationship type between DEPARTMENT and PROJECT. The participation of PROJECT is total, whereas that of DEPARTMENT is determined to be partial, after consultation with the users indicates that some departments may control no projects.
- SUPERVISION, a 1:N relationship type between EMPLOYEE (in the supervisor role) and EMPLOYEE (in the supervisee role). Both participations are determined to be partial, after the users indicate that not every employee is a supervisor and not every employee has a supervisor.
- WORKS_ON, determined to be an M:N relationship type with attribute, Hours, after the users indicate that a project can have several employees working on it. Both participations are determined to be total.
- DEPENDENTS_OF, a 1:N relationship type between EMPLOYEE and DEPENDENT, which is also the identifying relationship for the weak entity type DEPENDENT. The participation of EMPLOYEE is partial, whereas that of DEPENDENT is total.

ER Diagram of a Company Database

- **Employees works in Departments –**
  Many employee works in one Department but one employee can not work in many departments.
- **Manager controls a Department –**
  employee works under the manager of the Department and the manager records the date of joining of employee in the department.
- **Department has many Projects –**
  One department has many projects but one project can not come under many departments.
- **Employee works on project –**
  One employee works on several projects and the number of hours worked by the employee on a single project is recorded.
- **Employee has dependents –**
  Each Employee has dependents. Each dependent is dependent of only one employee.

ER Diagram of a University Database



ER Diagram of a University Database

The university database stores details about university students, courses, the semester a student took a particular course (and his mark and grade if he completed it), and what degree program each student is enrolled in.

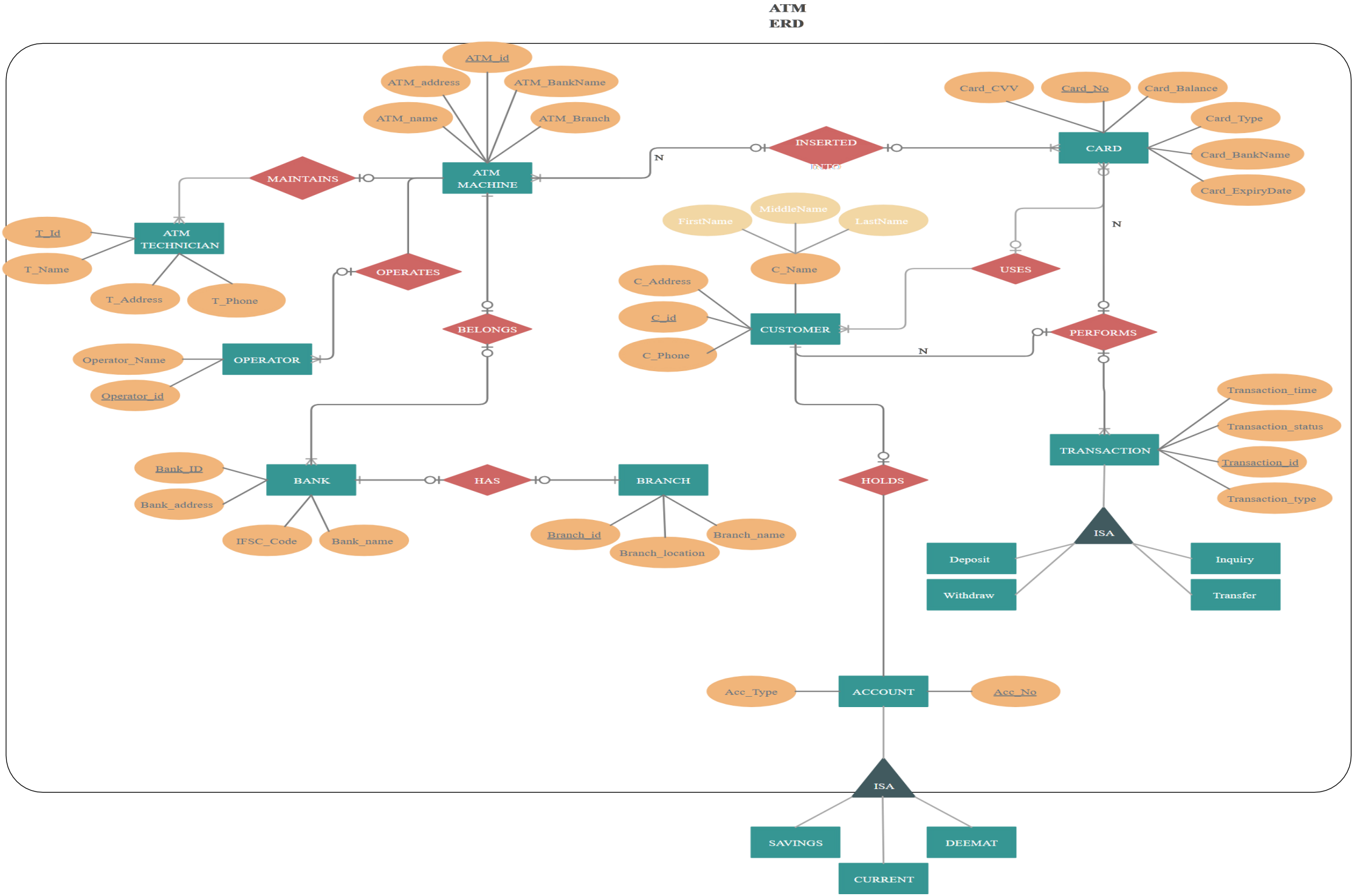
Consider the following requirements list:

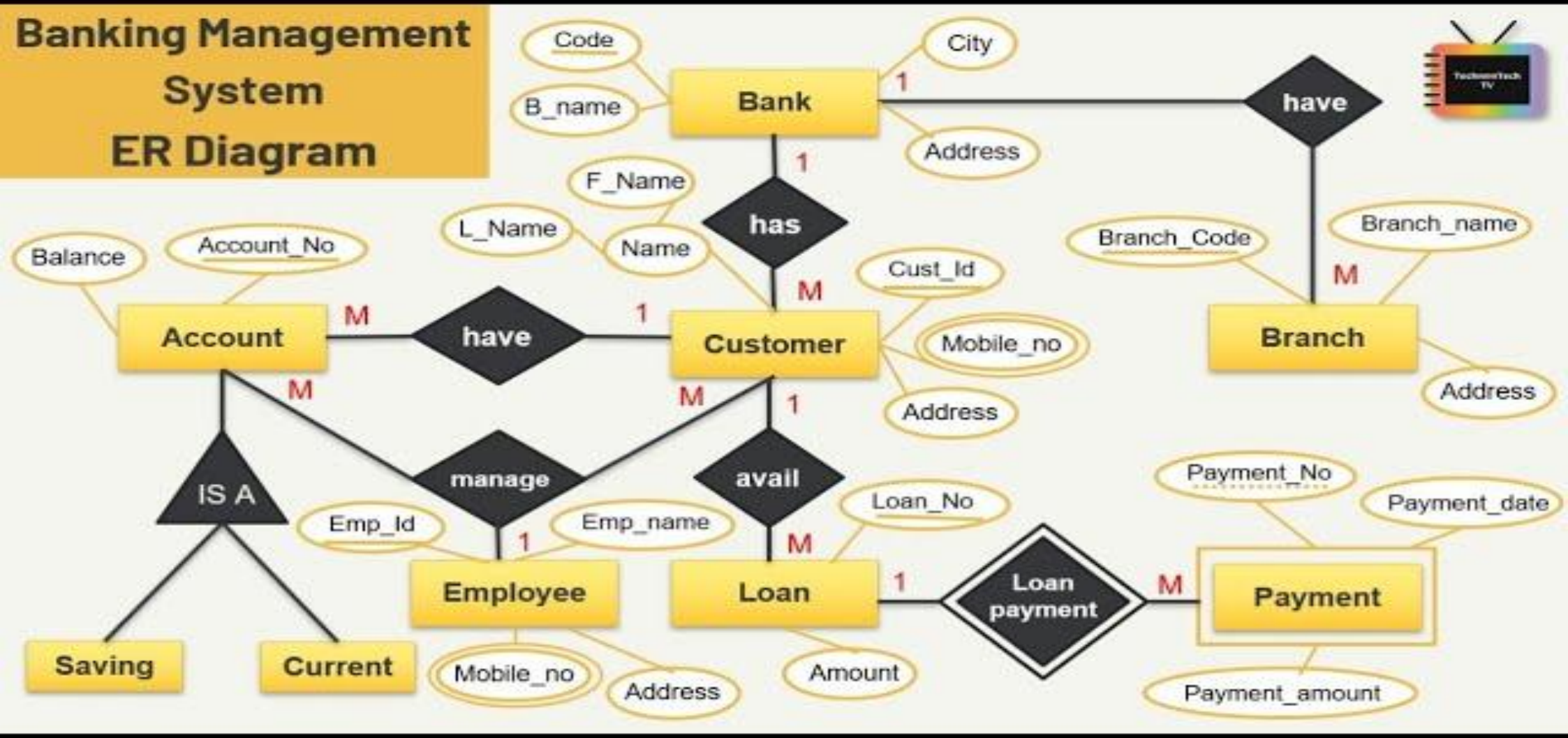
- The university offers one or more programs.
- A program is made up of one or more courses.
- A student must enroll in a program.
- A student takes the courses that are part of her program.
- A program has a name, a program identifier, the total credit points required to graduate, and the year it commenced.
- A course has a name, a course identifier, a credit point value, and the year it commenced.
- Students have one or more given names, a surname, a student identifier, a date of birth, and the year they first enrolled. We can treat all given names as a single object—for example, “John Paul.”
- When a student takes a course, the year and semester he attempted it are recorded. When he finishes the course, a grade (such as A or B) and a mark (such as 60 percent) are recorded.
- Each course in a program is sequenced into a year (for example, year 1) and a semester (for example, semester 1).

ER Diagram of a University Database

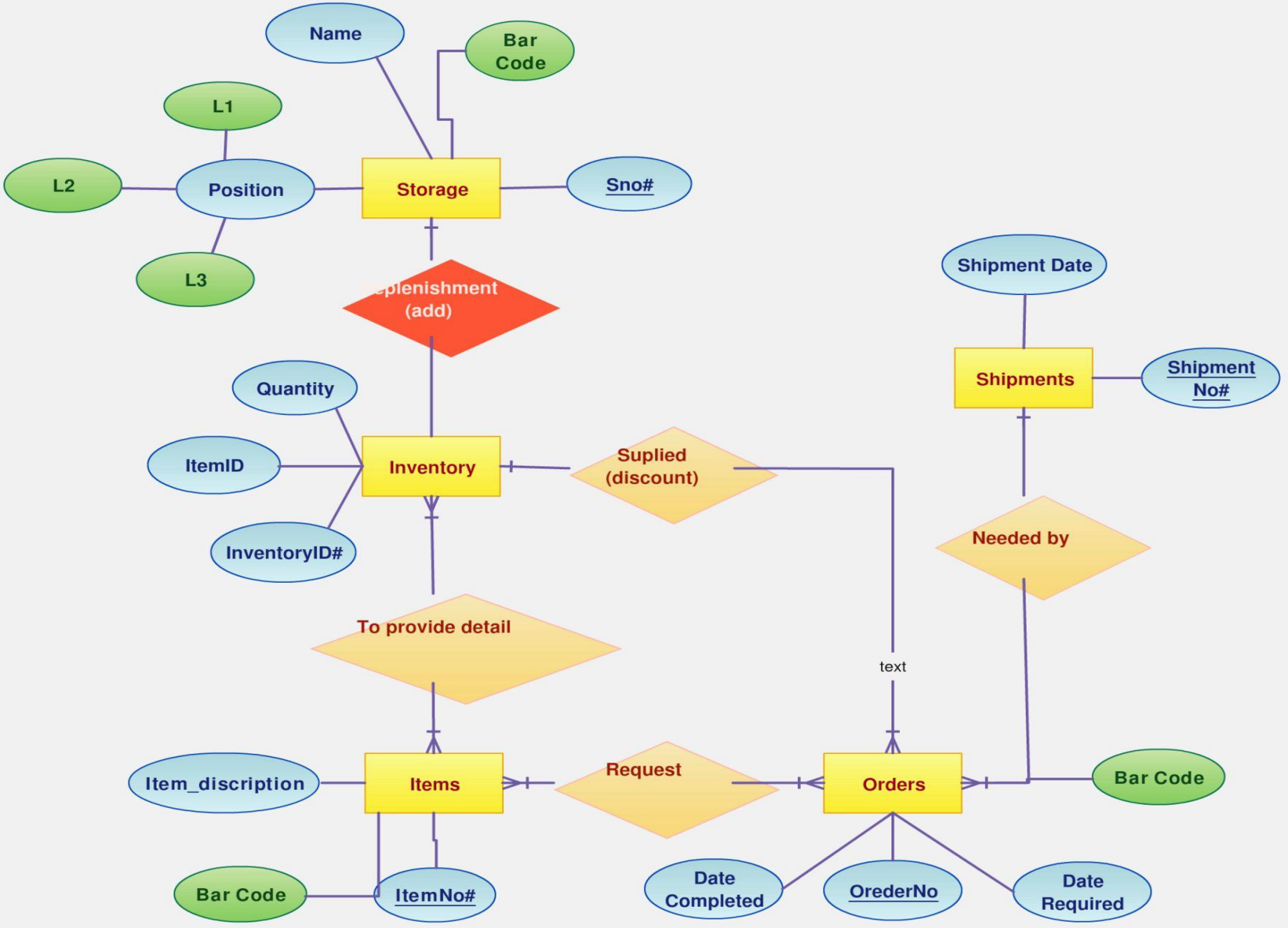
- Student is a strong entity, with an identifier, student_id, created to be the primary key used to distinguish between students (remember, we could have several students with the same name).
- Program is a strong entity, with the identifier program_id as the primary key used to distinguish between programs.
- Each student must be enrolled in a program, so the Student entity participates totally in the many-to-one Enrolls In relationship with Program. A program can exist without having any enrolled students, so it participates partially in this relationship.
- A Course has meaning only in the context of a Program, so it's a weak entity, with course_id as a weak key. This means that a Course is uniquely identified using its course_id and the program_id of its owning program.
- As a weak entity, Course participates totally in the many-to-one identifying relationship with its owning Program. This relationship has Year and Semester attributes that identify its sequence position.
- Student and Course are related through the many-to-many Attempts relationships; a course can exist without a student, and a student can be enrolled without attempting any courses, so the participation is not total.
- When a student attempts a course, there are attributes to capture the Year and Semester, and the Mark and Grade.

Draw an E-R Diagram for ATM System for Bank System

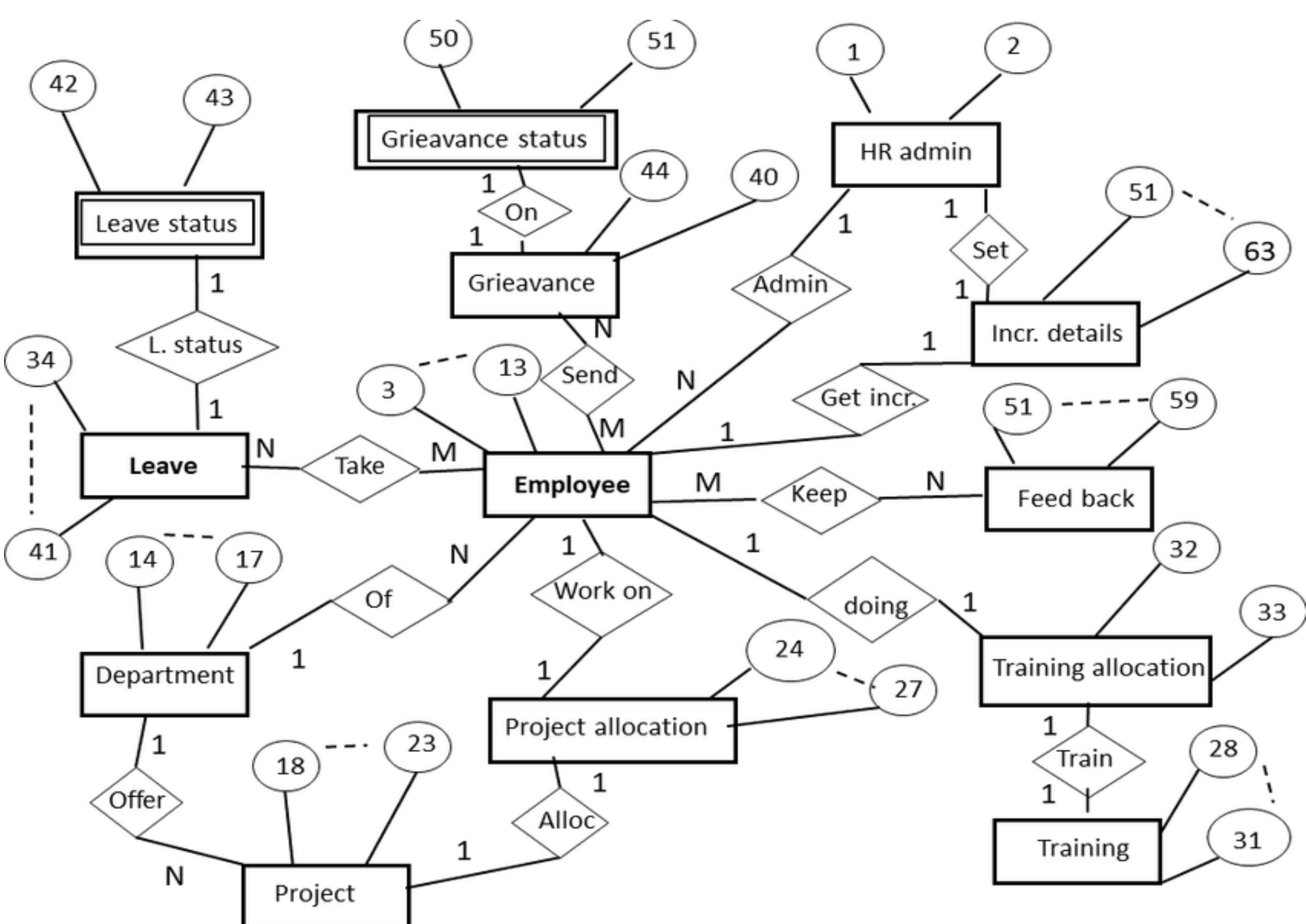


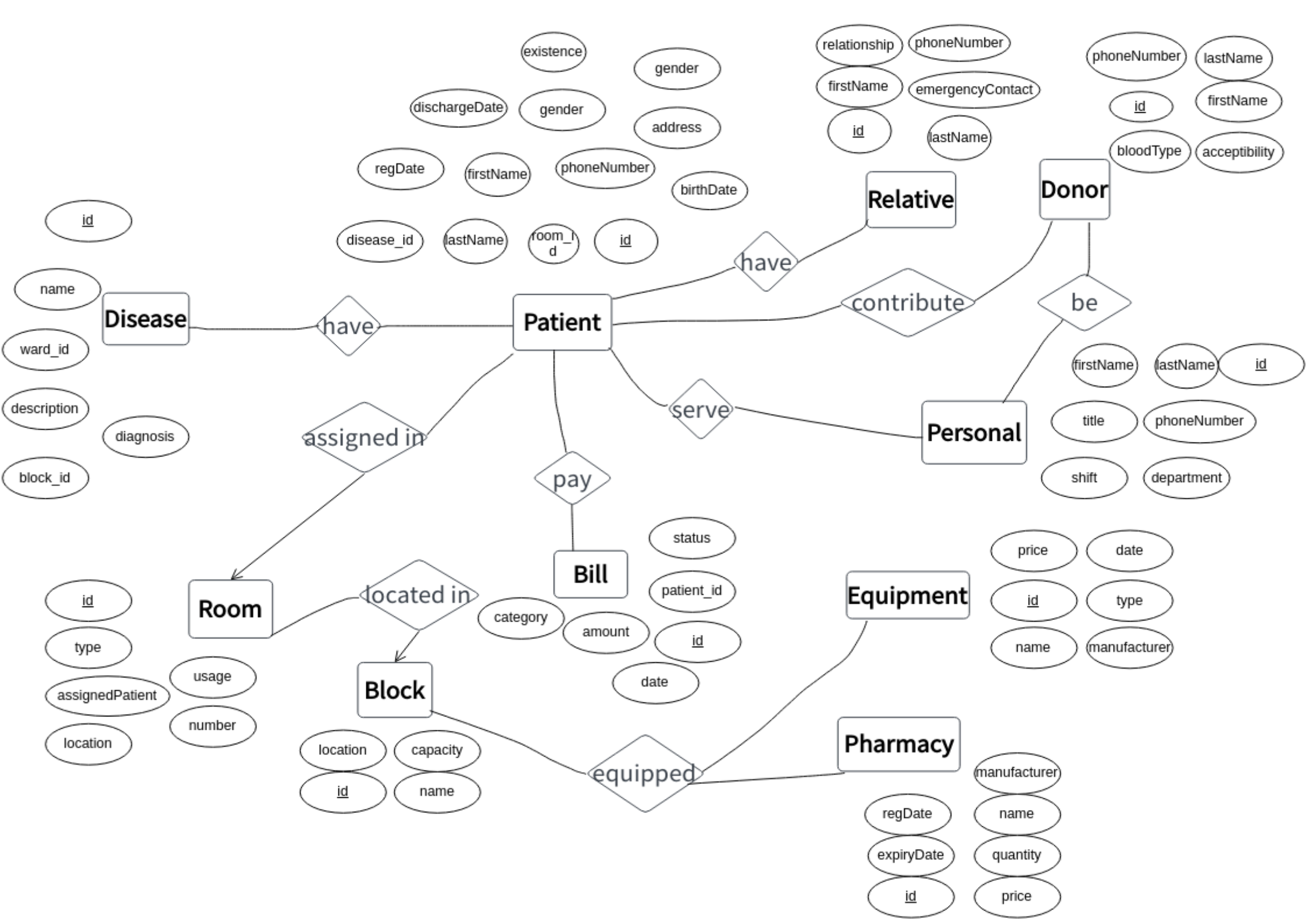

Banking Management System ER Diagram
Code
B_name
Bank
City
Address
1
1
have
F_Name
L_Name
Name
has
M
Cust_Id
Mobile_no
Address
Customer
Balance
Account_No
Account
M
have
1
M
M
1
IS A
manage
avail
Emp_Id
Emp_name
1
M
Employee
Loan_No
Loan
1
Loan payment
M
Payment
Payment_No
Payment_date
Payment_amount
Amount
Saving
Current
Mobile_no
Address
Branch_Code
Branch_name
M
Branch
Address

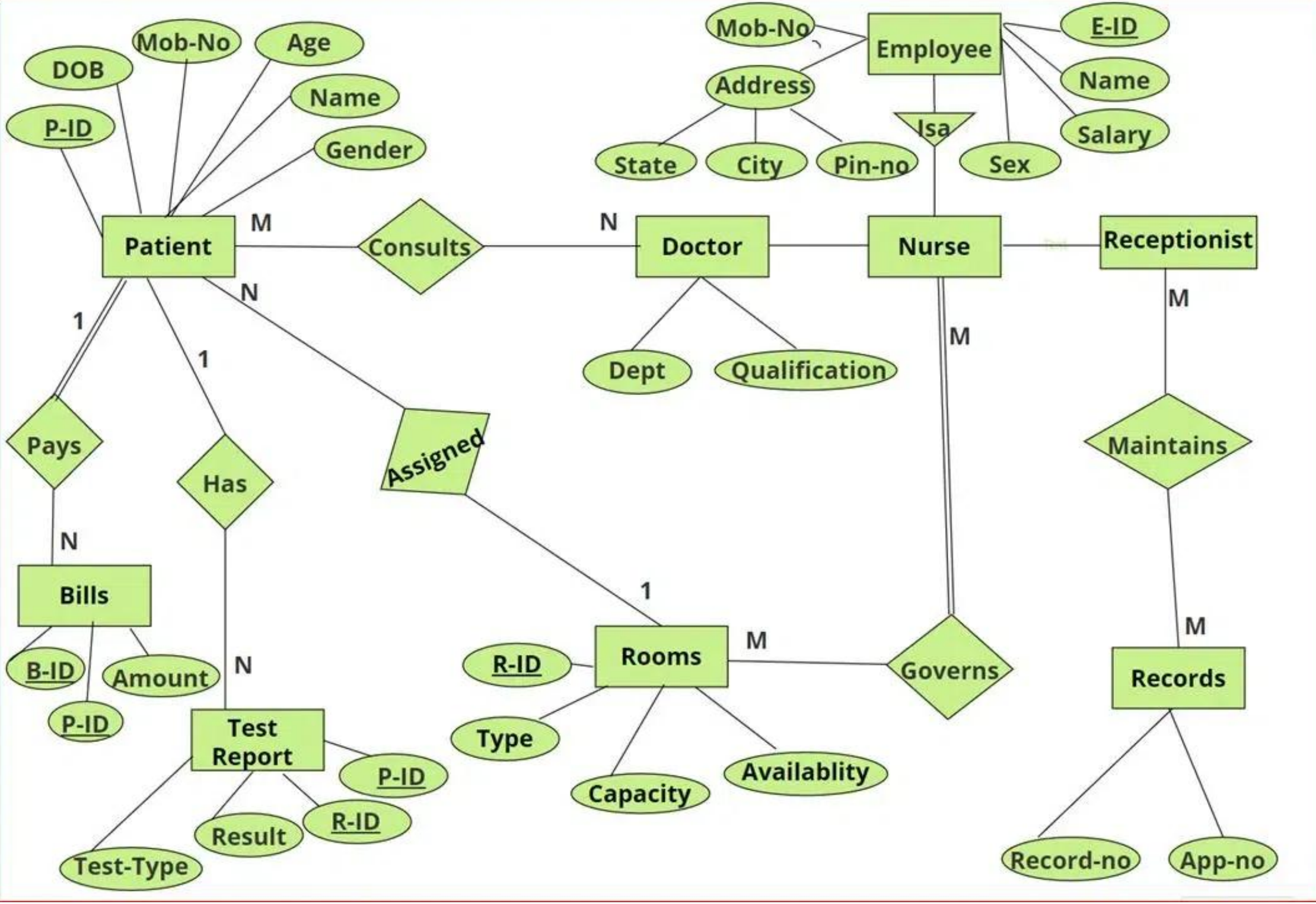
E-R Diagram for Inventory Management System



creately Diagramming


50
51
42
43
Grieavance status
1
On
1
Leave status
44
40
Grieavance
1
L. status
N
34
3
13
Send
M
Leave
N
Take
M
Employee
1
Get incr.
1
Incr. details
1
Set
1
HR admin
1
2
Admin
1
N
51
63
51
59
Feed back
M
Keep
N
41
14
17
N
Of
1
Work on
1
1
doing
1
32
33
Department
1
24
27
Project allocation
Training allocation
1
Train
1
Offer
18
23
1
Alloc
1
28
31
Training
N
Project


existence
gender
dischargeDate
gender
address
regDate
firstName
phoneNumber
birthDate
id
disease_id
lastName
room_id
id
relationship
phoneNumber
firstName
emergencyContact
id
lastName
phoneNumber
lastName
id
firstName
bloodType
acceptibility
Relative
Donor
have
contribute
be
Disease
have
Patient
name
ward_id
description
diagnosis
block_id
assigned in
pay
serve
Personal
firstName
lastName
id
title
phoneNumber
shift
department
status
Bill
patient_id
category
amount
id
date
Room
located in
Equipment
price
date
id
type
name
manufacturer
id
type
assignedPatient
location
usage
number
Block
location
capacity
id
name
equipped
Pharmacy
manufacturer
name
quantity
price
regDate
expiryDate
id


Mob-No
Age
DOB
Name
P-ID
Gender
Patient
M
Consults
N
Doctor
Mob-No
Employee
E-ID
Address
Name
Isa
Salary
State
City
Pin-no
Sex
Nurse
Receptionist
1
N
M
M
Dept
Qualification
1
Pays
Has
Assigned
Maintains
N
1
M
M
Bills
Rooms
Governs
Records
B-ID
Amount
R-ID
N
P-ID
Test Report
Type
P-ID
Capacity
Availablity
R-ID
Result
Test-Type
Record-no
App-no

Thank You